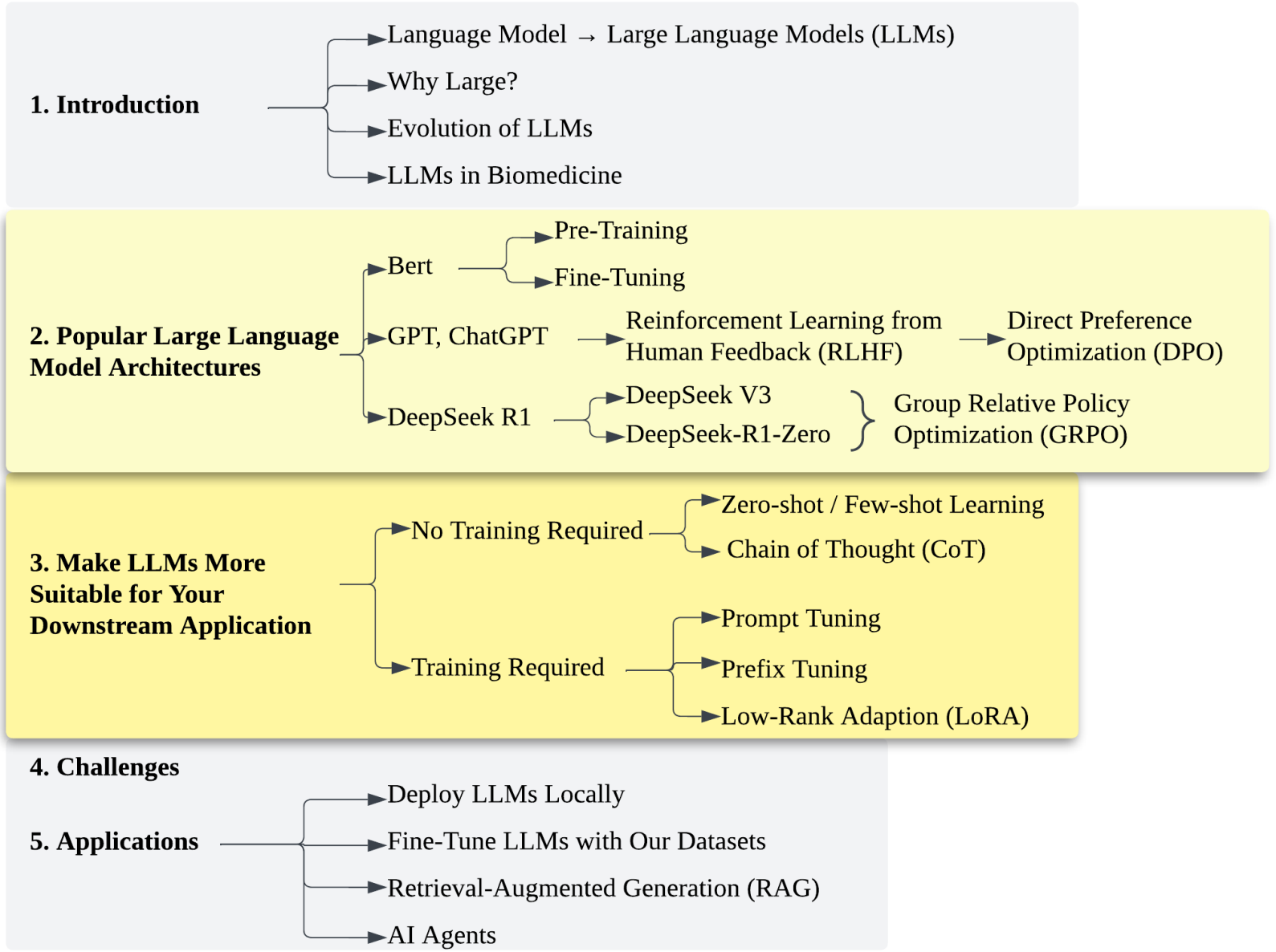
Bios 740- Chapter 8. Large Language Models

Acknowledgement: Many thanks to Dr. Shan Gao and Mr. Runpeng Dai for preparing some of these slides. I also drew on material from the lecture presentations of Stanford CS224w and StanfordCS231n as well as content generated by ChatGPT.

Overview of This Chapter

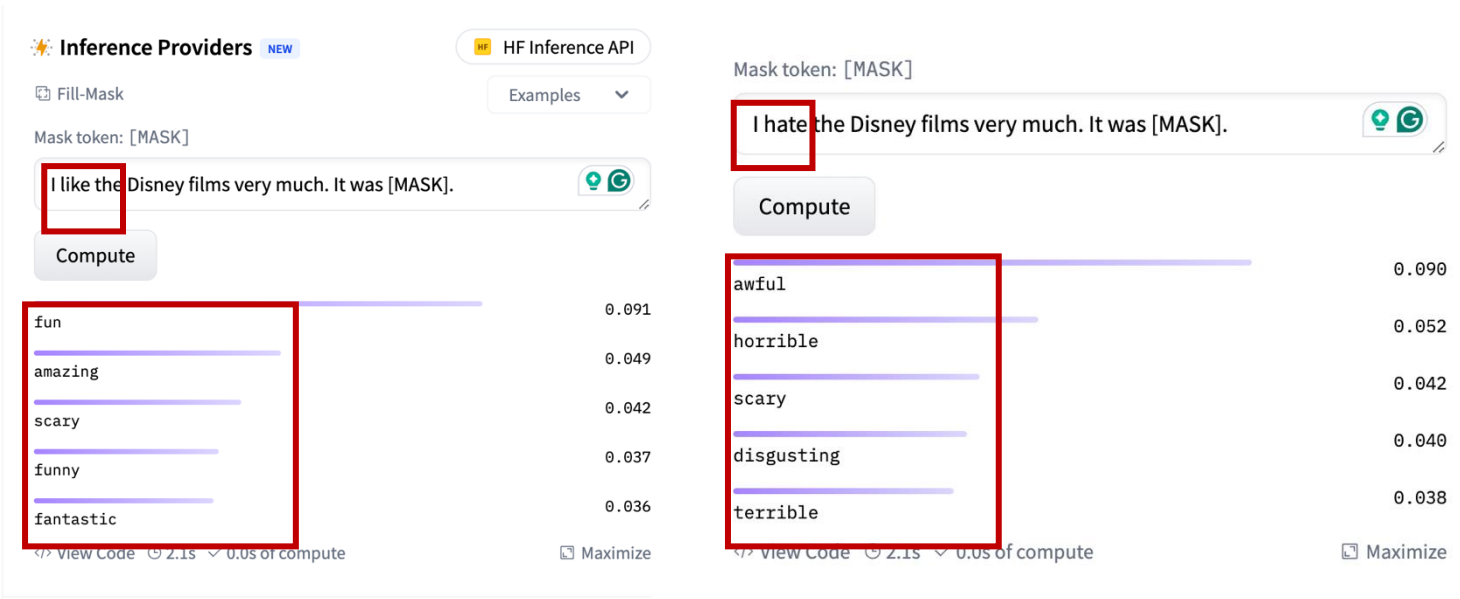


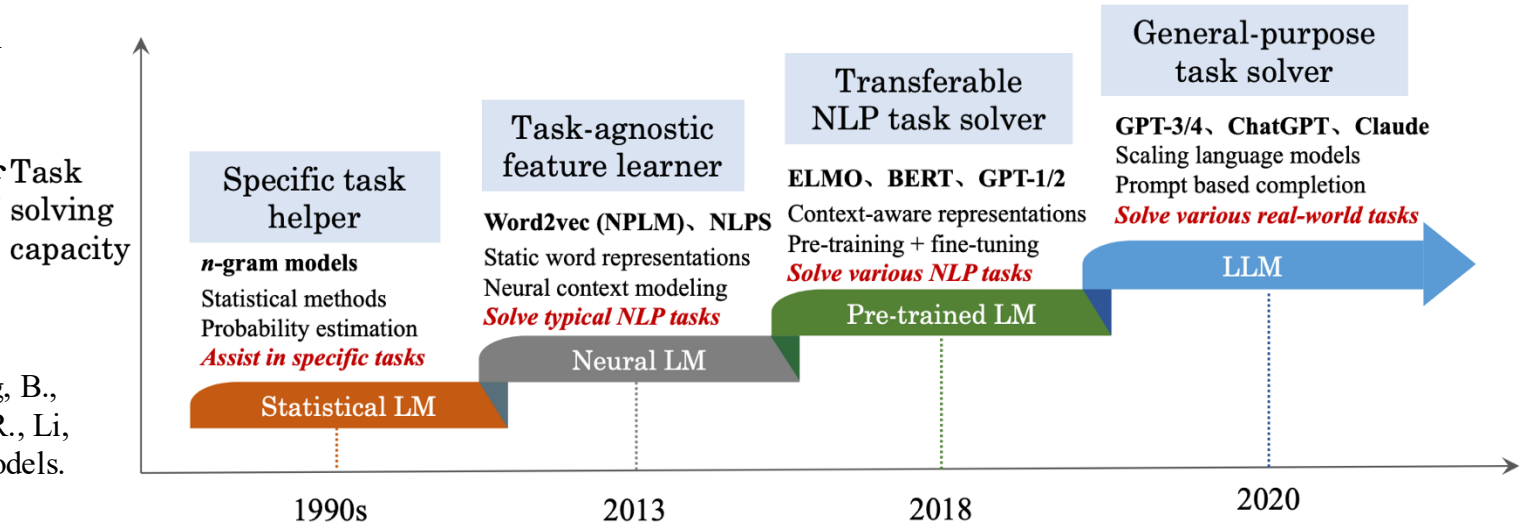
Content

1. Introduction to Large Language Models

Language Models >>Large LMs (LLMs)

- **Definition:** A language model aims to predict the probability of the occurrence of a token or a sequence of tokens.
- The probability prediction of a language model is closely related to **context** and **corpus** information.
- Language model is **not a new technical** concept specially for LLMs, but has evolved with the advance of artificial intelligence over the decades.
- **Definition:** LLMs have billions of parameters, trained on massive corpora.



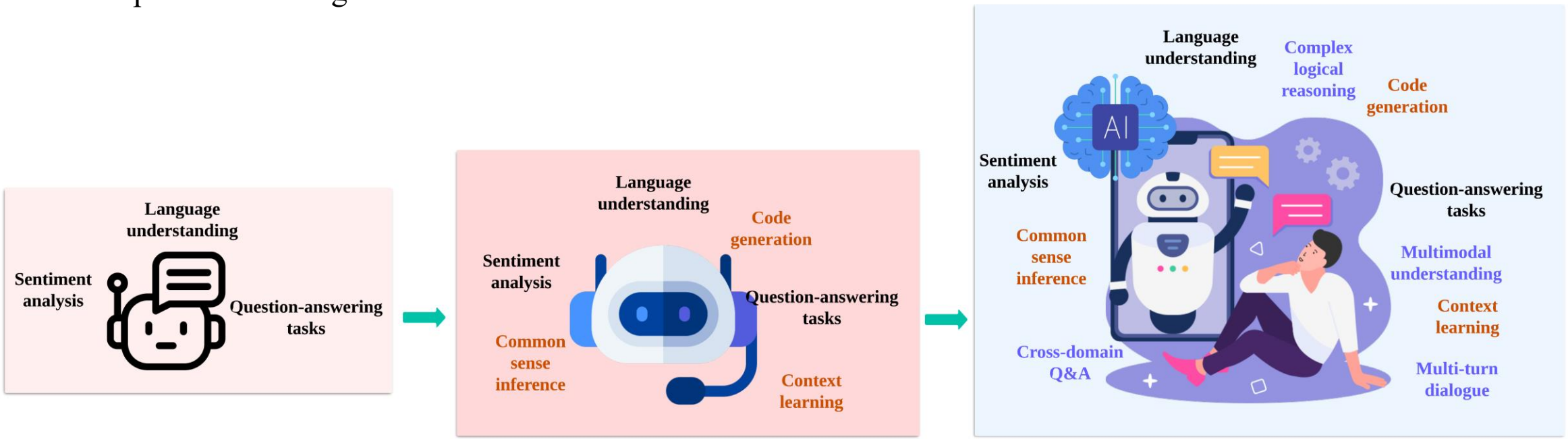


Zhao, W. X., Zhou, K., Li, J., Tang, T., Wang, X., Hou, Y., Min, Y., Zhang, B., Zhang, J., Dong, Z., Du, Y., Yang, C., Chen, Y., Chen, Z., Jiang, J., Ren, R., Li, Y., Tang, X., Liu, Z., . . . Wen, J. (2023). A Survey of Large Language Models. *ArXiv*. https://arxiv.org/abs/2303.18223

Evolution brings new abilities

Key Point: With the continuous iteration and updates of LLMs, the range of problems they can solve has become increasingly rich, demonstrating some new abilities.

- **Growing Complexity:** Each new LLM version shows improvements in reasoning, creativity, and context handling.
- **Emergent Properties:** Larger, more diverse training corpora lead to surprising capabilities (e.g., zero-shot translation, chain-of-thought prompting).
- **Broadening Applications:** Beyond text generation, LLMs now assist in code completion, legal drafting, and even scientific problem-solving.

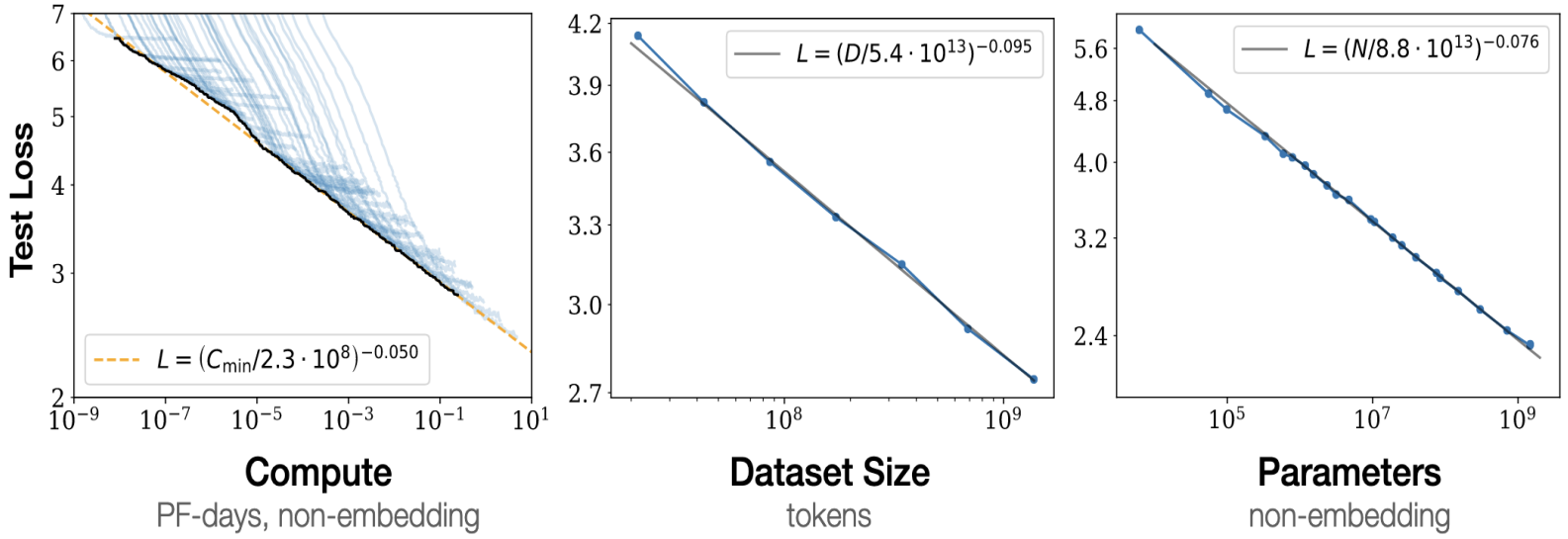


20 billion parameters

130 billion parameters

1750 billion parameters

Scaling Law



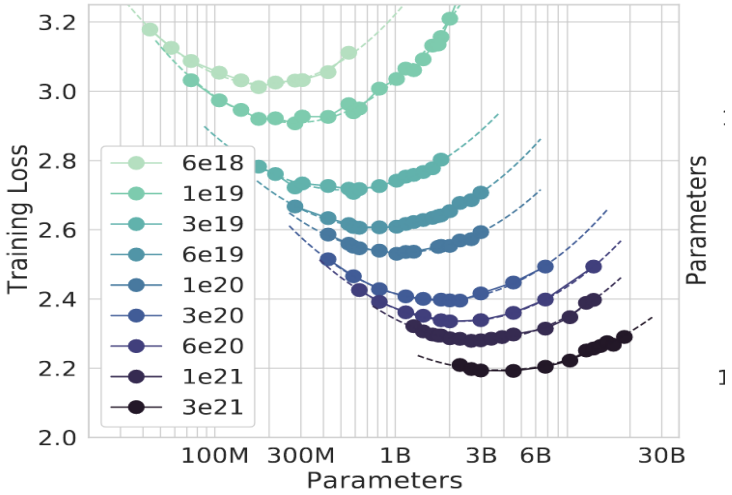
OpenAI

Scaling Law:

Empirical finding that language modeling performance improves continuously with larger model size, bigger datasets, and more compute.

▶ Context: Introduced prominently by OpenAI (2020) —"bigger is better" trend.

▶ Beyond a certain scale, emergent behaviors (complex reasoning, zero-shot skills) often appear.



DeepMind

Conducted large-scale experiments on Transformer-based models.

▶ Found that data and model size are equally crucial.

▶ **Balanced approach:** Sufficiently large dataset and model scale are needed for robust performance across tasks.

▶ **Reinforced the synergy:** more data =model can generalize; bigger model = can leverage that data better

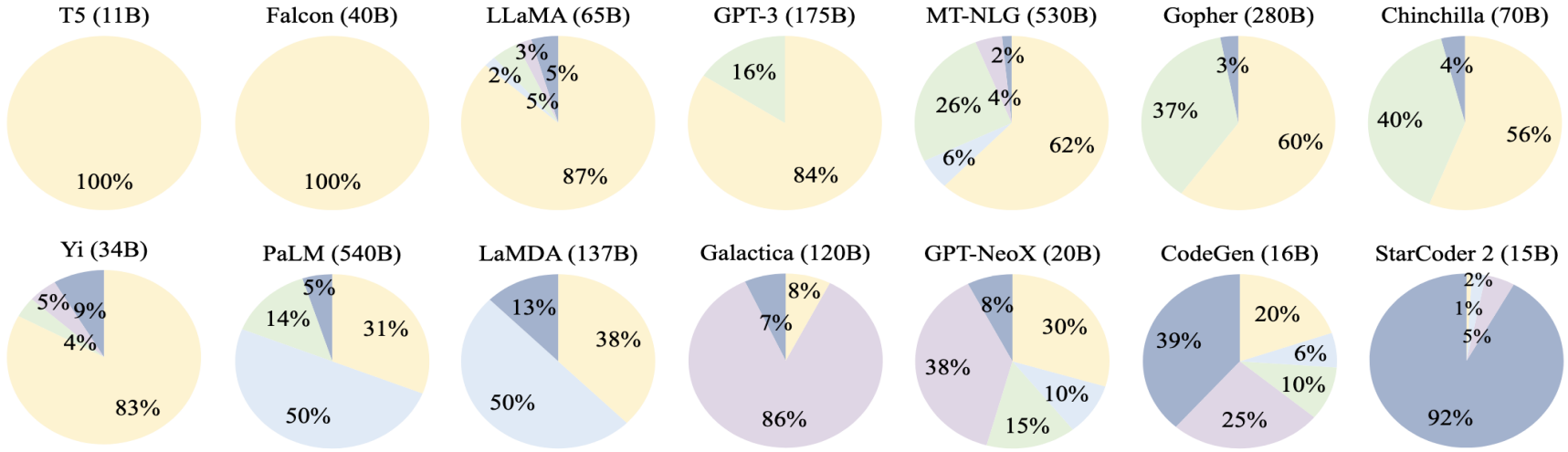
Kaplan, J., McCandlish, S., Henighan, T., Brown, T. B., Chess, B., Child, R., Gray, S., Radford, A., Wu, J., & Amodei, D. (2020). Scaling Laws for Neural Language Models. *ArXiv*. https://arxiv.org/abs/2001.08361

Hoffmann, J., Borgeaud, S., Mensch, A., Buchatskaya, E., Cai, T., Rutherford, E., Casas, D. D., Hendricks, L. A., Welbl, J., Clark, A., Hennigan, T., Noland, E., Millican, K., Driessche, G. V., Damoc, B., Guy, A., Osindero, S., Simonyan, K., Elsen, E., . . . Sifre, L. (2022). Training Compute-Optimal Large Language Models. *ArXiv*. https://arxiv.org/abs/2203.15556

Large-Scale Data for Pretraining

What data do LLMs use?

Typical data sources include text from the internet, books, scientific data, code repositories, and more.



Webpages	C4 (800G, 2019), OpenWebText (38G, 2023), Wikipedia (21G, 2023)
Conversation Data	the Pile - StackExchange (41G, 2020)
Books & News	BookCorpus (5G, 2015), Gutenberg (-, 2021), CC-Stories-R (31G, 2019), CC-NEWES (78G, 2019), REALNEWs (120G, 2019)
Scientific Data	the Pile - ArXiv (72G, 2020), the Pile - PubMed Abstracts (25G, 2020)
Code	BigQuery (-, 2023), the Pile - GitHub (61G, 2020)

Data preprocessing pipeline

Raw Corpus

Filtering & Selection

- Language Filtering
- Metric Filtering
- Statistic Filtering
- Keyword Filtering

Alice is writing a paper about LLMs. ~~#$^&~~ Alice is writing a paper about LLMs.

De-duplication

- Sentence-level
- Document-level
- Set-level

Alice is writing a paper about LLMs. ~~Alice is writing a paper about LLMs.~~

Privacy Reduction

- Detect Personality Identifiable Information (PII)
- Remove PII

`Replace('Alice')` is writing a paper about LLMs.

Tokenization

- Reuse Existing Tokenizer
- SentencePiece
- Byte-level BPE

`Encode('[Somebody] is writing a paper about LLMs.')`

Ready to pre-train!

32, 145, 66, 79, 12, 56, ...

Zhao, W. X., Zhou, K., Li, J., Tang, T., Wang, X., Hou, Y., Min, Y., Zhang, B., Zhang, J., Dong, Z., Du, Y., Yang, C., Chen, Y., Chen, Z., Jiang, J., Ren, R., Li, Y., Tang, X., Liu, Z., . . . Wen, J. (2023). A Survey of Large Language Models. *ArXiv*. https://arxiv.org/abs/2303.18223

Timeline of Existing LLMs

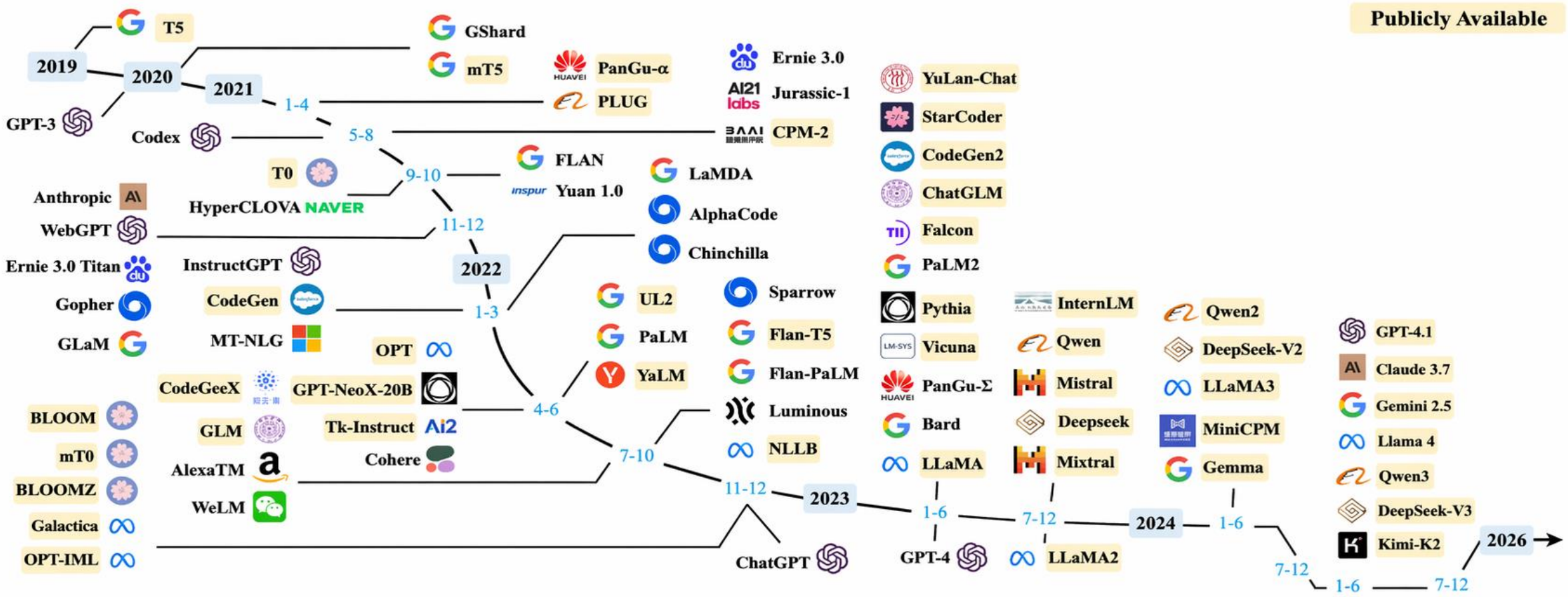


Fig. 3: A timeline of existing large language models (having a size larger than 10B) in recent years. The timeline was established mainly according to the release date (*e.g.*, the submission date to arXiv) of the technical paper for a model. If there was no corresponding paper, we set the date of a model as the earliest time of its public release or announcement. We mark the LLMs with publicly available model checkpoints in yellow color. Due to the space limit of the figure, we only include the LLMs with publicly reported evaluation results.

Zhao, W. X., Zhou, K., Li, J., Tang, T., Wang, X., Hou, Y., Min, Y., Zhang, B., Zhang, J., Dong, Z., Du, Y., Yang, C., Chen, Y., Chen, Z., Jiang, J., Ren, R., Li, Y., Tang, X., Liu, Z., . . . Wen, J. (2023). A Survey of Large Language Models. *ArXiv*. https://arxiv.org/abs/2303.18223

OpenAI LLM Series Timeline

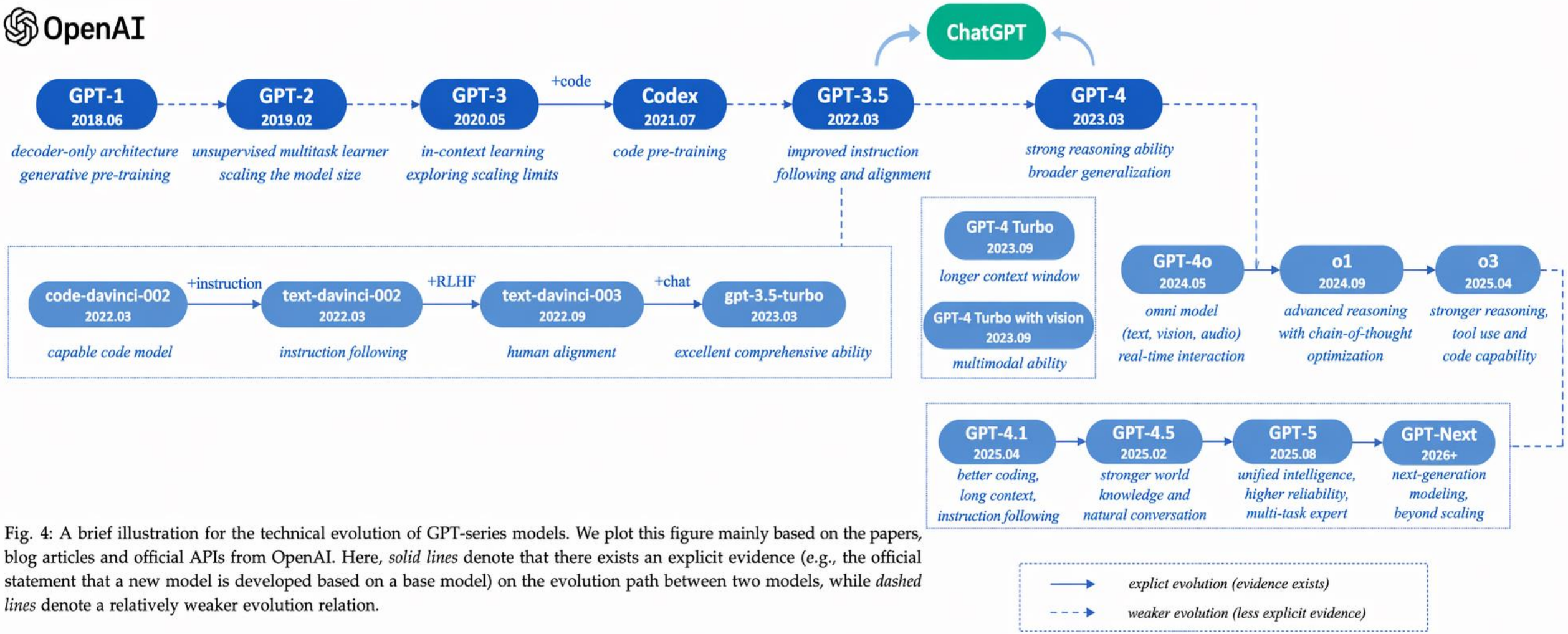


Fig. 4: A brief illustration for the technical evolution of GPT-series models. We plot this figure mainly based on the papers, blog articles and official APIs from OpenAI. Here, *solid lines* denote that there exists an explicit evidence (e.g., the official statement that a new model is developed based on a base model) on the evolution path between two models, while *dashed lines* denote a relatively weaker evolution relation.

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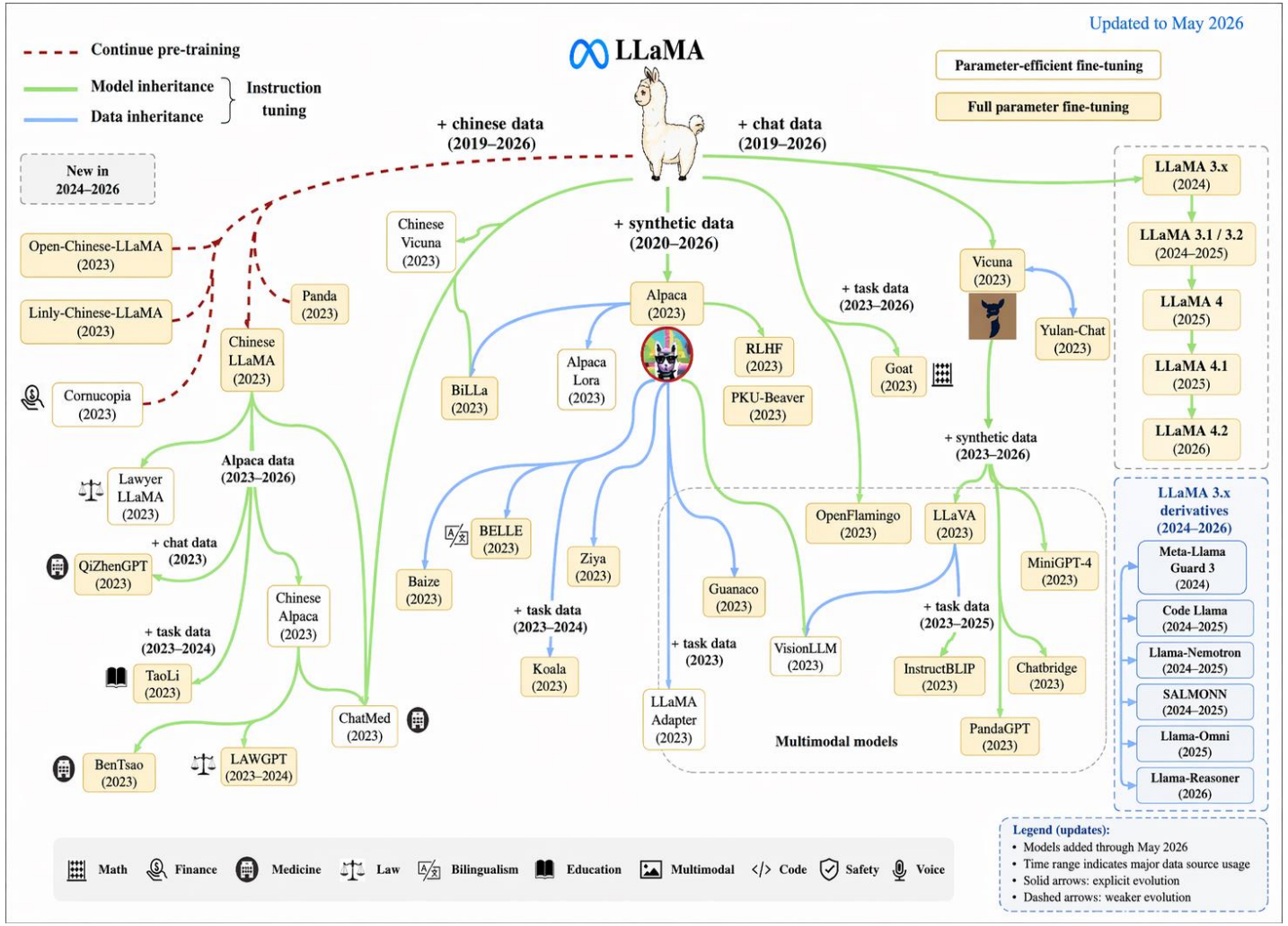
Large Language Model Meta AI (LLaMA)

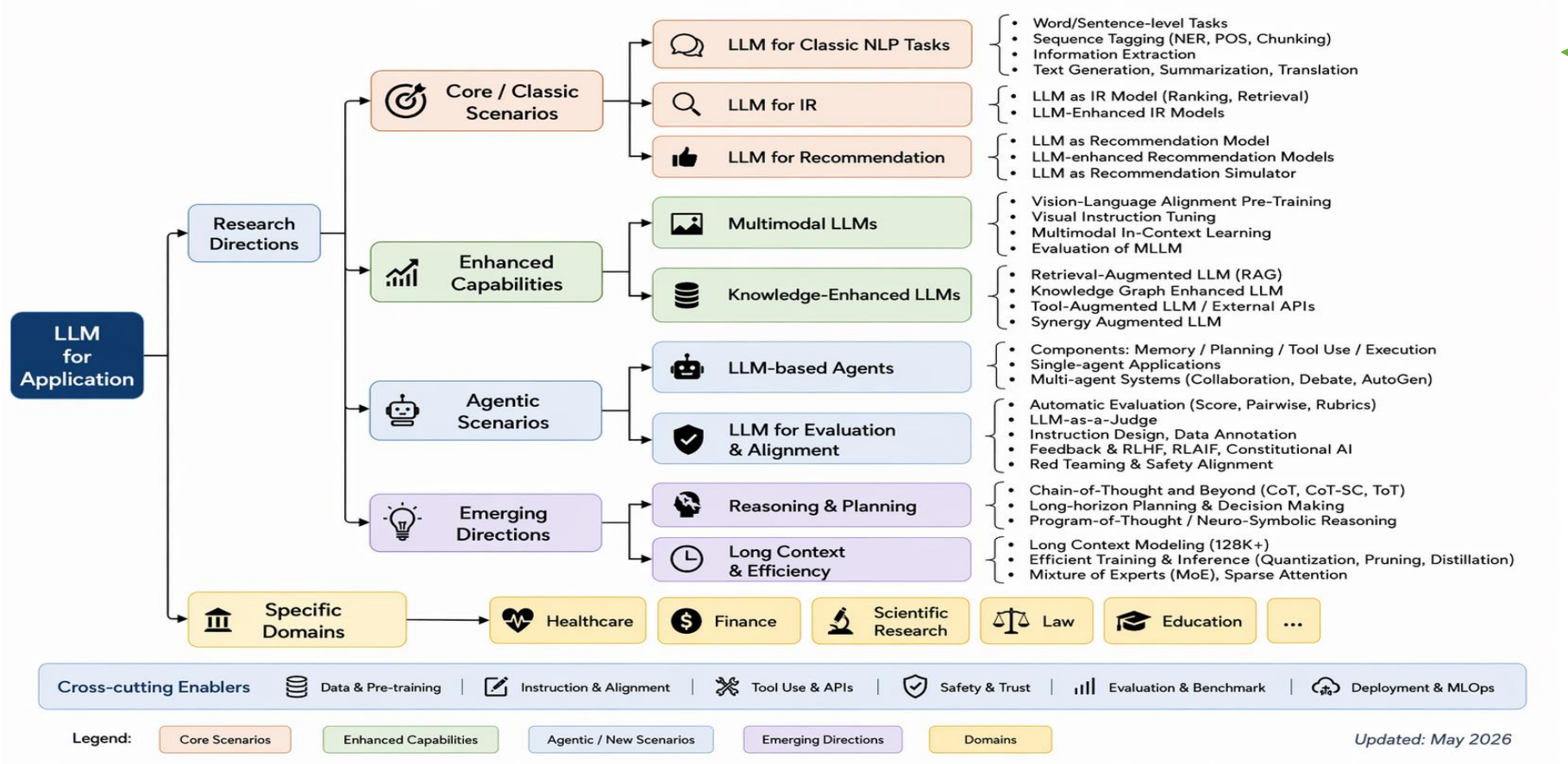
An Evolutionary Graph of the Research Work Conducted on LLaMA

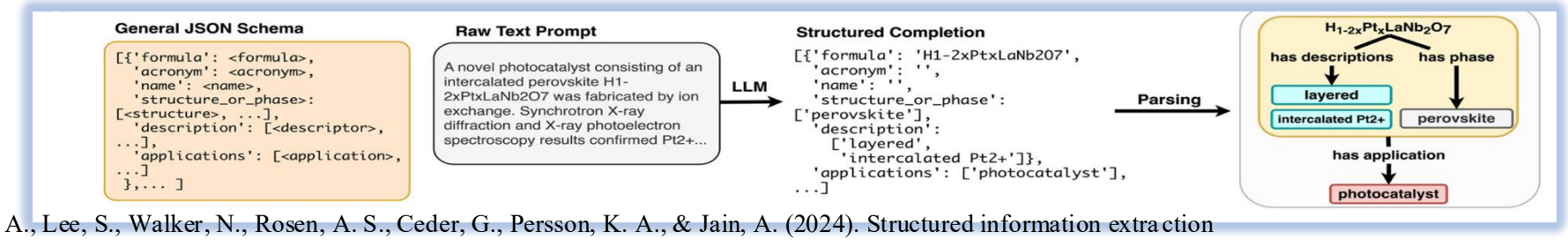


- **Purpose:** To provide a robust, scalable language model architecture.
- **Capabilities:** Natural language understanding, text generation, reasoning, and coding.
- Known for High efficiency and scalability compared to previous models.
- 2023: Initial release of LLaMA with a focus on efficient language modeling.
- 2024: LLaMA 2 introduced, featuring improved accuracy and efficiency.
- 2024: Integration of fine-tuning techniques, including instruction tuning.
- 2024--2025:The ecosystem expanded with derivatives such as Code Llama, Meta-Llama Guard, Llama-Nemotron, SALMONN, Llama-Omni, and Llama-Reasoner.
- 2025: LLaMA 4 series released, with stronger reasoning, multimodality, and efficiency.
- Ongoing (2026+): Research continues on advanced reasoning, real-world applications, and next-generation multimodal intelligence.

Zhao, W. X., Zhou, K., Li, J., Tang, T., Wang, X., Hou, Y., Min, Y., Zhang, B., Zhang, J., Dong, Z., Du, Y., Yang, C., Chen, Y., Chen, Z., Jiang, J., Ren, R., Li, Y., Tang, X., Liu, Z., . . . Wen, J. (2023). A Survey of Large Language Models. *ArXiv*. https://arxiv.org/abs/2303.18223

What LLMs Can Do?





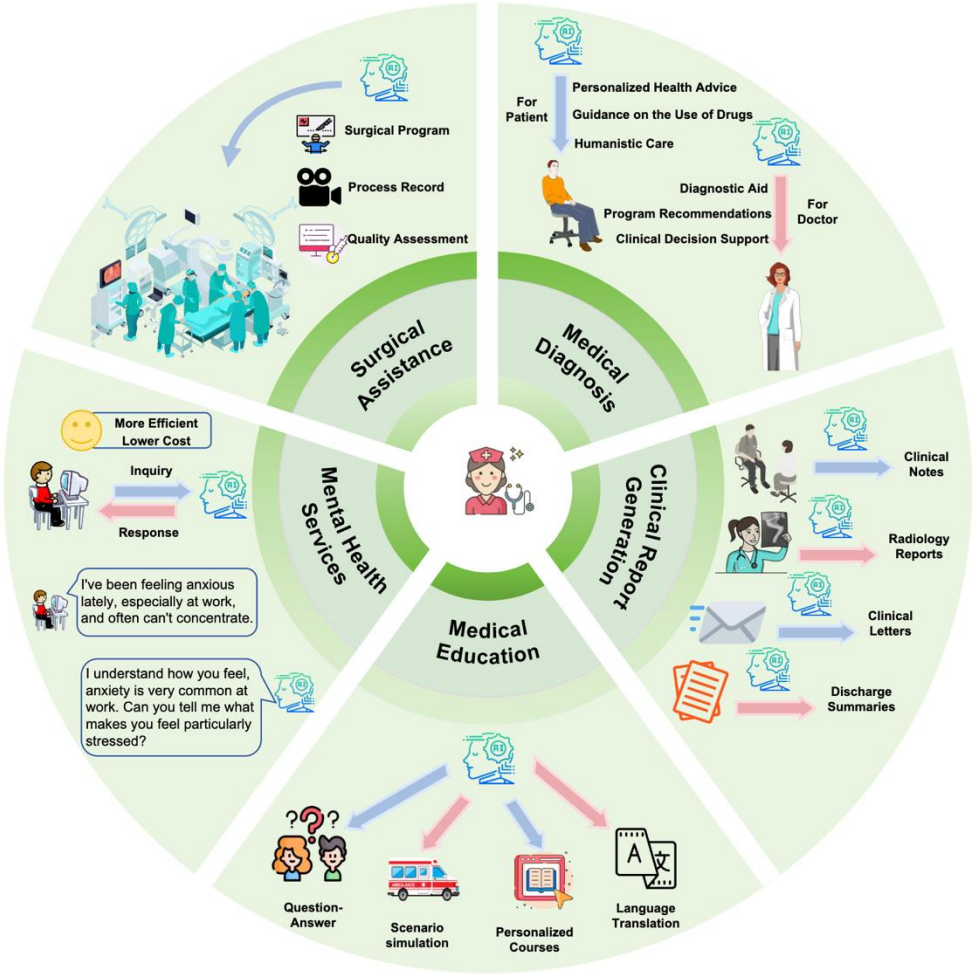
Information Retrieval

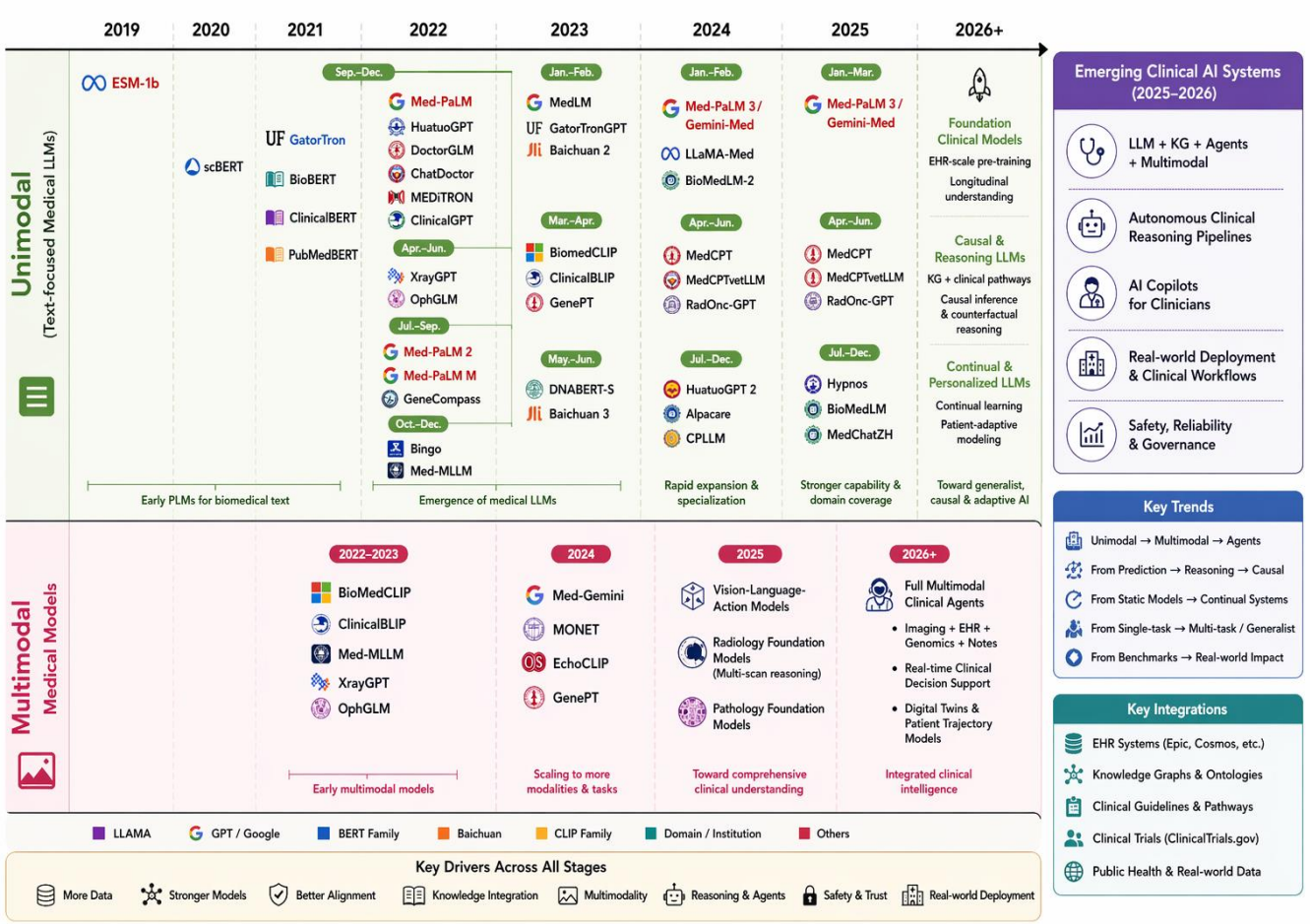
Dagdelen, J., Dunn, A., Lee, S., Walker, N., Rosen, A. S., Ceder, G., Persson, K. A., & Jain, A. (2024). Structured information extraction from scientific text with large language models. *Nature Communications*, *15*(1), 1-14. https://doi.org/10.1038/s41467-024-45563-x

Zhao, W. X., Zhou, K., Li, J., Tang, T., Wang, X., Hou, Y., Min, Y., Zhang, B., Zhang, J., Dong, Z., Du, Y., Yang, C., Chen, Y., Chen, Z., Jiang, J., Ren, R., Li, Y., Tang, X., Liu, Z., . . . Wen, J. (2023). A Survey of Large Language Models. *ArXiv*. https://arxiv.org/abs/2303.18223

LLMs in Biomedicine

Overview of potential applications of LLMs in medicine.

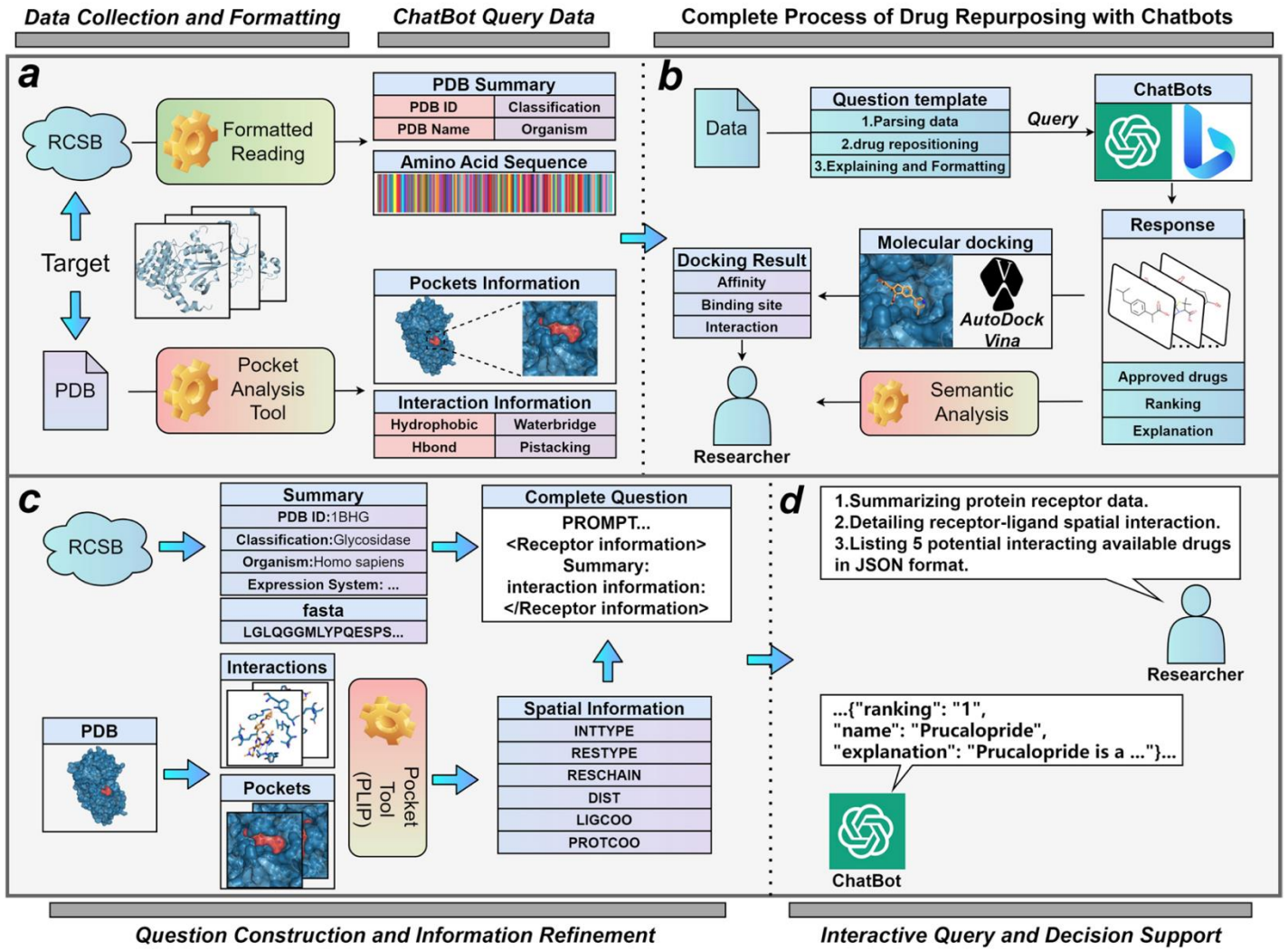




Chronological overview of LLMs and their variants in biomedical applications from 2019 to 2026. The timeline illustrates the evolution of both unimodal (top) and multimodal (bottom) models.

Wang, C., Li, M., He, J., Wang, Z., Darzi, E., Chen, Z., Ye, J., Li, T., Su, Y., Ke, J., Qu, K., Li, S., Yu, Y., Liò, P., Wang, T., Wang, Y. G., & Shen, Y. (2024). A Survey for Large Language Models in Biomedicine. *ArXiv*. https://arxiv.org/abs/2409.00133

LLMs in Biomedicine



DrugReAlign leverages LLMs and multi-source prompts to efficiently **repurpose existing drugs** by integrating extensive knowledge on targets and drugs.

DrugReAlign Framework and Detailed **Flowchart.**

- a. Construction of multisource prompts for targets
- b. Screening of potential drugs interacting with targets based on LLMs and multisource prompt information.
- c. Example of Prompt Construction for Specific Targets
- d. Interactive querying and decision support using LLMs

Wei, J., Zhuo, L., Fu, X. *et al*. DrugReAlign: a multisource prompt framework for drug repurposing based on large language models. *BMC Biol* **22**, 226 (2024). https://doi.org/10.1186/s12915-024-02028-3

Content

1. Introduction to Large Language Models

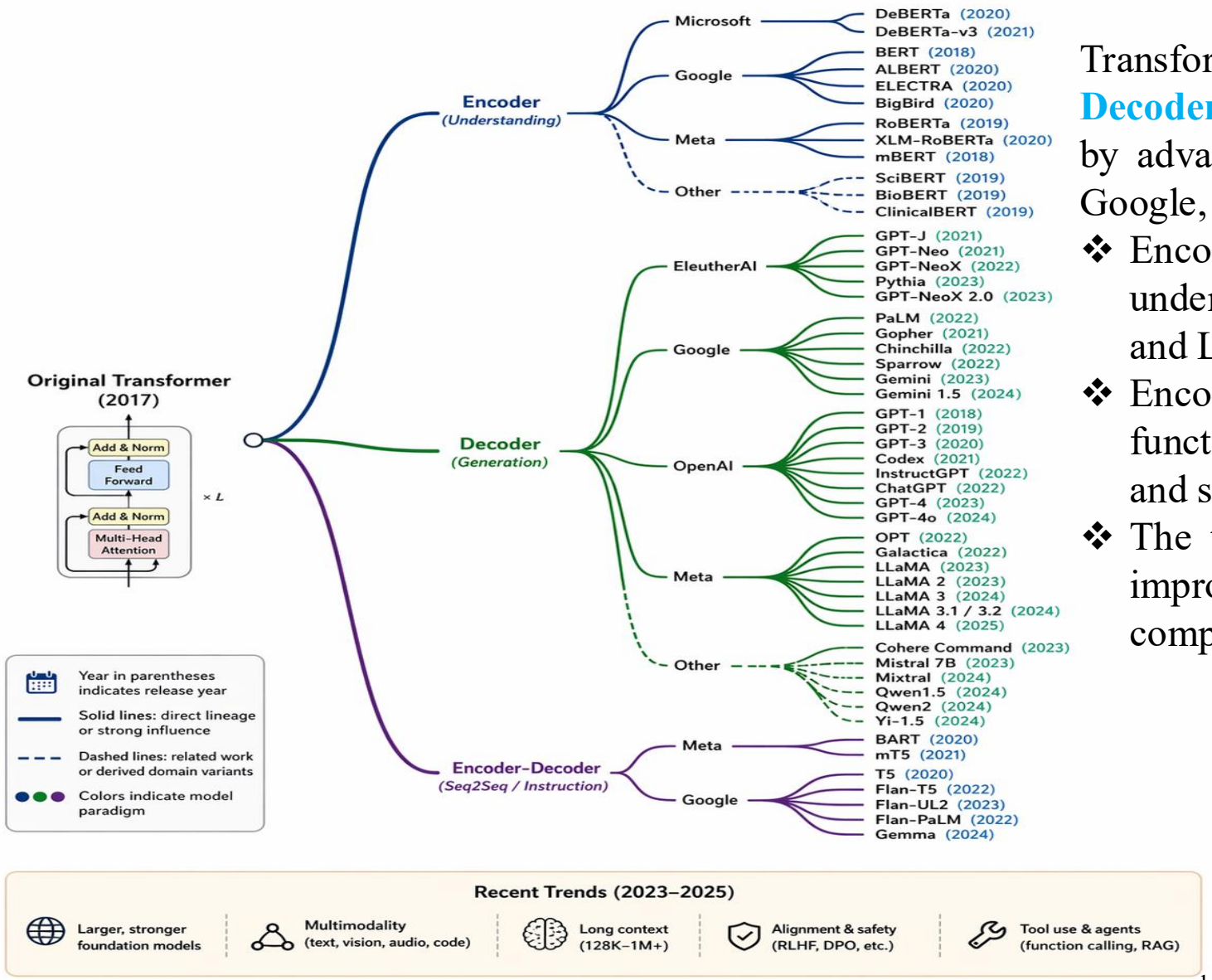
2. Popular Large Language Model Architectures

3. Make LLMs More Suitable for Your Downstream Application

4. Challenges

5. Applications

The core of LLMs --- Transformer



Transformer models have diversified into **Encoder, Decoder, and Encoder-Decoder branches**, driven by advancements from companies like Microsoft, Google, Meta, OpenAI, and Eleuther AI.

- Encoder models like BERT focus on language understanding, while Decoder models like GPT and LLaMA are aiming at generation.
- Encoder-Decoder models combine both functionalities, excelling in tasks like translation and summarization.
- The trend shows increasing specialization and improvement in both generative and comprehension tasks as the models evolve.

https://magazine.sebastianraschka.com/p/understanding-encoder-and-decoder

Encoder-only models —— BERT

What is BERT?

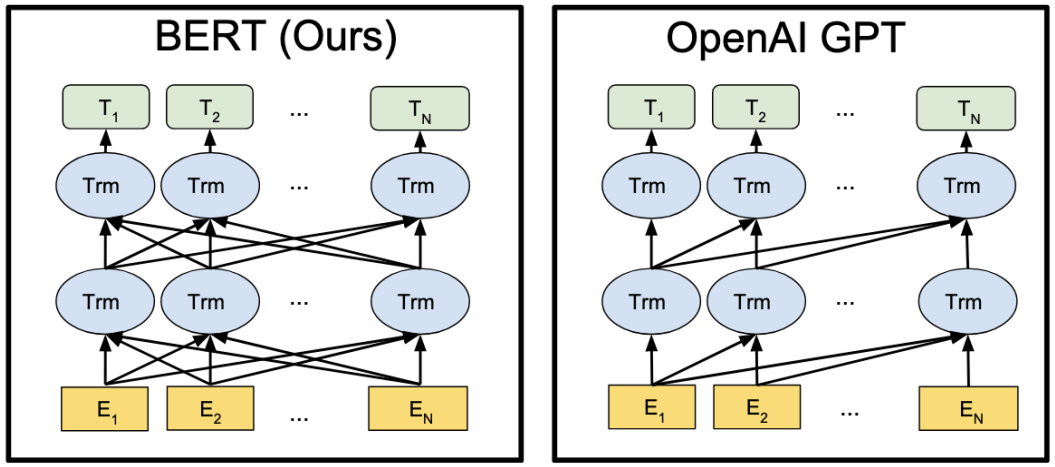
BERT (Bidirectional Encoder Representations from Transformers) is a pre-trained language representation model with an encoder-only architecture based on the Transformer's encoder module.

Key Features:

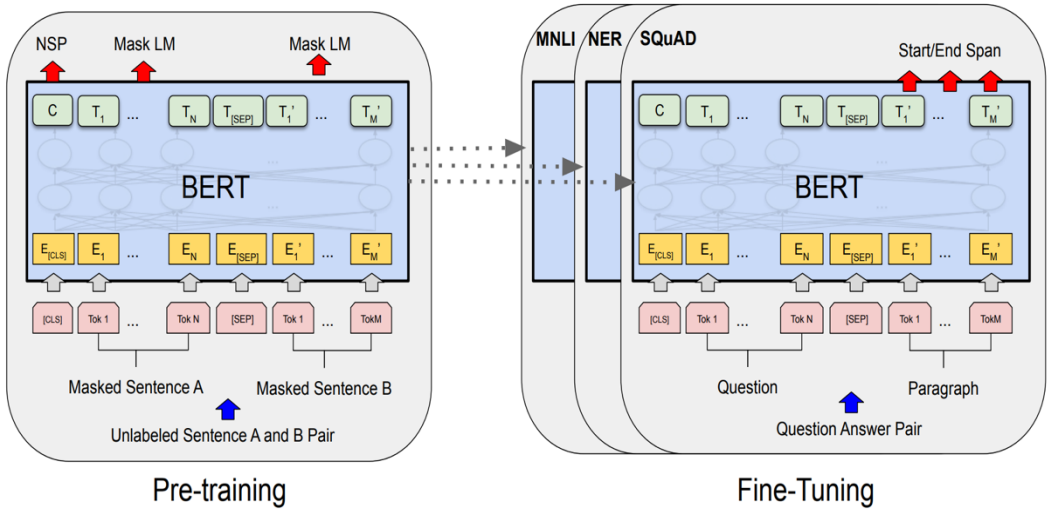
- **Bidirectional Encoding**: Uses bidirectional attention to understand **context from both directions**. Suitable for **understanding-based tasks** like classification and question answering.
- **Pre-training + Fine-tuning**: Pre-trained on large-scale unsupervised data (*Masked LM and Next Sentence Prediction (NSP)),* then fine-tuned on specific tasks.

Difference with OpenAI GPT

GPT: Uses unidirectional attention to predict the **next word** in a sequence. Suitable for **generation-based tasks** like text completion and conversation.



BERT uses a **bidirectional Transformer**.
OpenAI GPT uses a **left-to-right Transformer**.



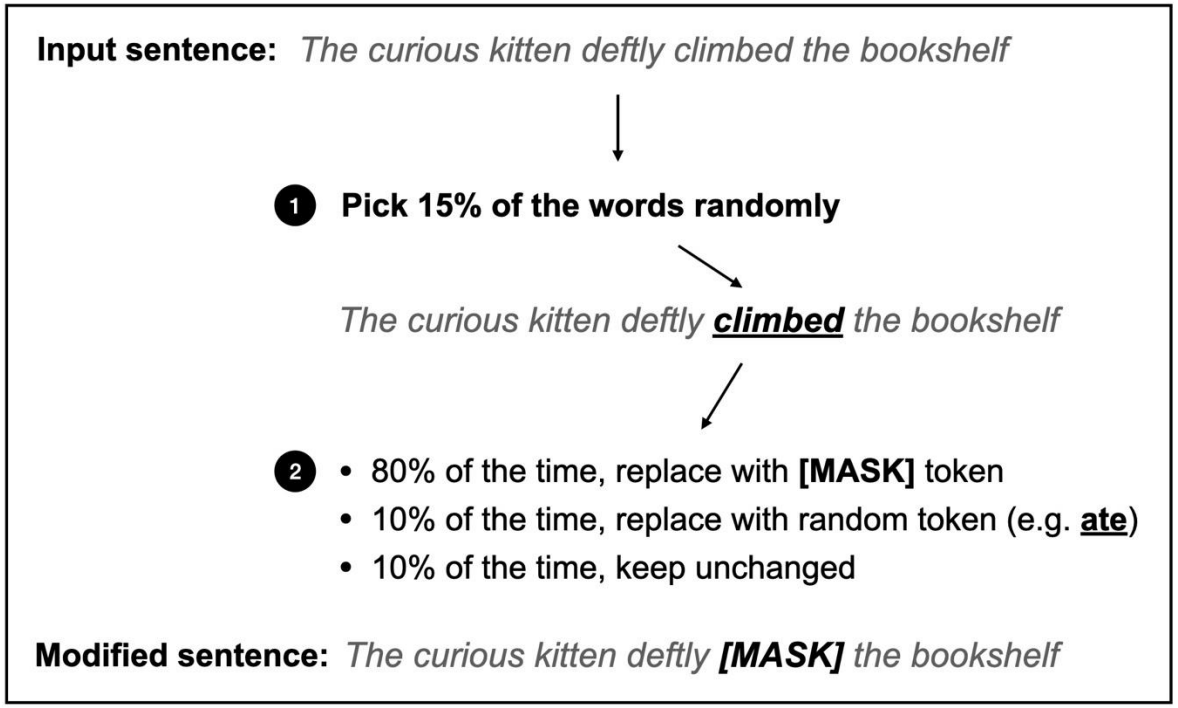
Devlin, J., Chang, M., Lee, K., & Toutanova, K. (2018). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. *ArXiv*. https://arxiv.org/abs/1810.04805

BERT Training Methods —— Pre-training

➢ Task1: Masked Language Model (MLM)

- **Randomly masks the input words** and predicts the original words.
- 80% of masked tokens are replaced with [MASK], 10% are replaced with a random word, and 10% remain unchanged to reduce pre-training and fine-tuning mismatch.

🔷The main idea behind masked language modeling is to mask (or replace) random word tokens in the input sequence and then train the model **to predict the original masked tokens based on the surrounding context**.



https://magazine.sebastianraschka.com/p/understanding-encoder-and-decoder

BERT Training Methods —— Pre-training

➢ Task2: Next Sentence Prediction (NSP)

Predicts whether two sentences are consecutive: 50% of examples are actual consecutive sentences (**IsNext**), 50% of examples are randomly chosen, non-consecutive sentences (**NotNext**).

Helps the model learn textual coherence and is particularly useful for tasks like **Question Answering (QA)** and **Natural Language Inference (NLI)**.

📌 **Example:**

✅ **IsNext:**

Sentence A: "She finished her homework."

Sentence B: "Then she went out to play with her friends."

Prediction: IsNext

The above two tasks do not require manually annotated data, allowing for a large corpus to be available for training.

❌ **NotNext:**

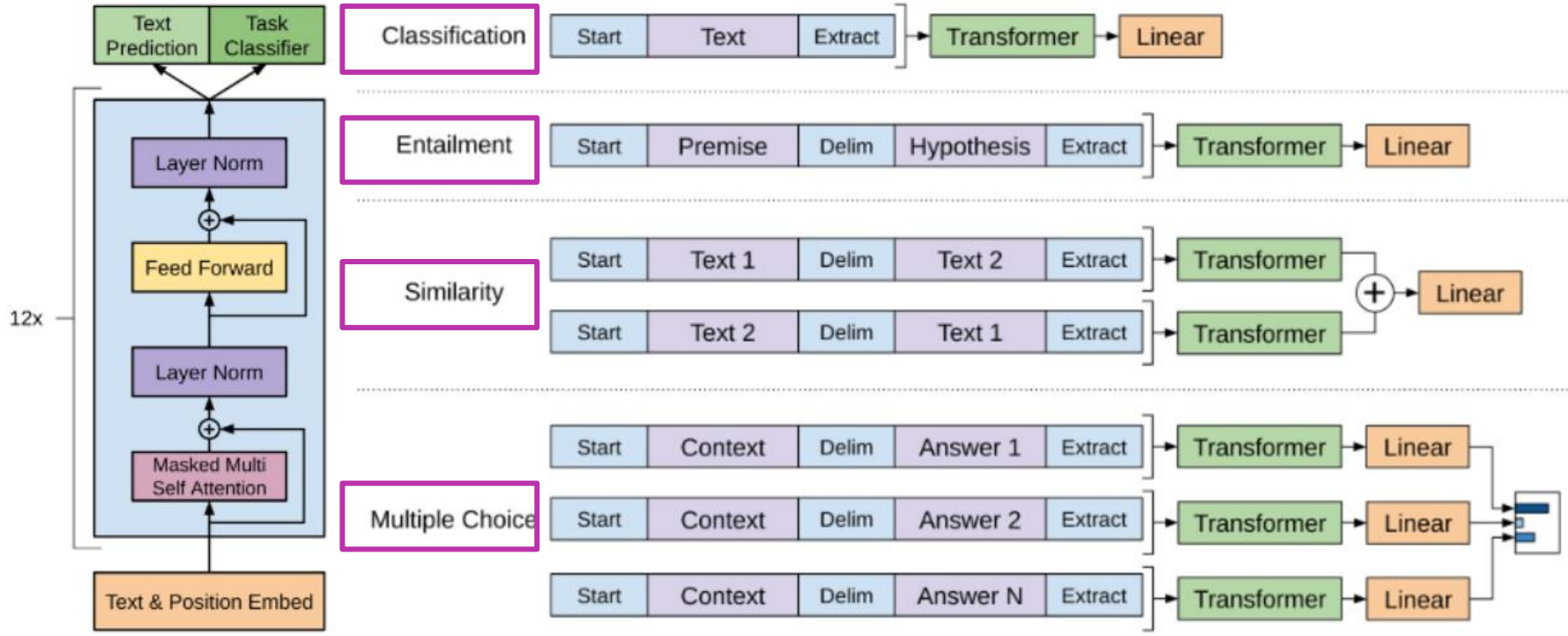
Sentence A: "She finished her homework."

Sentence B: "The stock market showed an unexpected rise today."

Prediction: NotNext

BERT Training Methods —— Fine-tuning

Same architecture, different tasks: A simple output layer is added for tasks such as classification, question answering (QA), and Named Entity Recognition (NER).



(i) Transformer architecture and training objectives used in this work. (ii) Input transformations for fine-tuning on different tasks. We convert all structured inputs into token sequences to be processed by our pre-trained model, followed by a linear+softmax layer.

Radford A, Narasimhan K, Salimans T, et al. Improving language understanding by generative pre-training [J]. 2018.

Sentiment classification using BERT

```python
import torch
from transformers import BertModel, BertTokenizer, BertForMaskedLM
# Load pre-trained BERT model and tokenizer
tokenizer = BertTokenizer.from_pretrained('bert-base-uncased')
model = BertModel.from_pretrained('bert-base-uncased')
masked_model = BertForMaskedLM.from_pretrained('bert-base-uncased')
# Example text
text = "BERT is a powerful language model that learns text representations through bidirectional Transformer architecture."
# Tokenize and convert to model input format
inputs = tokenizer(text, return_tensors="pt")
# Get BERT outputs
with torch.no_grad():
    outputs = model(**inputs)
# Get the [CLS] token representation (commonly used for classification tasks)
last_hidden_states = outputs.last_hidden_state
cls_embedding = last_hidden_states[:, 0, :]
print(f"CLS embedding shape: {cls_embedding.shape}")
# Simple sentiment analysis example (assuming we have a pre-trained classification head)
class SentimentClassifier(torch.nn.Module):
    def __init__(self, hidden_size=768):
        super().__init__()
        self.classifier = torch.nn.Linear(hidden_size, 2) # Binary classification: positive/negative
    def forward(self, embeddings):
        return self.classifier(embeddings)
# Initialize classifier
classifier = SentimentClassifier()
# Use [CLS] embedding for sentiment prediction
logits = classifier(cls_embedding)
predicted_class = torch.argmax(logits, dim=1).item()
print(f"Predicted sentiment: {'Positive' if predicted_class == 1 else 'Negative'}")
```

Decoder-only models GPT family

Transformer-based architecture for all GPT models, with variations in model depth and embedding size.

Key Objectives of GPT Models:

1. Improve performance on NLP tasks.
2. Enhance generalization to non-NLP tasks (e.g., coding, math).

OpenAI GPT Model Evolution

2018 · 2019 · 2020 · 2023 · 2024 · 2025

Model	Release Date	Layers	Heads	Embedding Size	Parameters	Training Data Size / Notes	Key Highlights
GPT-1	June 2018	12	12	768	117M	BookCorpus (size not explicitly summarized)	• First GPT model demonstrating the power of unsupervised pre-training
GPT-2	Feb 2019	48	25	1,600	1.5B	WebText ~40 GB	• Scale-up model showing strong zero-shot capabilities
GPT-3	May 2020	96	96	12,288	175B	Filtered corpus ~570 GB text (from much larger raw corpora)	• Large-scale language model • Strong few-shot learning performance
GPT-4	Mar 2023	N/D	N/D	N/D	N/D	Not disclosed (multimodal: text + image input, text output)	• High performance across reasoning, coding, and safety • Multimodal capabilities
GPT-4 Turbo	Nov 2023	N/D	N/D	N/D	N/D	Not disclosed 128K context window	• Improved efficiency and longer context window
GPT-4o	May 2024	N/D	N/D	N/D	N/D	Not disclosed (omni: text, vision, audio)	• Native multimodal (text, vision, audio) • Real-time and end-to-end
GPT-4.1	Apr 2025	N/D	N/D	N/D	N/D	Not disclosed 1M-token context window	• Optimized for coding and instruction following • 1 million token context window

Training data sizes for earlier models are approximate and based on published information. OpenAI has not disclosed details for newer models.

N/D: Not Disclosed
OpenAI has not publicly released architecture details (layers, heads, embedding size, parameters) for GPT-4 and later models.

General Trends

- **Larger Scale** — More parameters
- **Stronger Capabilities** — Better reasoning & generalization
- **Longer Context** — From 2K → 128K → 1M tokens
- **Multimodality** — Text → Vision → Audio
- **Safer & More Aligned** — Improved safety, alignment & reliability

How to improve GPT?

Challenge 1: Pretrained models **lack transparency**, raising concerns about **bias**.

Challenge 2: Need alignment with **human preferences.**

Instruct Learning vs. Prompt Learning

Prompt learning is the process of designing **input prompts** to coax a pre-trained language model into performing a desired task **without retraining the model**.

Key Features:

- Uses **carefully crafted text prompts** (manual or learned) to elicit behavior.
- Relies on the model's **pre-existing capabilities**.
- Examples include **zero-shot**, **few-shot**, and **prefix-tuning**.

Example: Prompt: *"Translate this sentence to French: 'Good morning'"* The model completes: *"Bonjour"*

Instruct learning refers to **fine-tuning** a language model using **instruction-response pairs**, so that the model better understands and follows **natural language commands**.

Key Features:

- The model is explicitly trained on how to **follow instructions**.
- Focuses on **alignment with human intent**.
- Better at following **user commands**.

Example: Instruction: *"Summarize the following article in three bullet points."* Model: Provides a clear 3-point summary.

Goal: Make the model **more helpful, honest, and harmless** by aligning it with human feedback and instruction-following behavior.

Key Advantages of Instruct Learning

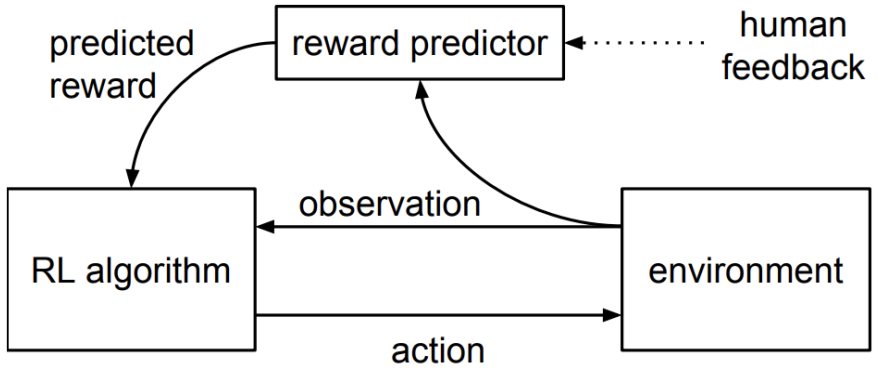
- ✓ **Better generalization**: Enables **zero-shot** learning after multi-task fine-tuning.
- ✓ **More effective than prompt learning**, which is **task-specific** and lacks adaptability.

Wei, J., Bosma, M., Zhao, V. Y., Guu, K., Yu, A. W., Lester, B., Du, N., Dai, A. M., & Le, Q. V. (2021). Finetuned Language Models Are Zero-Shot Learners. *ArXiv*. https://arxiv.org/abs/2109.01652

RL from Human Feedback (RLHF)

For Challenge 2: Focus on **3H principles:**

- **Helpful** (Improves usability)
- **Honest** (Enhances trustworthiness)
- **Harmless** (Mitigates harmful outputs)



Reinforcement Learning Basics

- Uses **rewards** as feedback, similar to loss functions in traditional training.
- Rewards can be **more flexible** than loss functions but are **not differentiable**, requiring an approximation.

RLHF Process

1. **Human annotations** provide reward signals.
2. **Model samples multiple outputs**, and **human preferences** guide reward learning.
3. **Reinforcement Learning optimizes the model** to maximize human-aligned responses.

>>>> **InstructGPT & ChatGPT**

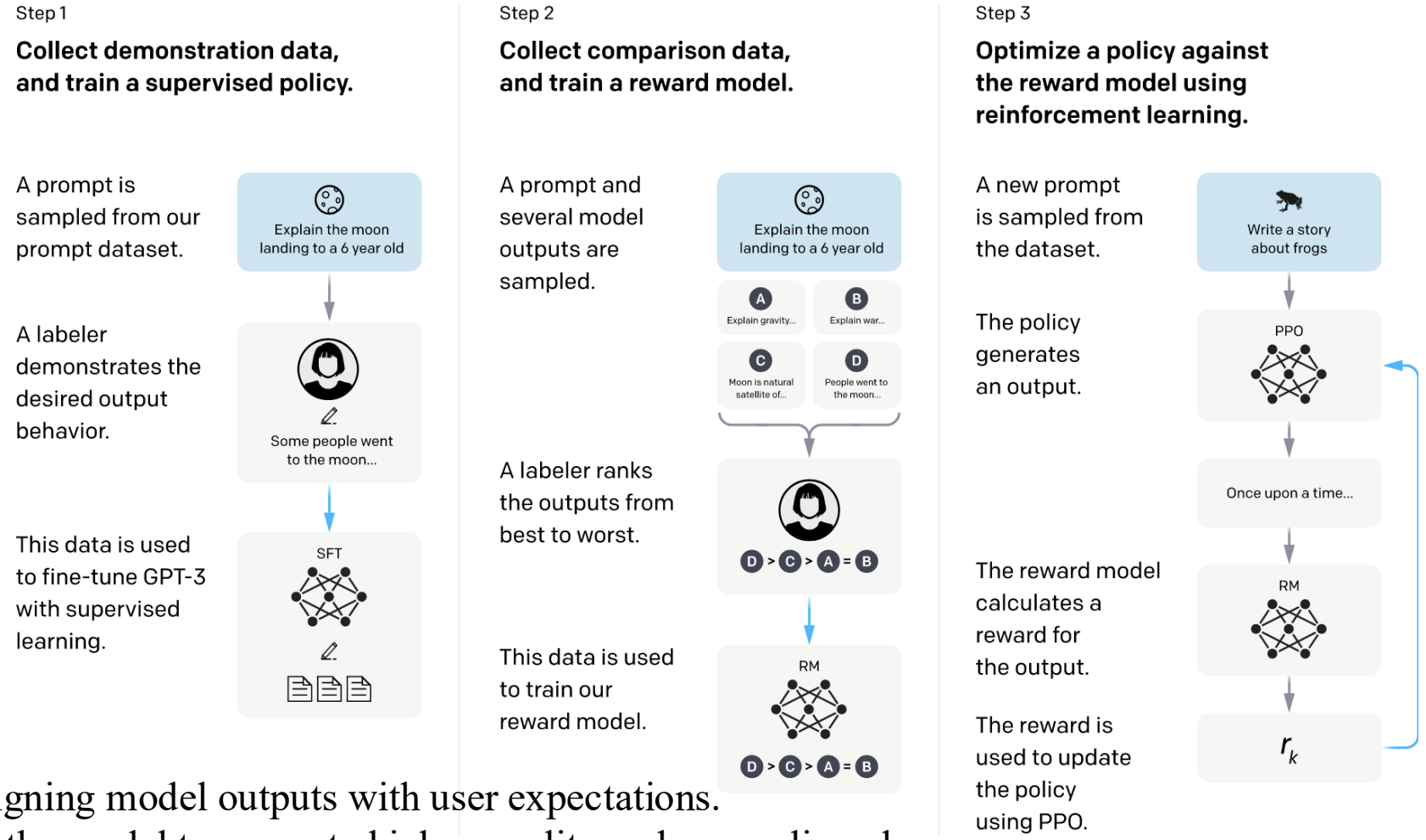
Christiano P F, Leike J, Brown T, et al. Deep reinforcement learning from human preferences[J]. Advances in neural information processing systems, 2017, 30.

InstructGPT/ChatGPT

- Training language models to follow instructions with human feedback. (**RLHF**)

- A diagram illustrating the three steps of our method:

1. Supervised fine-tuning (SFT),
2. Reward model (RM) training,
3. Reinforcement learning via proximal policy optimization (PPO) on this reward model.



Human Feedback: Essential for aligning model outputs with user expectations.
Reinforcement Learning: Enables the model to generate higher-quality and more aligned responses.
Iterative Process: Combines supervised fine-tuning with reinforcement learning for enhanced model performance.

Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C. L., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., Schulman, J., Hilton, J., Kelton, F., Miller, L., Simens, M., Askell, A., Welinder, P., Christiano, P., Leike, J., . . . Lowe, R. (2022). Training language models to follow instructions with human feedback. *ArXiv*. https://arxiv.org/abs/2203.02155

Dataset Collection

1. Supervised Fine-Tuning (SFT) Dataset
Purpose: Fine-tune GPT-3 using **instruction-response pairs**.
Sources:

- **User data** from OpenAI's Playground.
- **40 trained labelers** hired by OpenAI.

2. Reward Model (RM) Dataset
Purpose: Train a **reward model** to guide content alignment.
Process:

- Model generates candidate responses.
- **Labelers rank responses** based on quality & bias.
- **Lower scores for biased/harmful content**, discouraging unwanted outputs.

3. PPO (Proximal Policy Optimization) Dataset
Purpose: Used in the third stage of RLHF training.
Source: **User-generated queries from GPT-3's API.**

Table 1: Distribution of use case categories from our API prompt dataset.

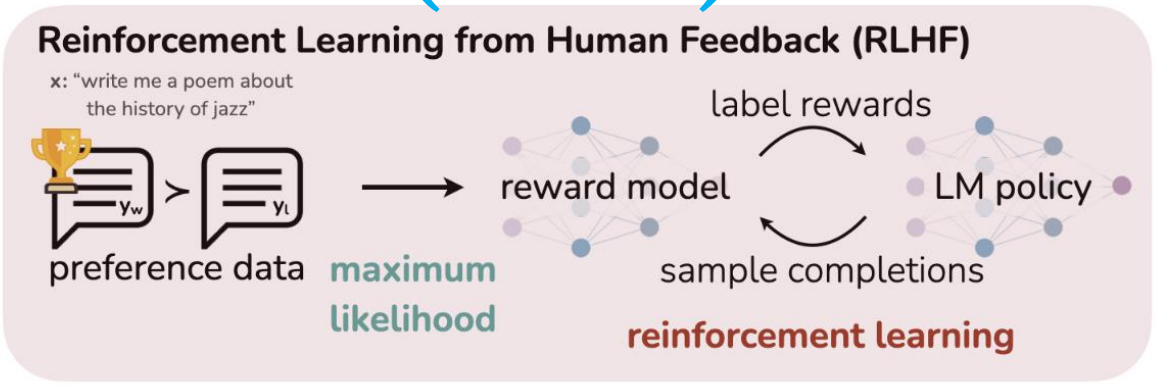
Use-case	(%)
Generation	45.6%
Open QA	12.4%
Brainstorming	11.2%
Chat	8.4%
Rewrite	6.6%
Summarization	4.2%
Classification	3.5%
Other	3.5%
Closed QA	2.6%
Extract	1.9%

Table 6: Dataset sizes, in terms of number of prompts.

SFT Data			RM Data			PPO Data		
split	source	size	split	source	size	split	source	size
train	labeler	11,295	train	labeler	6,623	train	customer	31,144
train	customer	1,430	train	customer	26,584	valid	customer	16,185
valid	labeler	1,550	valid	labeler	3,488			
valid	customer	103	valid	customer	14,399			

Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C. L., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., Schulman, J., Hilton, J., Kelton, F., Miller, L., Simens, M., Askell, A., Welinder, P., Christiano, P., Leike, J., . . . Lowe, R. (2022). Training language models to follow instructions with human feedback. *ArXiv*. https://arxiv.org/abs/2203.02155

Direct Preference Optimization (DPO)



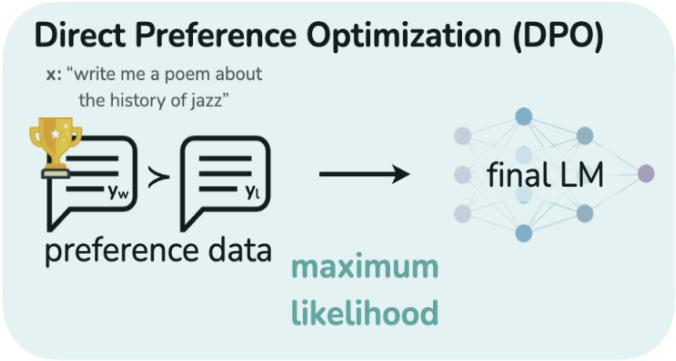
Challenges with RLHF:

- **Complexity:** Requires multiple LM training steps and continuous policy sampling.
- **High computational costs.**

What is DPO?

Avoiding RL

- **A simpler alternative to RLHF** that directly optimizes a language model for human preferences.
- **No explicit reward modeling or reinforcement learning required.**



Key Advantages of DPO

- **Optimizes the same objective as RLHF** (reward maximization with KL-divergence constraint).
- **Updates model probability distribution**:
  - ➢ **Increases log probability of preferred responses.**
  - ➢ **Uses per-example importance weighting** to prevent model degeneration.
- **Relies on a theoretical preference model** (e.g., **Bradley-Terry model**) to align reward functions with empirical data.

Rafailov, R., Sharma, A., Mitchell, E., Ermon, S., Manning, C. D., & Finn, C. (2023). Direct Preference Optimization: Your Language Model is Secretly a Reward Model. *ArXiv*. https://arxiv.org/abs/2305.18290

DeepSeek-R1

Why DeepSeek-R1 is important?

- **Cost-Effective Excellence:** DeepSeek-R1 achieves reasoning performance comparable to OpenAI's o1 model but at a significantly reduced cost.
- **Open-Source Ecosystem Contributions:**
  - **Reinforcement Learning (RL)**
  - **Supervised Fine-Tuning (SFT)**
  - **GRPO Algorithm**: Eliminates the need for a separate critic model, enhancing RL efficiency.
  - **Integrated Reward Mechanism**: Combines metrics such as accuracy, language consistency, helpfulness, and harmlessness to ensure well-rounded model performance.
  - **Multi-Domain Data Mixing Strategy**: Improves adaptability across diverse tasks, enabling the model to perform effectively in various scenarios.

Guo, D., Yang, D., Zhang, H., Song, J., Zhang, R., Xu, R., Zhu, Q., Ma, S., Wang, P., Bi, X., Zhang, X., Yu, X., Wu, Y., Wu, Z. F., Gou, Z., Shao, Z., Li, Z., Gao, Z., Liu, A., . . . Zhang, Z. (2025). DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning. *Nature*. https://arxiv.org/abs/2501.12948

Article

DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning

https://doi.org/10.1038/s41586-025-09422-z
Received: 14 February 2025
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Open access
Check for updates

General reasoning represents a long-standing and formidable challenge in artificial intelligence (AI). Recent breakthroughs, exemplified by large language models (LLMs)[1,2] and chain-of-thought (CoT) prompting[3], have achieved considerable success on foundational reasoning tasks. However, this success is heavily contingent on extensive human-annotated demonstrations and the capabilities of models are still insufficient for more complex problems. Here we show that the reasoning abilities of LLMs can be incentivized through pure reinforcement learning (RL), obviating the need for human-labelled reasoning trajectories. The proposed RL framework facilitates the emergent development of advanced reasoning patterns, such as self-reflection, verification and dynamic strategy adaptation. Consequently, the trained model achieves superior performance on verifiable tasks such as mathematics, coding competitions and STEM fields, surpassing its counterparts trained through conventional supervised learning on human demonstrations. Moreover, the emergent reasoning patterns exhibited by these large-scale models can be systematically used to guide and enhance the reasoning capabilities of smaller models.

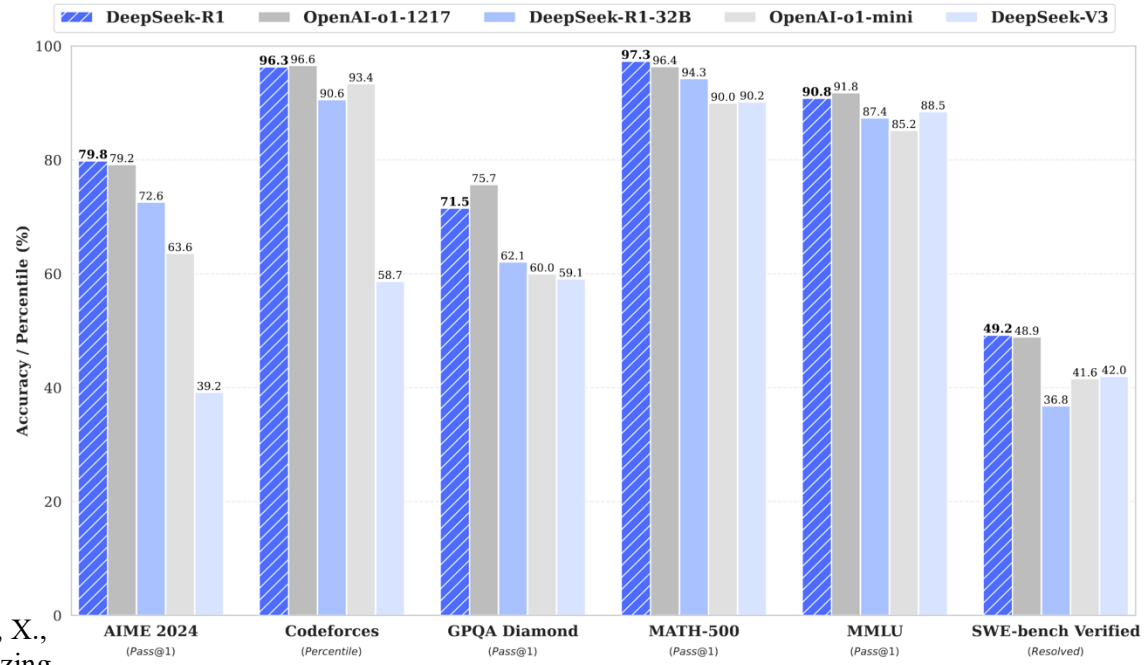
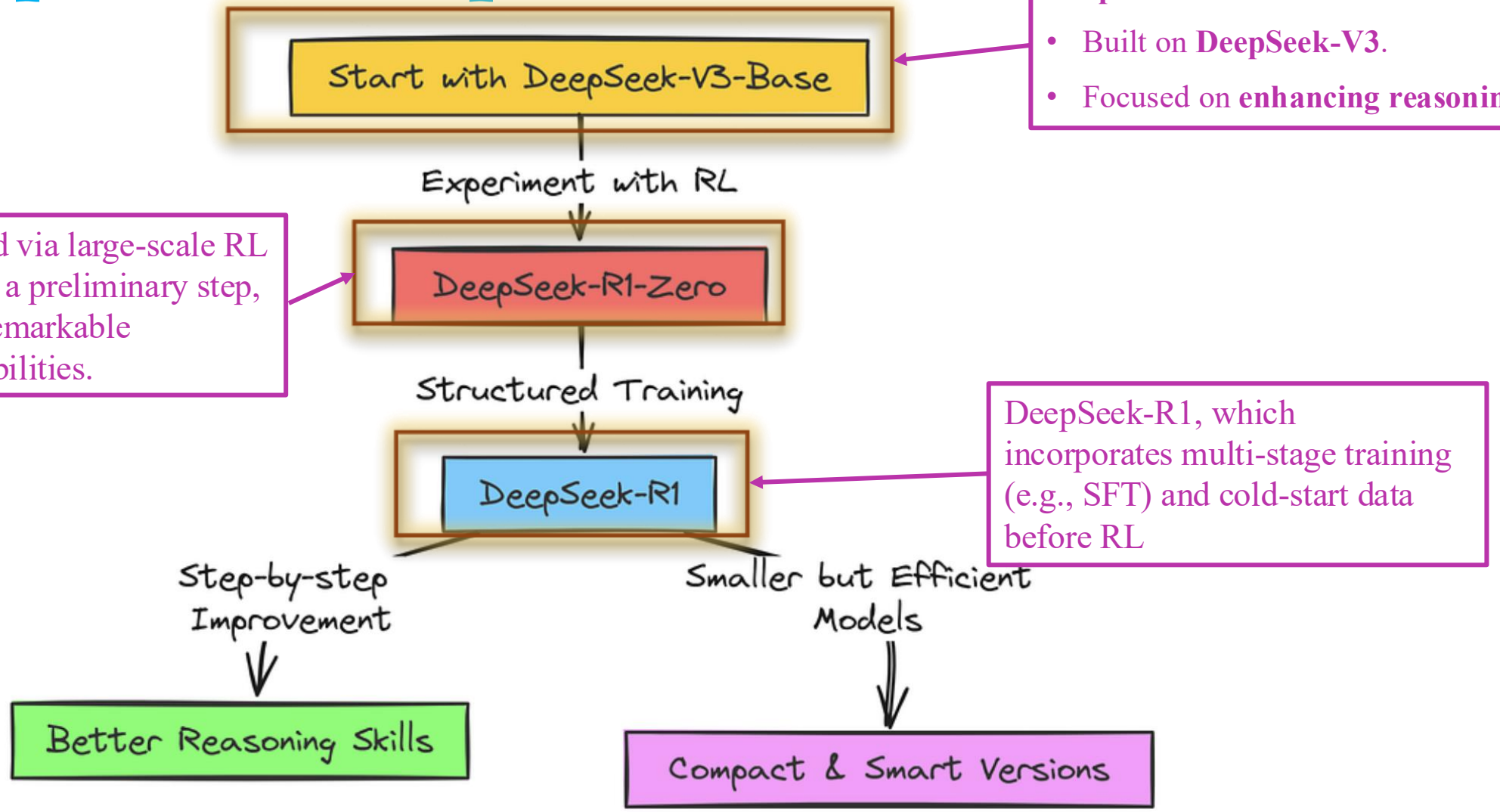


Figure 1 | Benchmark performance of DeepSeek-R1.

DeepSeek-R1 Pipeline



DeepSeek-R1 is not from scratch.

- Built on **DeepSeek-V3**.
- Focused on **enhancing reasoning**.

A model trained via large-scale RL without SFT as a preliminary step, demonstrates remarkable reasoning capabilities.

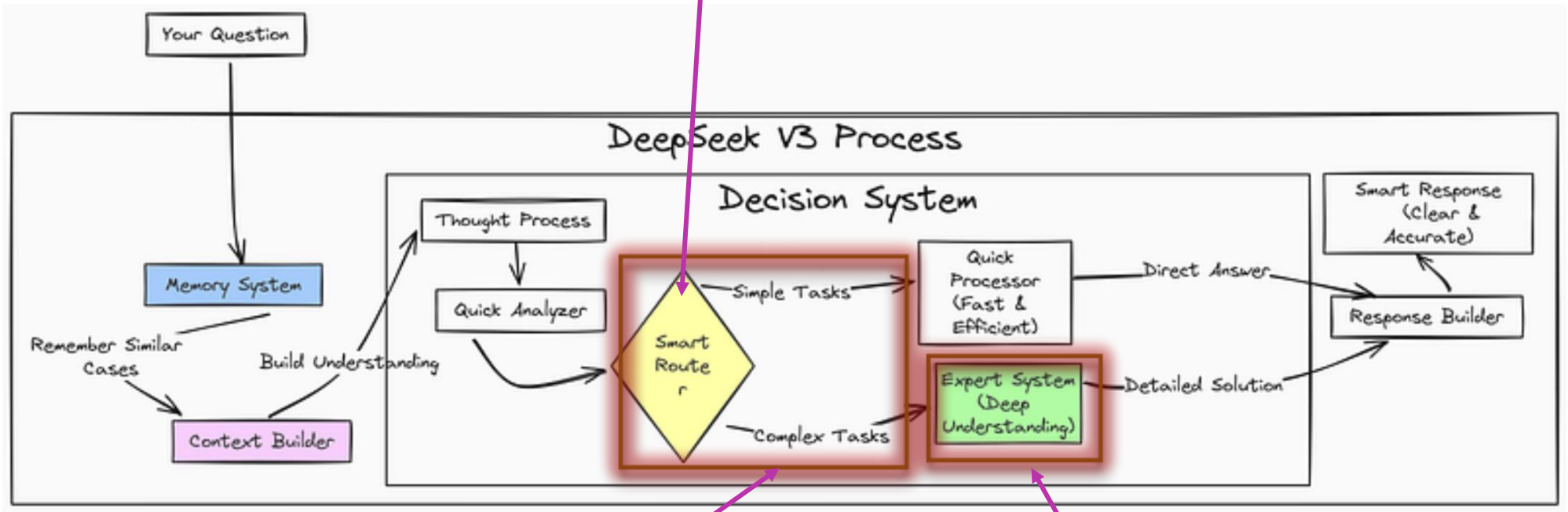
DeepSeek-R1, which incorporates multi-stage training (e.g., SFT) and cold-start data before RL

DeepSeek Implementation Quick Overview (Created by Fareed Khan)

https://levelup.gitconnected.com/drawing-deepseek-r1-architecture-and-training-process-from-scratch-72043da33955

How DeepSeek V3 Thinks?

Uses a **router** to combine responses into clear outputs.



DeepSeek V3 Architecture (Created by Fareed Khan)

DeepSeek V3 follows two paths:

- **Quick processor** for simple tasks.
- **Expert system** for complex problems.

A **Mixture of Experts (MoE)** model, dynamically selecting the best expert for each task.

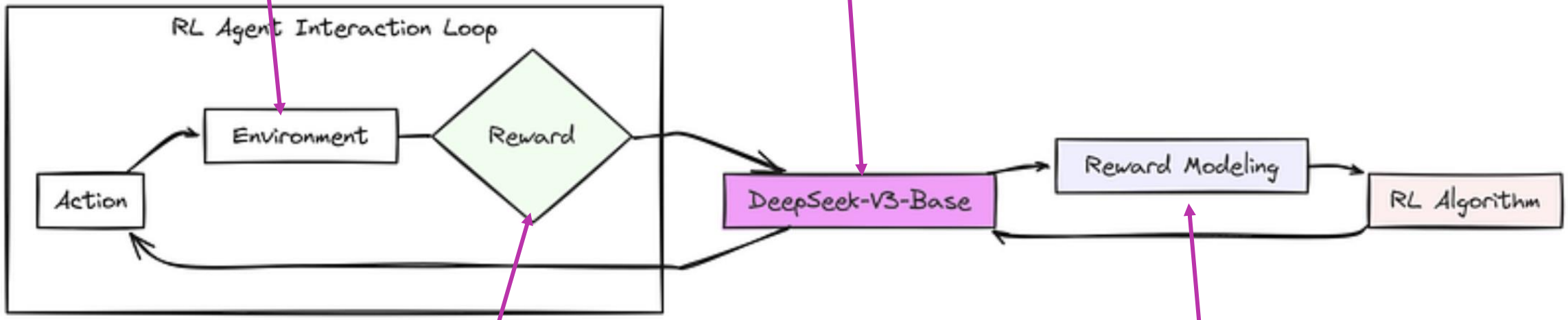
https://levelup.gitconnected.com/drawing-deepseek-r1-architecture-and-training-process-from-scratch-72043da33955

DeepSeek-R1-Zero

DeepSeek V3 is the Policy Model (Actor) In RL Setup

Reasoning Task as the Environment

DeepSeek V3 as an RL Agent: It acts as an "actor" that generates answers and reasoning.



V3 as an agent workflow (Created by Fareed Khan)

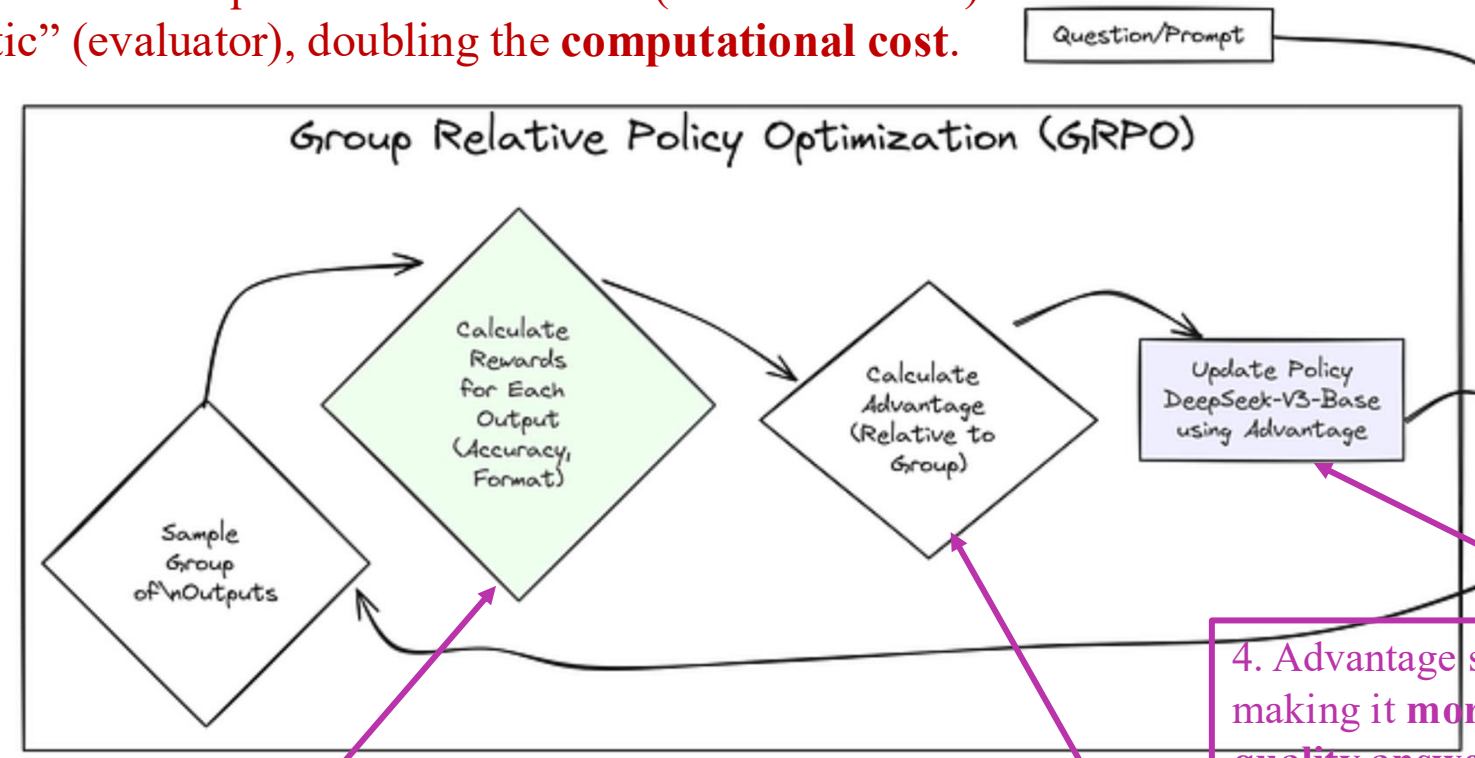
Rewards as Feedback: After generating an answer, the model receives a reward indicating how well it performed.

Learning from Rewards: Positive feedback reinforces good reasoning, helping the model refine its decision-making.

https://levelup.gitconnected.com/drawing-deepseek-r1-architecture-and-training-process-from-scratch-72043da33955

GRPO (Group Relative Policy Optimization)

Traditional RL requires both an "actor" (decision-maker) and a "critic" (evaluator), doubling the **computational cost**.

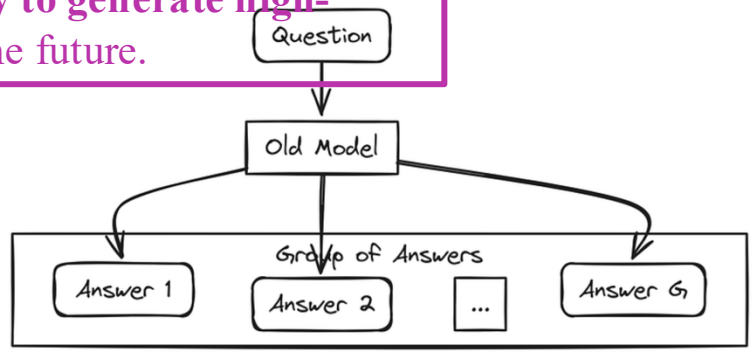


1. The model (Old Policy) generates multiple answers for the same prompt

4. Advantage scores update the Old Policy, making it **more likely to generate high-quality answers** in the future.

2. GRPO Eliminates the Critic Model: Instead of a **separate** critic, it determines a **baseline reference** from a group of generated answers.

3. Advantage Calculation: Answers are compared to the group's average; better-than-average answers get **positive advantages**, while worse ones get **negative advantages**—without needing a critic.



What averaging means (Created by Fareed Khan)

https://levelup.gitconnected.com/drawing-deepseek-r1-architecture-and-training-process-from-scratch-72043da33955

GRPO

- In order to save the training costs of RL, it adopts GRPO, which foregoes the critic model that is typically the same size as the policy model, and estimates the baseline from group scores instead.

Specifically, for each question q, GRPO samples a group of outputs $\{o_1, o_2, \cdots, o_G\}$ from the old policy $\pi_{\theta old}$ and then optimizes the policy model π_θ by maximizing the following objective:

$$\mathcal{J}_{GRPO}(\theta) = \mathbb{E}[q \sim P(Q), \{o_i\}_{i=1}^{G} \sim \pi_{\theta_{old}}(O|q)]$$

$$\frac{1}{G}\sum_{i=1}^{G}\left(\min\left(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{old}}(o_i|q)}A_i, \text{clip}\left(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{old}}(o_i|q)}, 1-\varepsilon, 1+\varepsilon\right)A_i\right) - \beta\mathbb{D}_{KL}\left(\pi_\theta||\pi_{ref}\right)\right), \quad (1)$$

$$\mathbb{D}_{KL}\left(\pi_\theta||\pi_{ref}\right) = \frac{\pi_{ref}(o_i|q)}{\pi_\theta(o_i|q)} - \log\frac{\pi_{ref}(o_i|q)}{\pi_\theta(o_i|q)} - 1, \quad (2)$$

where ε and β are hyper-parameters, and A_i is the advantage, computed using a group of rewards $\{r_1, r_2, \ldots, r_G\}$ corresponding to the outputs within each group:

$$A_i = \frac{r_i - mean(\{r_1, r_2, \cdots, r_G\})}{std(\{r_1, r_2, \cdots, r_G\})}. \quad (3)$$

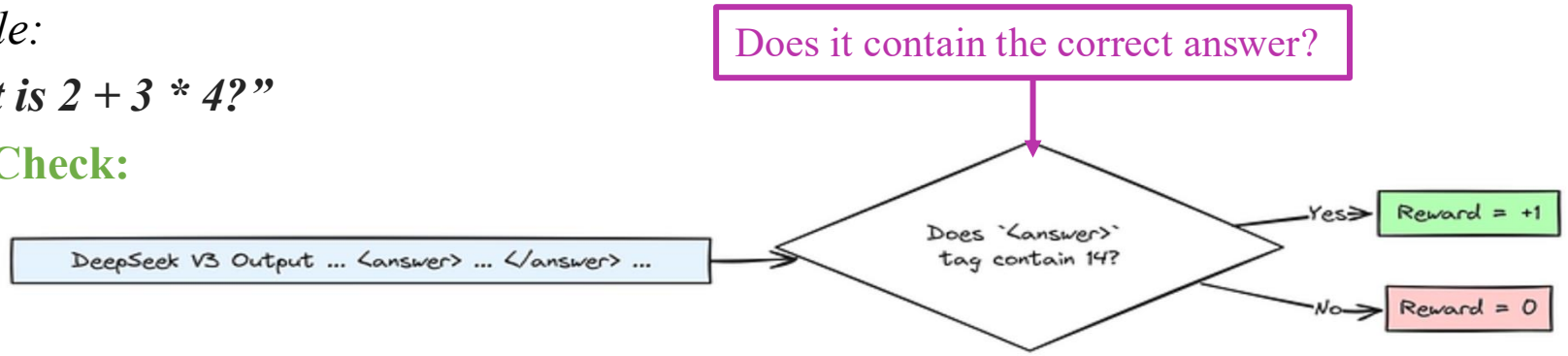
Reward Modeling for DeepSeek R1 Zero

- The reward is the source of the training signal, which decides the optimization direction of RL (Simple Rule-Based Rewards).
- **Accuracy rewards**: The accuracy reward model evaluates whether the response is correct.
- **Format Rewards:** enforces the model to put its thinking process between '<think>' and '</think>' tags.

📌 *For example:*

prompt: ***"What is 2 + 3 * 4?"***

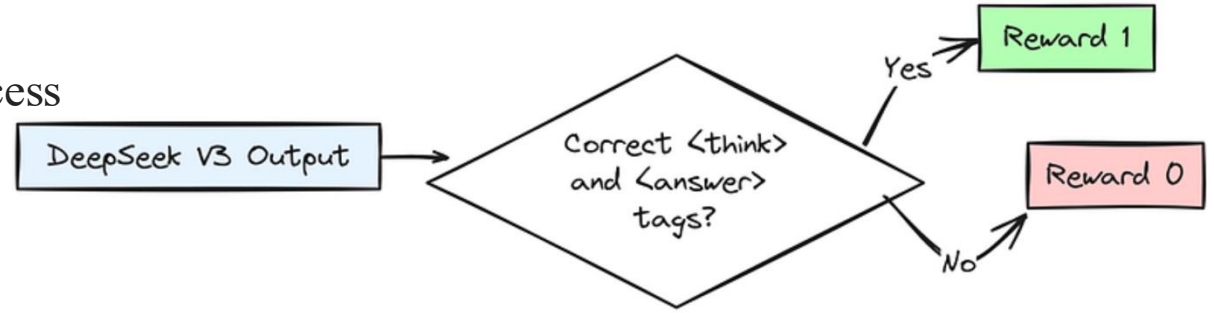
1. **Accuracy Check:**



Rule based check (Created by Fareed Khan)

2. **Format Check:**

Check if the model output *correctly* encloses the reasoning process within **<think>…</think>** and the final answer within **<answer>…</answer>**.



Format Rewards Process (Created by Fareed Khan)

https://levelup.gitconnected.com/drawing-deepseek-r1-architecture-and-training-process-from-scratch-72043da33955

Training Template for Reward

This template acts as a blueprint, instructing **DeepSeek-V3-Base** on how to structure its responses during the Reinforcement Learning process.

📌 For the example problem, the **input** would look like:

```
A conversation between User and Assistant. The
user asks a question, and the Assistant solves it.
The assistant first thinks about the reasoning
process in the mind and then provides the user
with the answer. The reasoning process and answer
are enclosed within <think> </think> and <answer>
</answer> tags respectively, i.e., <think>
reasoning process here </think> <answer> answer
here </answer>. User: What is 2 + 3 * 4?.
Assistant:
```

We expect the model to generate an **output** that conforms to the template, like:

```
<think>
Order of operations: multiply before
add. 3 * 4 = 12. 2 + 12 = 14
</think>
<answer>
14
</answer>
```

The DeepSeek team intentionally kept this template simple and focused on structure,
not on telling the model how to reason.

An Example

1. Generating Outputs Using the Old Policy (DeepSeek-V3-Base)

Before RL updates, the model **samples multiple responses** (G = 4) for a given prompt.

📌 Example outputs for *What is 2 + 3 × 4?*:

o1: Incorrect order of operations. ❌

o2: Correct answer with reasoning. ✅

o3: Correct answer but missing reasoning tags. ✅

o4: Incorrect reasoning and answer. ❌

- **o1:** `<think> 2 + 3 = 5, 5 * 4 = 20 </think> <answer> 20 </answer>` *(Incorrect order of operations)*
- **o2:** `<think> 3 * 4 = 12, 2 + 12 = 14 </think> <answer> 14 </answer>` *(Correct)*
- **o3:** `<answer> 14 </answer>` *(Correct, but missing `<think>` tags)*
- **o4:** `<think> ...some gibberish reasoning... </think> <answer> 7 </answer>` *(Incorrect and poor reasoning)*

2. Reward Assignment (Rule-Based System)

Each output is evaluated based on:

- **Accuracy Reward:** Whether the final answer is correct.
- **Format Reward:** Proper reasoning structure with and tags.

Correct and well-structured outputs receive **higher rewards**, while incorrect or incomplete outputs receive **lower rewards**.

Output	Accuracy Reward	Format Reward	Total Reward
o1 (Incorrect reasoning)	0	0.1	0.1
o2 (Correct with reasoning)	1	0.1	1.1
o3 (Correct but missing tags)	1	0	1.0
o4 (Incorrect and poor reasoning)	0	0.1	0.1

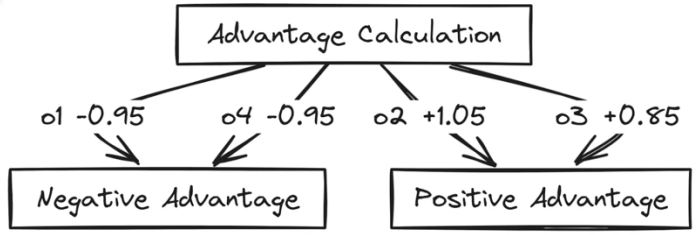
reward.md hosted with ❤ by GitHub view raw

3. Computing the Advantage Score

Mean reward and standard deviation are calculated across all sampled outputs.

- **Advantage score** is determined by comparing each output's reward against the average:

- **Positive advantage:** Encouraged (o2, o3). ✅
- **Negative advantage:** Discouraged (o1, o4). ❌

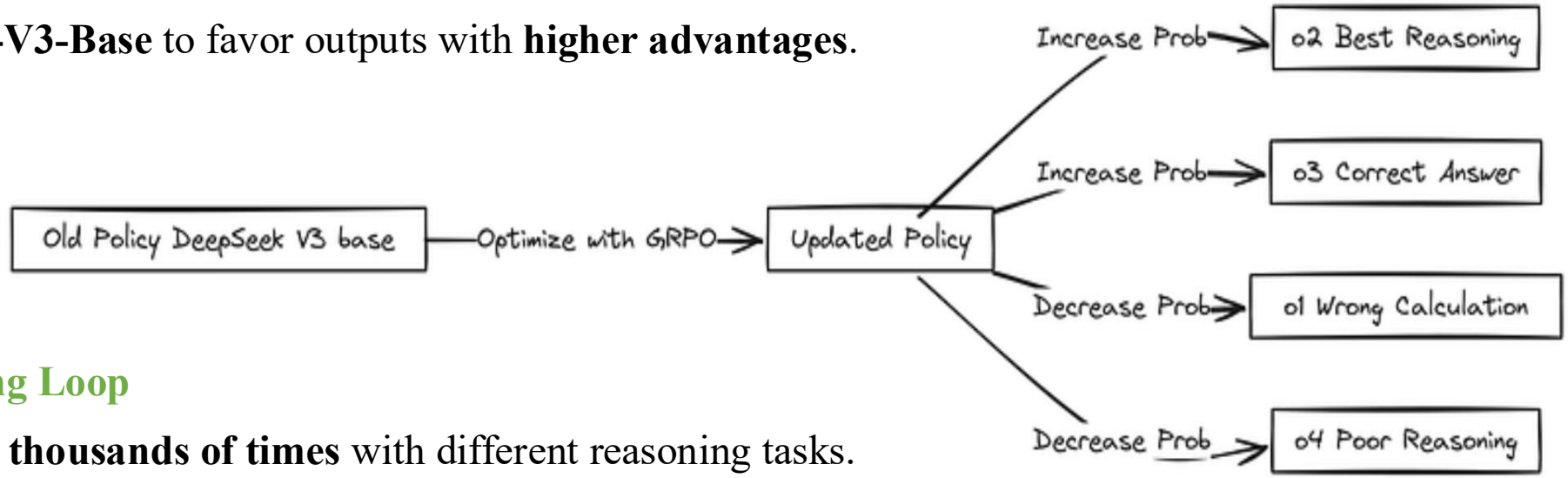


Visualizing the advantage calculation (Created by Fareed Khan)

An Example

4. Policy Update via GRPO

Updates **DeepSeek-V3-Base** to favor outputs with **higher advantages**.



GRPO working (Created by Fareed Khan)

5. Iterative Training Loop

The process repeats **thousands of times** with different reasoning tasks.

Over time, DeepSeek-R1 Zero improves by:

- **Applying correct order of operations**.
- **Providing logical reasoning steps**.
- **Consistently using the correct format**.

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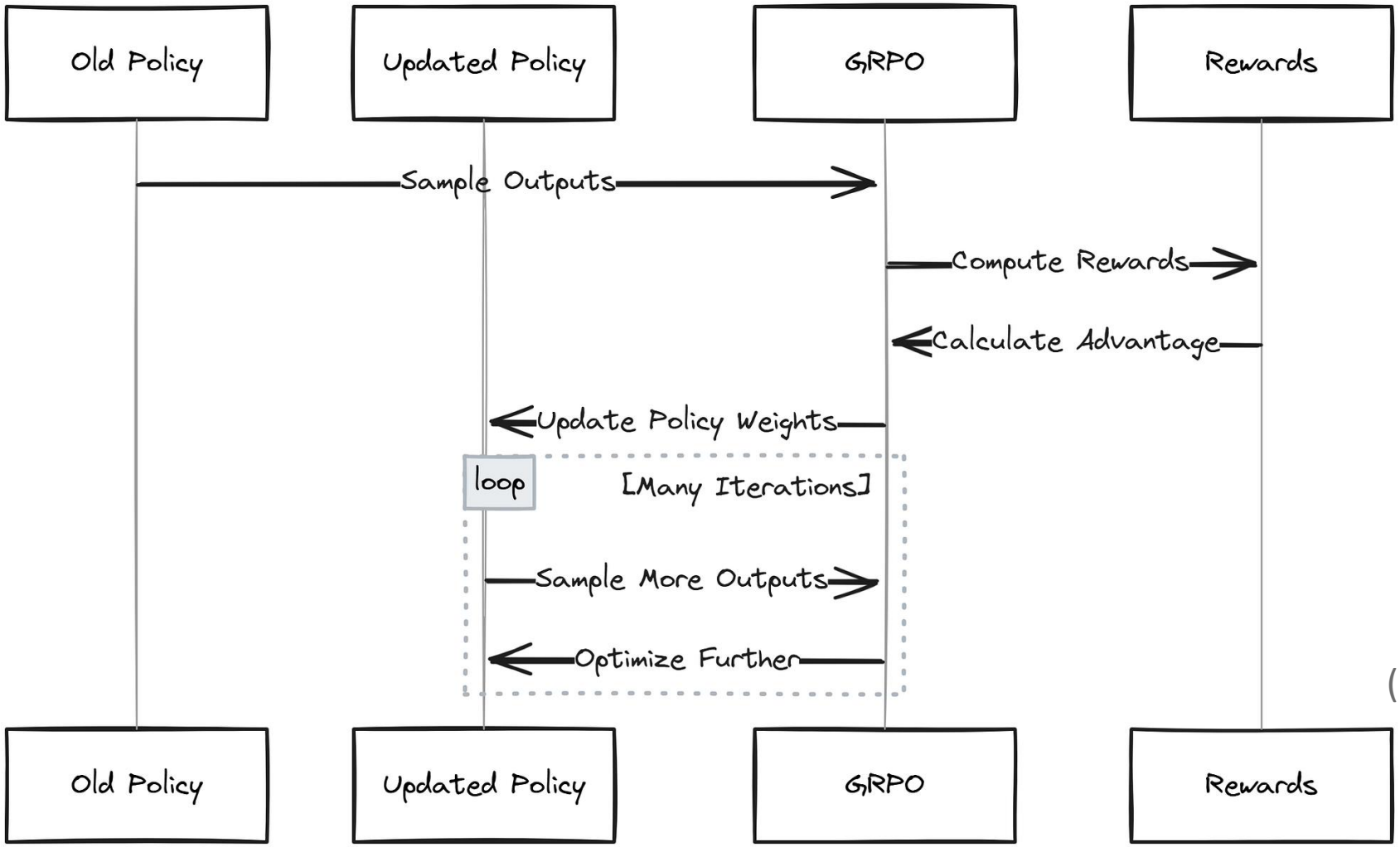
DeepSeek simplified training process



(Created by Fareed Khan)

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DeepSeek-R1 Zero to DeepSeek-R1

Two main problems with DeepSeek-R1 Zero:

1. The tags captured reasoning, but the output was often **difficult for humans to read and analyze**.
2. When given multi-lingual prompts, the model sometimes **mixed languages within the same response**.

Two key questions arise:

1. Can reasoning performance be further improved with a small amount of high-quality data as a cold start?
2. How to train a user-friendly model with clear Chains of Thought (CoT) and strong general capabilities?

➢ **Next Steps:**

- To **fix these issues**, researchers refined **DeepSeek-R1 Zero** into **DeepSeek-R1** 🚀
- **Goal: Better reasoning + Better output**
- **Pipeline Structure: Four-stage training process**.

✓ Cold Start >> Ensures a structured, readable format with a summary at the end of each response.

✓ Reasoning-oriented Reinforcement Learning

✓ Rejection Sampling and Supervised Fine-Tuning

✓ Non-Reasoning data

Step 1: Cold Start Data Collection (1)

1. Few-Shot Prompting with Long CoT:

- Provided **step-by-step Chain-of-Thought (CoT) examples**.
- Used | **special_token** | markers to clearly structure reasoning and answers.

📌 For **2 + 3 * 4?**, they might show prompts like this:

```
Problem Examples with Solutions:

Problem: What's the square root of 9 plus 5?
Solution: | special_token | First, find the square root
of 9, which is 3. Then, add 5 to 3. 3 + 5 equals 8. |
special_token | Summary: The answer is 8.

Problem: Train travels at 60 mph for 2 hours, how far?
Solution: | special_token | Use the formula: Distance =
Speed times Time. Speed is 60 mph, Time is 2 hours.
Distance = 60 * 2 = 120 miles. | special_token |
Summary: Train travels 120 miles.

Problem: What is 2 + 3 * 4?
Solution:
```

Output:

```
| special_token | Following order of
operations (PEMDAS/BODMAS), do
multiplication before addition. So,
first calculate 3 * 4 = 12. Then,
add 2 to 12. 2 + 12 = 14. |
special_token | Summary: The answer
is 14.
```

Cold Start Data Collection (2)

2. **Direct Prompting**

Directly prompting the model to not only solve problems but also to explicitly show its reasoning step-by-step and then double-check its answer.

📌 For **What is 2 + 3 * 4?** the prompt could be:

```
Problem: Solve this, show reasoning step-by-step, and verify: What is 2 + 3 * 4?
```

And they do expect an output that includes both reasoning steps and a verification part:

```
| special_token | Reasoning: To solve 2 + 3 * 4, I need to use order of operations. Multiplication comes before addition. Step 1: Calculate 3 * 4 = 12. Step 2: Add 2 to the result from step 1: 2 + 12 = 14. Verification: Checking order of operations again, yes, multiplication is before addition. Calculation looks right. | special_token | Summary: The answer is 14.
```

3. Refining the Results Through Post-processing by Human Annotators

Step 2: Reasoning-Oriented RL

After fine-tuning on cold-start data, the model undergoes large-scale RL training similar to DeepSeek-R1-Zero (**Same GRPO algorithm**, but with an **upgraded Reward System**).

◆ **Focus areas**: reasoning-intensive tasks (coding, mathematics, science, logic).

- **Key Upgrade:** **Language Consistency Reward**

Fixes the R1 Zero issue where reasoning mixed multiple languages. (If a question is asked in English, the reasoning and answer **must also be in English**.)

- **RL Training Loop**

a. **Generate multiple outputs** using the model.

b. **Assign rewards**, now including **Language Consistency Reward**.

c. **Use GRPO** to calculate **advantage scores**.

d. **Train the model** to favor high-reward responses.

e. **Repeat** the process to refine reasoning and consistency.

Step 3: Rejection Sampling & SFT

SFT after RL Convergence:

Fine-tune DeepSeek-V3-Base again using the data generated by DeepSeek-R1-Zero.

- Uses the converged checkpoint to collect new SFT data.
- Unlike the initial cold-start data, which primarily focuses on reasoning, this stage incorporates data from other domains to enhance the model's capabilities in writing, role-playing, and other general-purpose tasks.

Reasoning Data Collection:

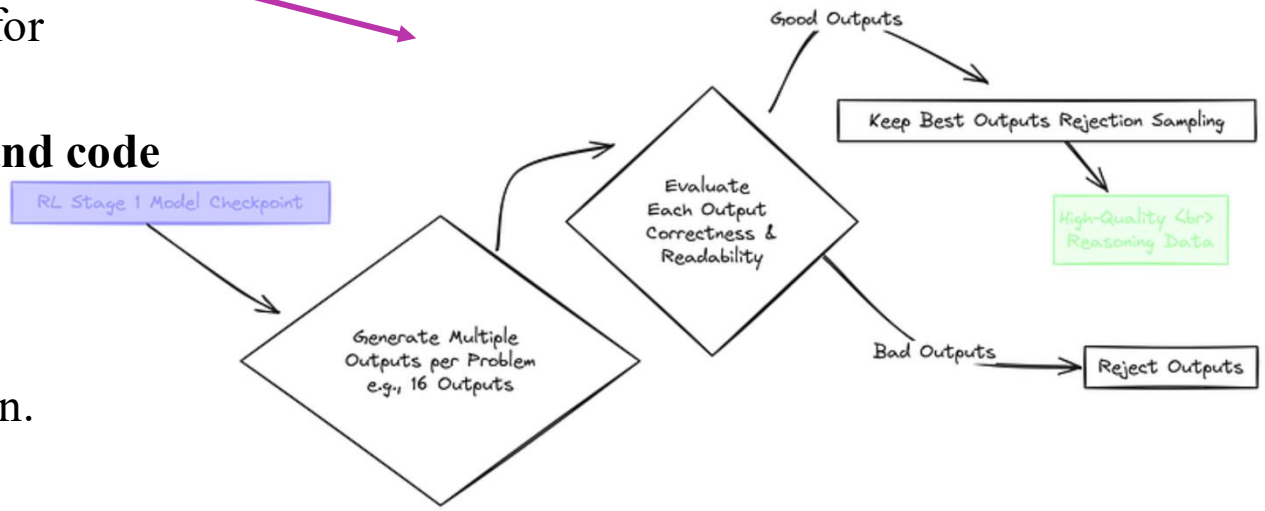
- Curated reasoning prompts with rejection sampling from the RL-trained checkpoint.
- Expanded dataset includes generative reward models for judgment.
- Filters out **mixed-language CoT, long paragraphs, and code blocks** for better readability.

>> **600k reasoning-related samples** collected.

Non-Reasoning Data Collection:

Covers writing, factual QA, self-cognition, and translation.

>> **200k non-reasoning samples** collected.



Rejection Sampling (Created by Fareed Khan)

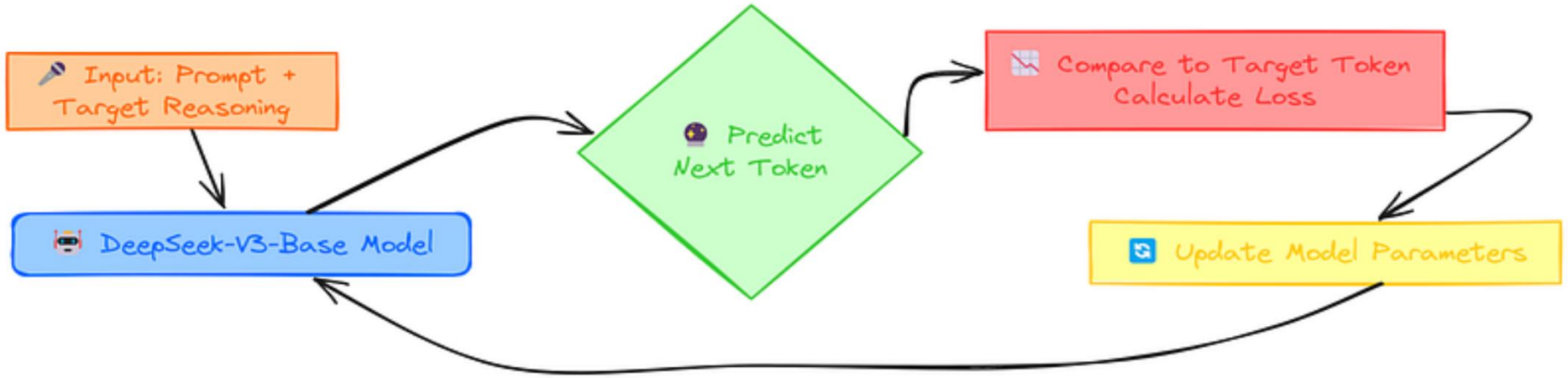
https://levelup.gitconnected.com/drawing-deepseek-r1-architecture-and-training-process-from-scratch-72043da33955

Supervised Fine-Tuning

800k total samples used for fine-tuning DeepSeek-V3-Base over **two epochs**.

Training Process (Step-by-Step):

1. **Input Prompt + Target Reasoning** → Given to the model as a training example.
2. **Predict Next Token** → Model generates the next word in the reasoning sequence.
3. **Compare to Target Token (Calculate Loss)** → Measures how far the prediction is from the correct token.
4. **Update Model Parameters** → Backpropagation + optimizer adjust model weights.
5. **Repeat Process** → Loops over thousands of examples, refining the model's structured reasoning skills.



SFT Process (Created by Fareed Khan)

https://levelup.gitconnected.com/drawing-deepseek-r1-architecture-and-training-process-from-scratch-72043da33955

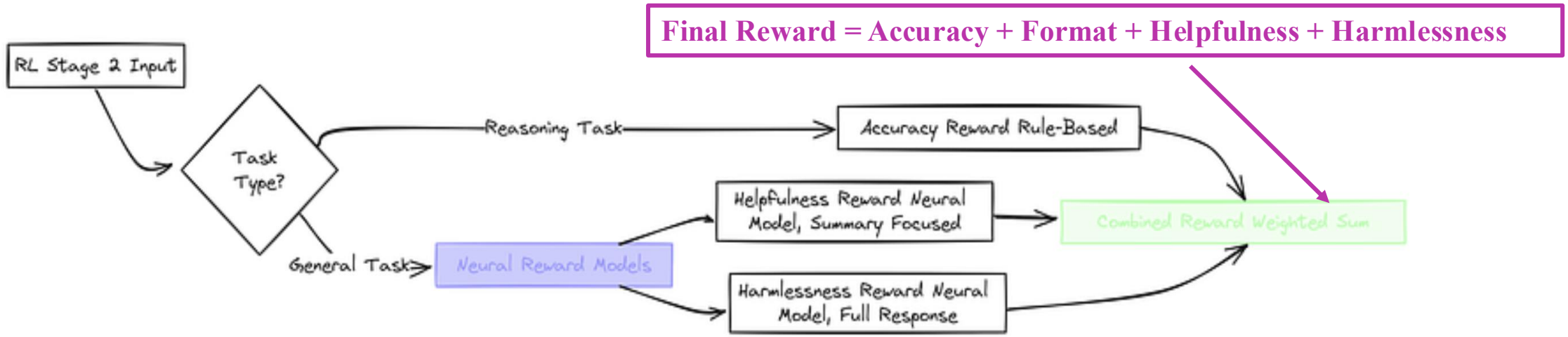
Step 4: RL for All Scenarios

- alignment with human Preferences

Implement a secondary RL stage aimed at improving the model's helpfulness and harmlessness while simultaneously refining its reasoning capabilities.

Expanded Reward System

- **Helpfulness:** Checks if the response provides **useful context**, beyond just the answer. Ensures that summaries (if included) are informative.
- **Harmlessness:** Evaluates if the response is **safe, unbiased, and appropriate**.



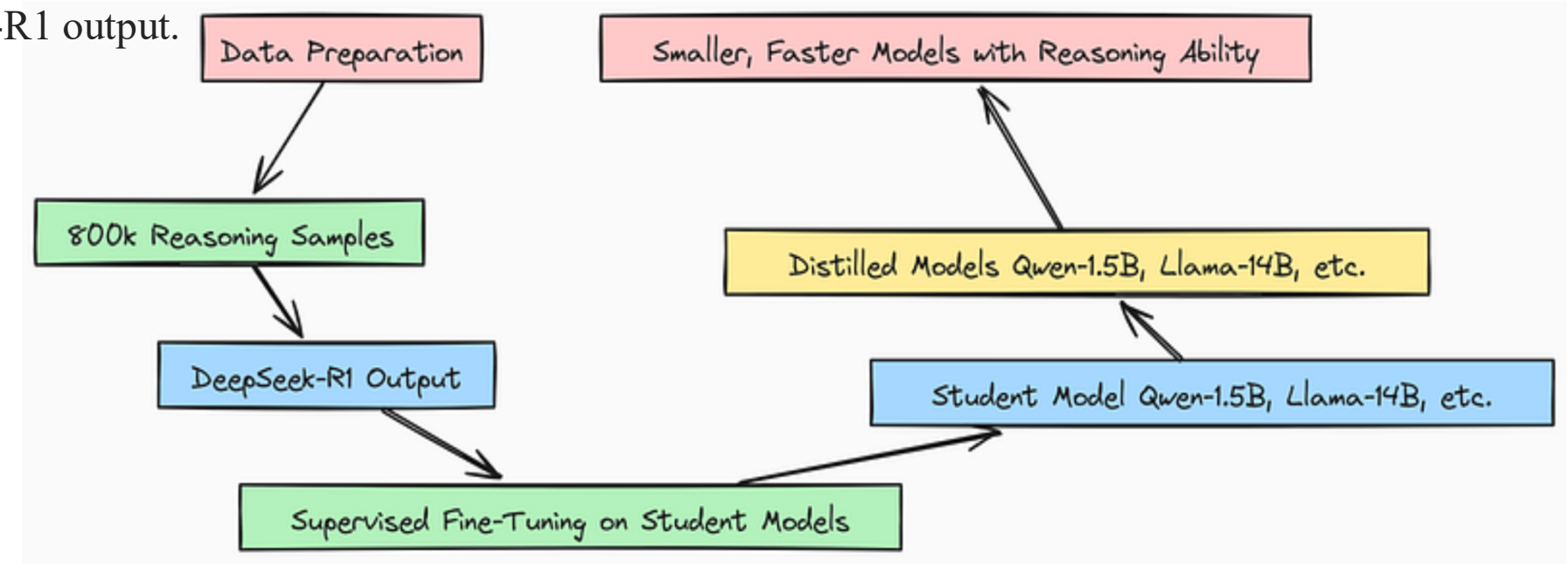
Final RL Step (Created by Fareed Khan)

https://levelup.gitconnected.com/drawing-deepseek-r1-architecture-and-training-process-from-scratch-72043da33955

Distillation

To equip more efficient smaller models with reasoning capabilities like DeepSeek-R1, we directly fine-tuned open-source models like **Qwen** and **Llama** using the 800k samples curated with DeepSeek-R1.

- **DeepSeek-R1 Output**: For each sample, the output from the teacher model (DeepSeek-R1) is used as the target for the student model.
- **Supervised Fine-Tuning (SFT)**: The student models (e.g., Qwen-1.5B, Llama-14B) are fine-tuned on these 800k samples to match the DeepSeek-R1 output.



Distillation of R1 (Created by Fareed Khan)

https://levelup.gitconnected.com/drawing-deepseek-r1-architecture-and-training-process-from-scratch-72043da33955

Content

1. Introduction to Large Language Models

2. Popular Large Language Model Architectures

3. Make LLMs More Suitable for Your Downstream Application

4. Challenges

5. Applications

Optimizing LLMs for Downstream Tasks

1. No Training Required (Prompt Engineering)

- **Zero-shot / Few-shot**: Providing descriptions and examples. Example: "Translate this sentence to French: 'Hello'."
- **Chain of Thought (CoT)**: Enhancing logical reasoning.
- **Prompt Optimization**: Iterative refinement of prompts.
- **Pros**: No computational cost, easy to implement.
- **Cons**: Limited effectiveness for complex tasks.

2. Training Required (Fine-Tuning)

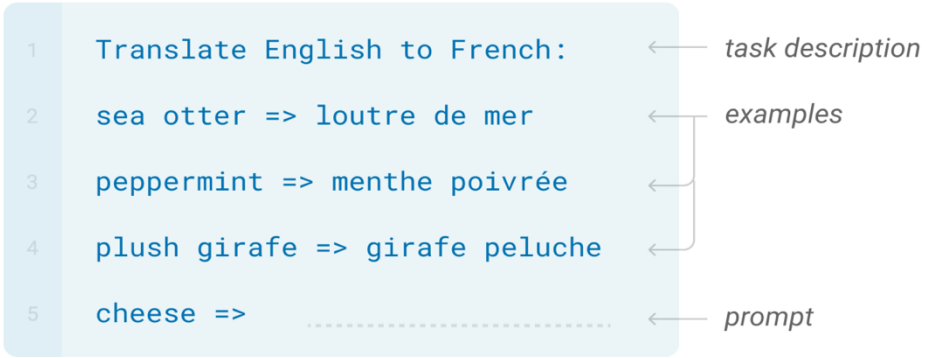
- **Full Fine-Tuning**: Adjusting all model parameters for task specificity.
- **Parameter-Efficient Fine-Tuning (PEFT, e.g., LoRA, Adapters)**: Modifying only small parts of the model.
- **Instruction Tuning**: Enhancing task adaptability.
- **Domain Adaptation**: Training on specialized datasets (e.g., biomedical data).
- **Pros**: High precision, better task adaptation.
- **Cons**: Requires computational resources, risk of overfitting.

In-Context Learning (ICL)

Technique	Definition	Key Feature
In-Context Learning (ICL)	LLMs learn tasks from provided examples or instructions within the prompt, without retraining.	No weight updates, adapts based on context.
Few-Shot Prompting	A small set of examples guides the model's response.	Demonstrations help model understand task format.
Zero-Shot Prompting	The model performs a task without any examples.	Relies solely on generalization capability.

Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.



Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

- Few-shot improves accuracy, while zero-shot relies on model's pre-learned knowledge

Brown, T. B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., Agarwal, S., Krueger, G., Henighan, T., Child, R., Ramesh, A., Ziegler, D. M., Wu, J., Winter, C., Hesse, C., . . . Amodei, D. (2020). Language Models are Few-Shot Learners. *ArXiv*. https://arxiv.org/abs/2005.14165

Useful tips for designing prompts (1)

Ingredient	Collected Prompts
Task Description	T1. Make your prompt **as detailed as possible**, *e.g., "Summarize the article into a short paragraph within 50 words. The major storyline and conclusion should be included, and the unimportant details can be omitted."* T2. It is helpful to let the LLM know that it is **an expert with a prefixed prompt**, *e.g., "You are a sophisticated expert in the domain of compute science."* T3. Tell the model **more what it should do**, but not what it should not do. T4. To avoid the LLM to generate too long output, you can just use the prompt: "*Question: Short Answer:* ". Besides, you can also use the following suffixes, "*in a or a few words*", "*in one of two sentences*".
Input Data	I1. For the question required factual knowledge, it is useful to first **retrieve relevant documents** via the search engine, and then **concatenate them into the prompt** as reference. I2. To highlight some important parts in your prompt, please **use special marks**, *e.g., quotation* ("") and *line break* (\n). You can also use both of them for emphasizing.
Contextual Information	C1. For complex tasks, you can **clearly describe the required intermediate steps** to accomplish it, *e.g., "Please answer the question step by step as: Step 1 - Decompose the question into several sub-questions, . . . "* C2. If you want LLMs to provide the score for a text, it is necessary to provide a **detailed description about the scoring standard** with examples as reference. C3. When LLMs generate text according to some context (*e.g.*, making recommendations according to purchase history), instructing them with **the explanation about the generated result** conditioned on context is helpful to improve the quality of the generated text. C4. An approach similar to **tree-of-thoughts** but can be **done in one prompt**: *e.g., Imagine three different experts are answering this question. All experts will write down one step of their thinking, then share it with the group of experts. Then all experts will go on to the next step, etc. If any expert realizes they're wrong at any point then they leave. The question is*

Zhao, W. X., Zhou, K., Li, J., Tang, T., Wang, X., Hou, Y., Min, Y., Zhang, B., Zhang, J., Dong, Z., Du, Y., Yang, C., Chen, Y., Chen, Z., Jiang, J., Ren, R., Li, Y., Tang, X., Liu, Z., . . . Wen, J. (2023). A Survey of Large Language Models. *ArXiv*. https://arxiv.org/abs/2303.18223

Useful tips for designing prompts (2)

Demonstration	D1. **Well-formatted in-context exemplars** are very useful, especially for producing the outputs with complex formats. D2. For few-shot chain-of-thought prompting, you can also use the prompt "*Let's think step-by-step*", and the few-shot examples should be **separated by "\n"** instead of full stop. D3. You can also **retrieve similar examples** in context to supply the useful task-specific knowledge for LLMs. To retrieve more relevant examples, it is useful to **first obtain the answer** of the question, and then concatenate it with the question for retrieval. D4. The **diversity of the in-context exemplars** within the prompt is also useful. If it is not easy to obtain diverse questions, you can also seek to keep the **diversity of the solutions** for the questions. D5. When using chat-based LLMs, you can **decompose in-context exemplars into multi-turn messages**, to better match the human-chatbot conversation format. Similarly, you can also decompose the reasoning process of an exemplars into multi-turn conversation. D6. **Complex and informative** in-context exemplars can help LLMs answer complex questions. D7. As a symbol sequence can typically be divided into multiple segments (*e.g.*, $i_1, i_2, i_3 \longrightarrow i_1, i_2$ and i_2, i_3), the preceding ones can be used **as in-context exemplars** to guide LLMs to predict the subsequent ones, meanwhile providing historical information. D8. **Order matters** for in-context exemplars and prompts components. For very long input data, the position of the question (first or last) may also affect the performance. D9. If you can not obtain the in-context exemplars from existing datasets, an alternative way is to use the **zero-shot generated ones** from the LLM itself.
Other Designs	O1. Let the **LLM check its outputs** before draw the conclusion, *e.g.*, "*Check whether the above solution is correct or not.*" O2. If the LLM can not well solve the task, you can **seek help from external tools** by prompting the LLM to manipulate them. In this way, the tools should be encapsulated into callable APIs with detailed description about their functions, to better guide the LLM to utilize the tools. O3. The prompt should be **self-contained**, and better not include pronouns (*e.g.*, it and they) in the context. O4. When using LLMs for comparing two or more examples, the order affects the performance a lot. O5. Before the prompt, **assigning a role for the LLM** is useful to help it better fulfill the following task instruction, *e.g.*, "*I want you to act as a lawyer*". O6. OpenAI models can perform a task better in English than other languages. Thus, it is useful to first **translate the input into English** and then feed it to LLMs. O7. For multi-choice questions, it is useful to **constrain the output space** of the LLM. You can use a more detailed explanation or just imposing constraints on the logits. O8. For sorting based tasks (*e.g.*, recommendation), instead of directly outputting the complete text of each item after sorting, one can **assign indicators** (*e.g.*, *ABCD*) to the unsorted items and instruct the LLMs to directly output the sorted indicators.

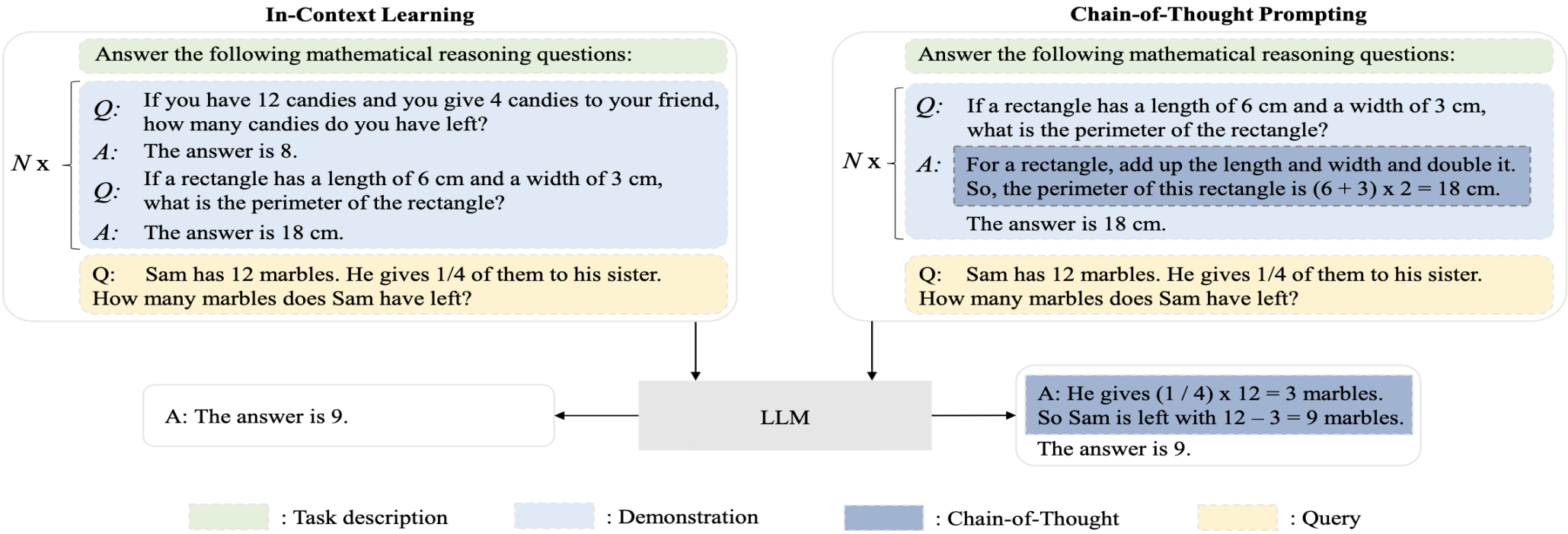
Zhao, W. X., Zhou, K., Li, J., Tang, T., Wang, X., Hou, Y., Min, Y., Zhang, B., Zhang, J., Dong, Z., Du, Y., Yang, C., Chen, Y., Chen, Z., Jiang, J., Ren, R., Li, Y., Tang, X., Liu, Z., . . . Wen, J. (2023). A Survey of Large Language Models. *ArXiv*. https://arxiv.org/abs/2303.18223

ICL vs. Chain-of-Thought (CoT)

Limitations of ICL:

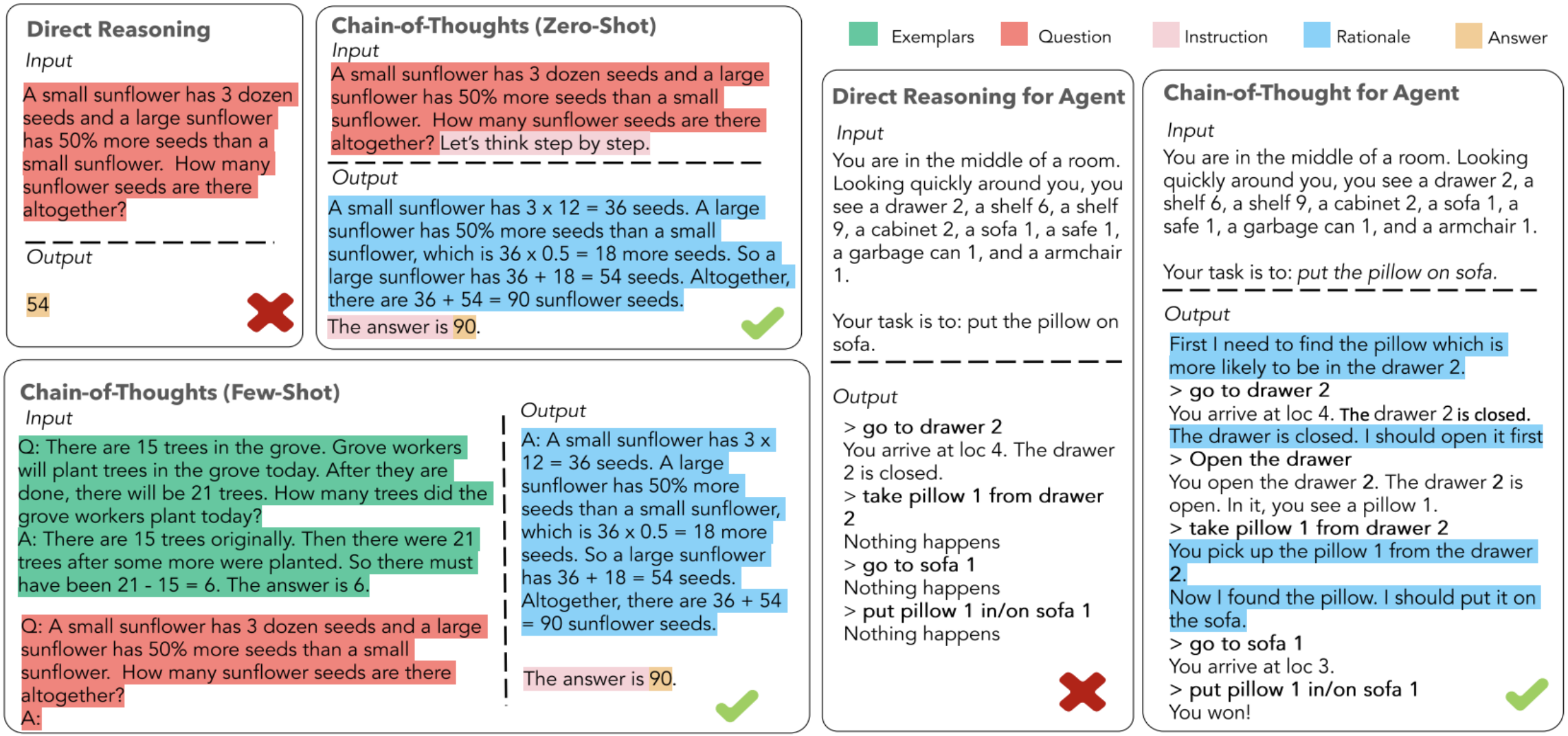
- **Poor Generalization for Complex Tasks**: Struggles with multi-step reasoning or problems requiring deeper logical connections.
- Performance depends on the number and quality of demonstration examples.

CoT: Instead of simply constructing the prompts with input-output pairs like ICL, CoT prompting further incorporates intermediate reasoning steps, which serve as the bridge between inputs and outputs.



Zhao, W. X., Zhou, K., Li, J., Tang, T., Wang, X., Hou, Y., Min, Y., Zhang, B., Zhang, J., Dong, Z., Du, Y., Yang, C., Chen, Y., Chen, Z., Jiang, J., Ren, R., Li, Y., Tang, X., Liu, Z., . . . Wen, J. (2023). A Survey of Large Language Models. *ArXiv*. https://arxiv.org/abs/2303.18223

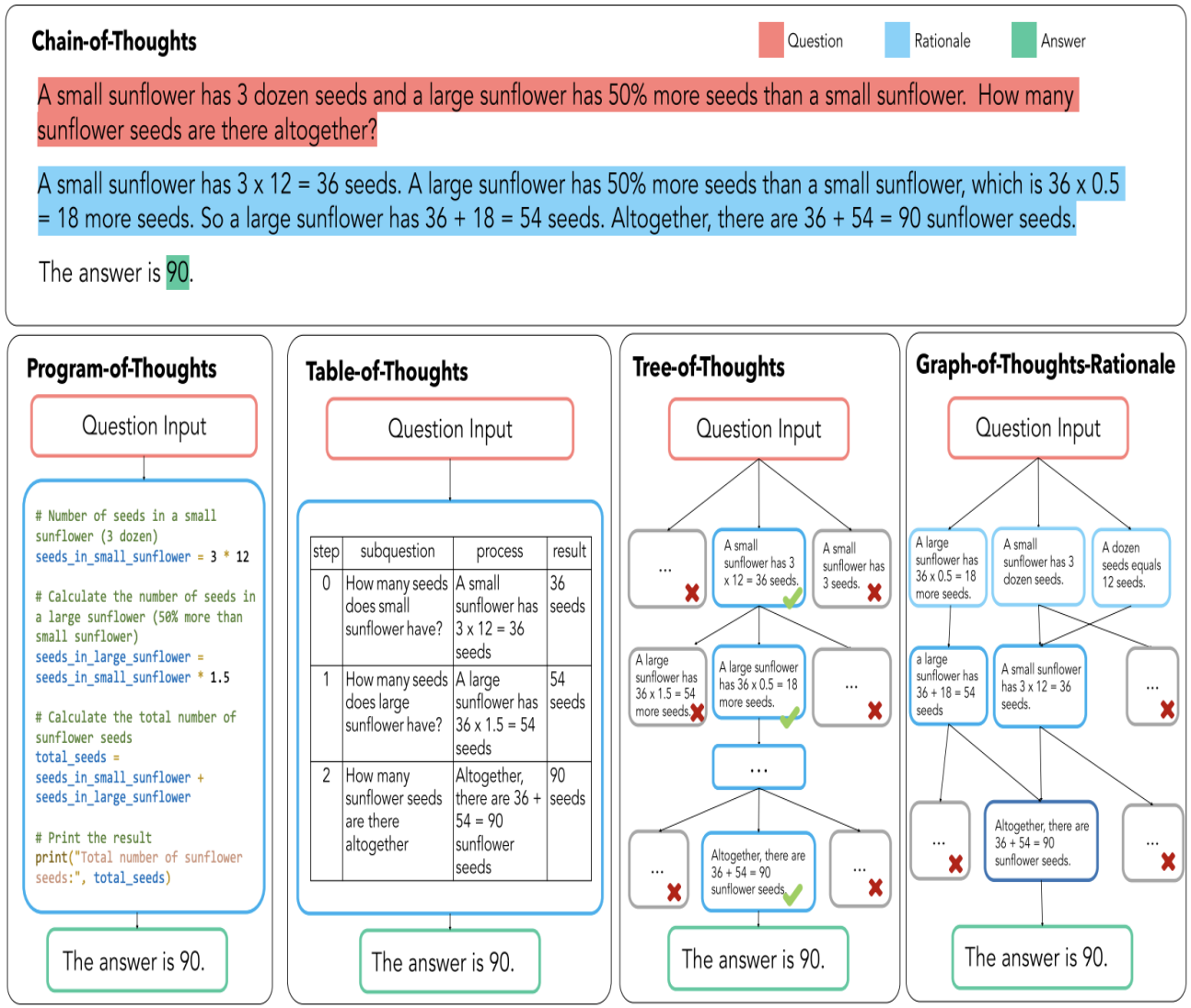
CoT vs. Direct Reasoning



CoT refers to a series of intermediate reasoning steps that are generated to solve a problem or arrive at an answer

Zhang, Z., Yao, Y., Zhang, A., Tang, X., Ma, X., He, Z., Wang, Y., Gerstein, M., Wang, R., Liu, G., & Zhao, H. (2023). Igniting Language Intelligence: The Hitchhiker's Guide From Chain-of-Thought Reasoning to Language Agents. *ArXiv*. https://arxiv.org/abs/2311.11797

Formulation Shifts of CoT



(i) Chain-of-Thoughts (CoT) (Zhang et al., 2023),
(ii) Programm-of-Thoughts (PoT) (Chenetal., 2022),
(iii) Table-of-Thoughts (Tab-CoT) (Ziqi&Lu, 2023),
(iv) Tree-of-Thoughts (ToT) (Yaoetal., 2023a),
(v) Graph-of-Thoughts-Rationale (GoT-Rationale) (Bestaetal., 2023).

Recent extensions (2024–2026):

Self-Consistency and Verifier-based Reasoning (Wang et al., 2023; extended in 2024–2025)

Tool-Augmented Reasoning / ReAct-style frameworks (ReAct: Synergizing Reasoning and Acting in Language Models; subsequent RAG + tool-use systems)

Agent-based Reasoning and Planning (multi-agent and autonomous workflows, 2024–2026)

Implicit / Distilled Reasoning (Post-CoT) (training-time reasoning via RLHF, RLAIF, and verifier models, 2024–2026)

Zhang, Z., Yao, Y., Zhang, A., Tang, X., Ma, X., He, Z., Wang, Y., Gerstein, M., Wang, R., Liu, G., & Zhao, H. (2023). Igniting Language Intelligence: The Hitchhiker's Guide From Chain-of-Thought Reasoning to Language Agents. *ArXiv*. https://arxiv.org/abs/2311.11797

Pre-training vs. Fine-Tuning

1. Definitions & Objectives

- **Pre-Training**: Learns general patterns (syntax, semantics) from large unlabeled datasets.

  Goal: Captures universal knowledge.

- **Fine-Tuning**: Adapts a pre-trained model for specific tasks using smaller labeled datasets.

  Goal: Enhances task-specific alignment (e.g., coding styles, domain jargon).

2. Technical Differences

- **Pre-Training**: Trains all parameters from scratch.
- **Fine-Tuning**:

➢ **Full Fine-Tuning**: Adjusts all parameters (risk of catastrophic forgetting).

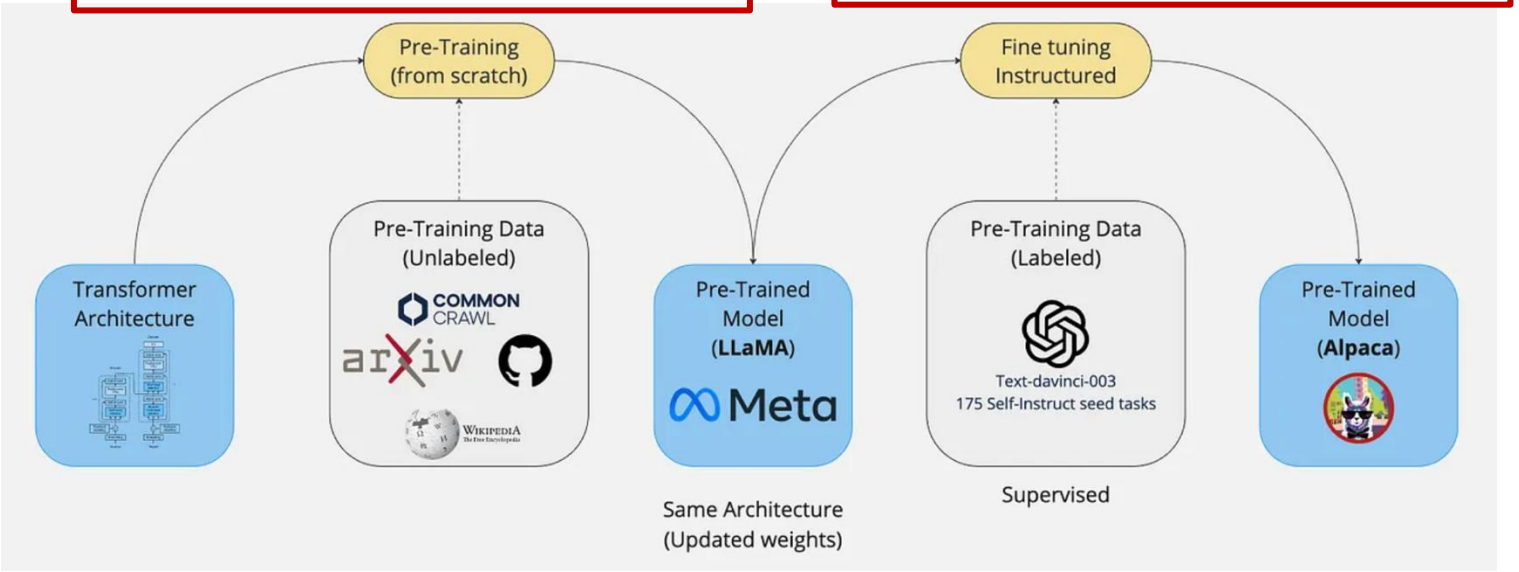
➢ **PEFT** (Parameter-Efficient Fine-Tuning): Updates only key subsets (e.g., LoRA, adapters) to retain core knowledge.

Challenges for Pre-training:

- High computational cost
- Risk of dataset-specific biases

Challenges for Fine-tuning:

- Balancing task-specific adaptation with knowledge retention
- Overfitting



https://medium.com/@eordaxd/fine-tuning-vs-pre-training-651d05186faf

PEFT - A Smarter Approach to Fine-Tuning

Why Parameter-Efficient Fine-Tuning (PEFT)?

- **Large models** achieve remarkable performance but demand **high computational resources**.
- **Full fine-tuning** updates billions of parameters, making **customization costly and impractical**.
- **PEFT** optimizes efficiency by modifying only a **small subset of parameters** while leveraging pre-trained knowledge.

Key Advantages:

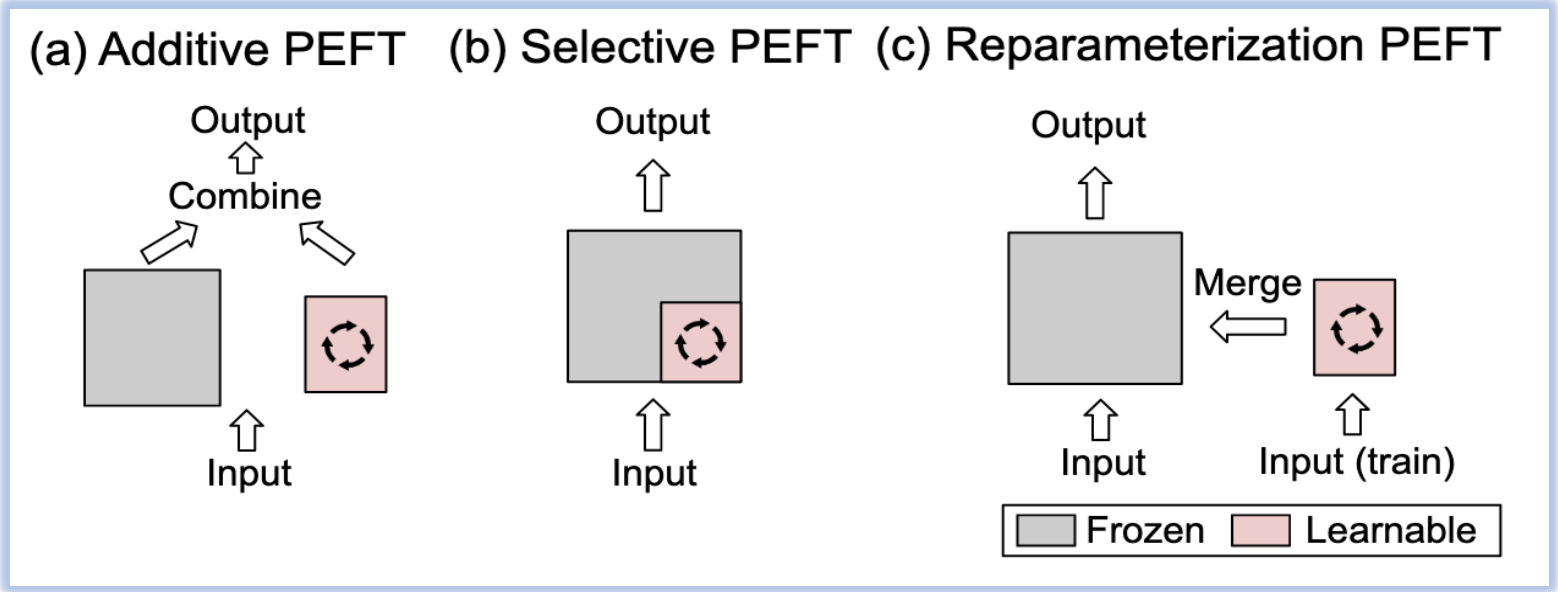
- **Lower Computational Cost:** Reduces GPU/TPU memory usage and processing power.
- **Faster Training:** Updates fewer parameters, enabling quicker adaptation.
- **Preserves Pre-Trained Knowledge:**

Prevents catastrophic forgetting while enhancing task-specific performance.

- **Balances Efficiency & Performance:**

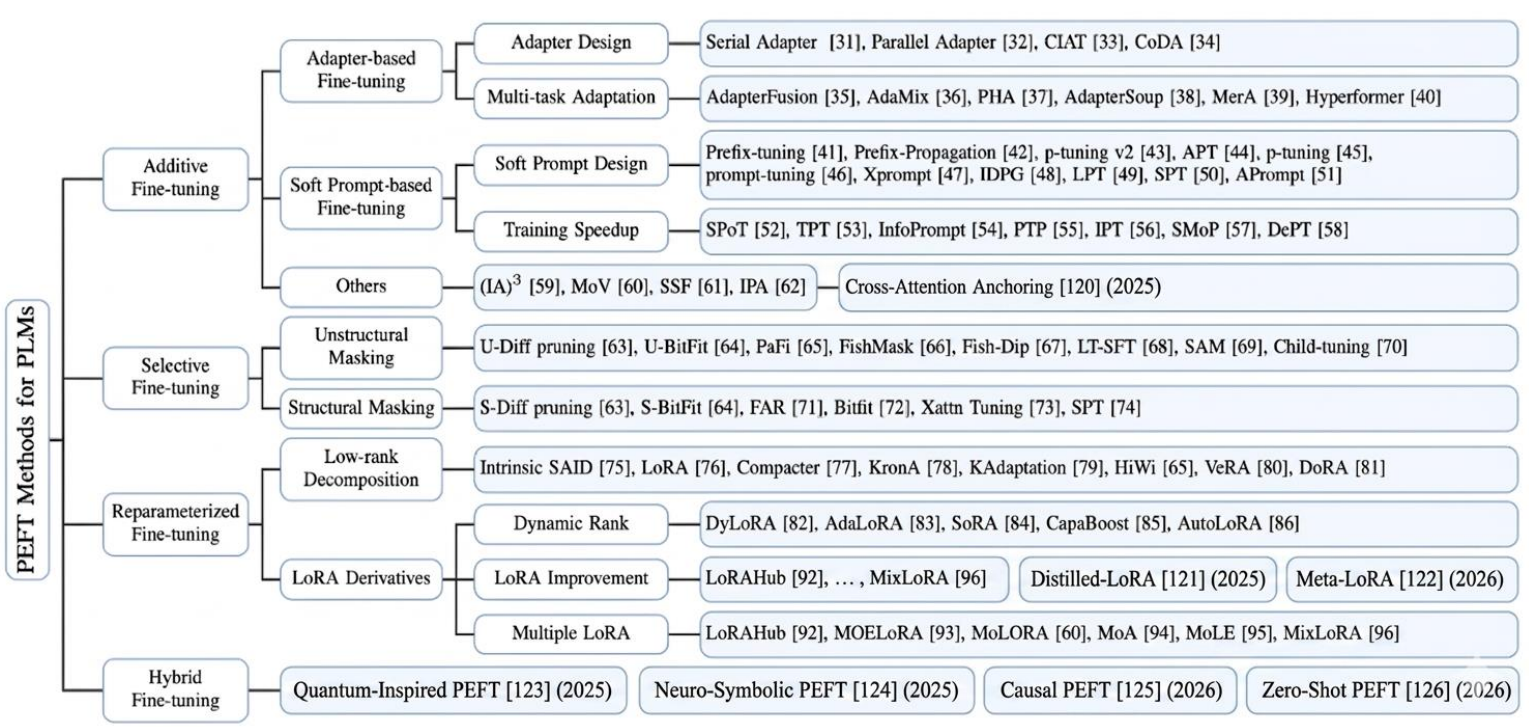
Adapts models effectively without full retraining.

Different types of PEFT algorithms



Han, Z., Gao, C., Liu, J., Zhang, J., & Zhang, S. Q. (2024). Parameter-Efficient Fine-Tuning for Large Models: A Comprehensive Survey. *ArXiv*. https://arxiv.org/abs/2403.14608

Taxonomy of PEFT Methods for Large Models

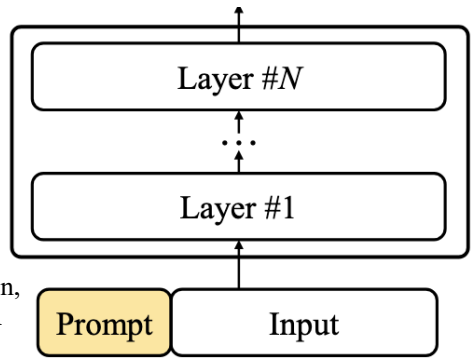


additive PEFT modifies the model architecture by injecting new trainable modules or parameters;
selective PEFT makes a subset of parameters trainable during fine-tuning;
reparameterized PEFT constructs a (low-dimensional) reparameterization of the original model parameters for training, then equivalently transforms it back for inference;
hybrid PEFT combines advantages from different PEFT methods to build a unified PEFT model.

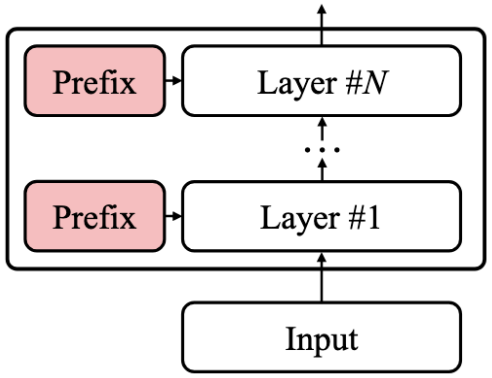
- The Three Types of **PEFT** We Will Introduce

Zhao, W. X., Zhou, K., Li, J., Tang, T., Wang, X., Hou, Y., Min, Y., Zhang, B., Zhang, J., Dong, Z., Du, Y., Yang, C., Chen, Y., Chen, Z., Jiang, J., Ren, R., Li, Y., Tang, X., Liu, Z., . . . Wen, J. (2023). A Survey of Large Language Models. *ArXiv*. https://arxiv.org/abs/2303.18223

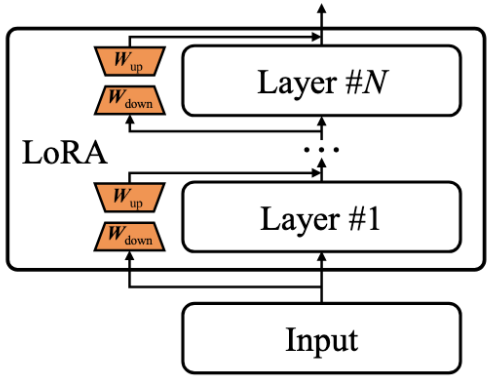
Prompt Tuning



Prefix Tuning



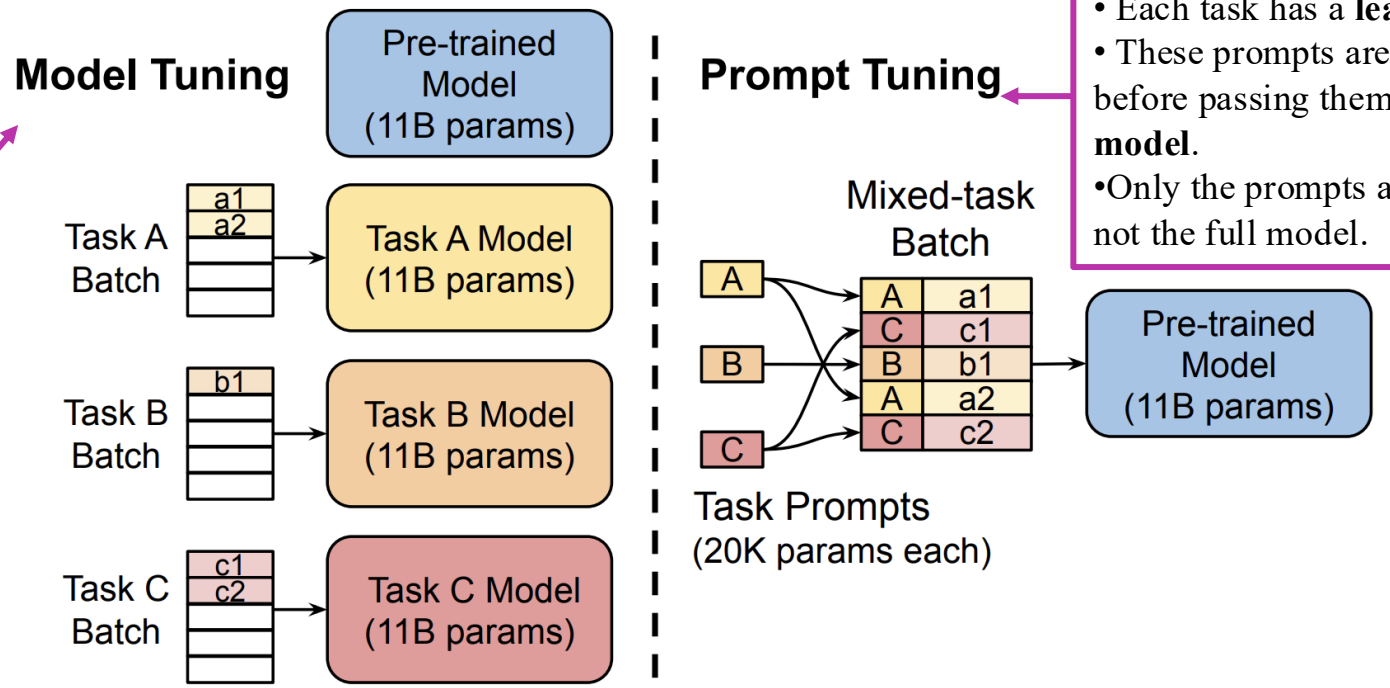
Low-Rank Adaption (Lora)



Han, Z., Gao, C., Liu, J., Zhang, J., & Zhang, S. Q. (2024). Parameter-Efficient Fine-Tuning for Large Models: A Comprehensive Survey. *ArXiv*. https://arxiv.org/abs/2403.14608

Prompt Tuning

Model tuning requires making a taskspecific copy of the entire pre-trained model for each downstream task and inference must be performed in separate batches.



- You create a **mixed-task batch** with examples from all tasks.
- Each task has a **learned prompt** (e.g., A, B, C).
- These prompts are **prepended to the inputs** before passing them into the **shared pre-trained model**.
- Only the prompts are updated during training—not the full model.

What is Prompt-Tuning?

- Prompt tuning adjusts a set of extra parameters, known as "**soft prompts**," which are integrated into the model's input processing.
- Functions as a form of **information augmentation** by explicitly specifying the task and expected output.

How Prompting Works?

- Tells the model what task to perform and what kind of response to generate.

Lester, B., & Constant, N. (2021). The Power of Scale for Parameter-Efficient Prompt Tuning. *ArXiv*. https://arxiv.org/abs/2104.08691

How Prompt Tuning Works?

Prompt-Tuning Pipeline Using T5 (Encoder-Decoder Model)

1. Given a sequence of n tokens $\{x_1, x_2, \ldots, x_n\}$, The T5 first embeds them into a matrix: $X_e \in R^{nxe}$, where e is the embedding dimension.
2. Soft-prompts are trainable parameters: $P_e \in R^{pxe}$, where p is the prompt length.
3. **How Soft-Prompting Works?**

- **Concatenation**: The soft prompt P_e is prepended to the embedded input: $[P_e;\ X_e] \in R^{\{(p+n)\times e\}}$
- **Encoding & Decoding**: This combined matrix flows through the encoder-decoder as usual.
- **Training Objective**: The model maximizes the probability of output Y, only the prompt parameters P_e are updated, keeping the pretrained model frozen.

Advantages:

- Higher Memory Efficiency
- Multi-Task Capability
- Scaling Properties

Prefix-Tuning: Optimizing Continuous Prompts for Generation

What is Prefix-Tuning?

- A **lightweight alternative** to fine-tuning for **natural language generation (NLG)** tasks.
- Inspired by **prompting**, but instead of real tokens, it uses **learnable task-specific vectors** (prefix).

Intuition Behind Prefix-Tuning

Inspired by Prompting: Proper context can steer a language model (LM) without changing its core parameters.

Example: To generate *Obama*, the model is more likely to predict it if given *Barack* as context.

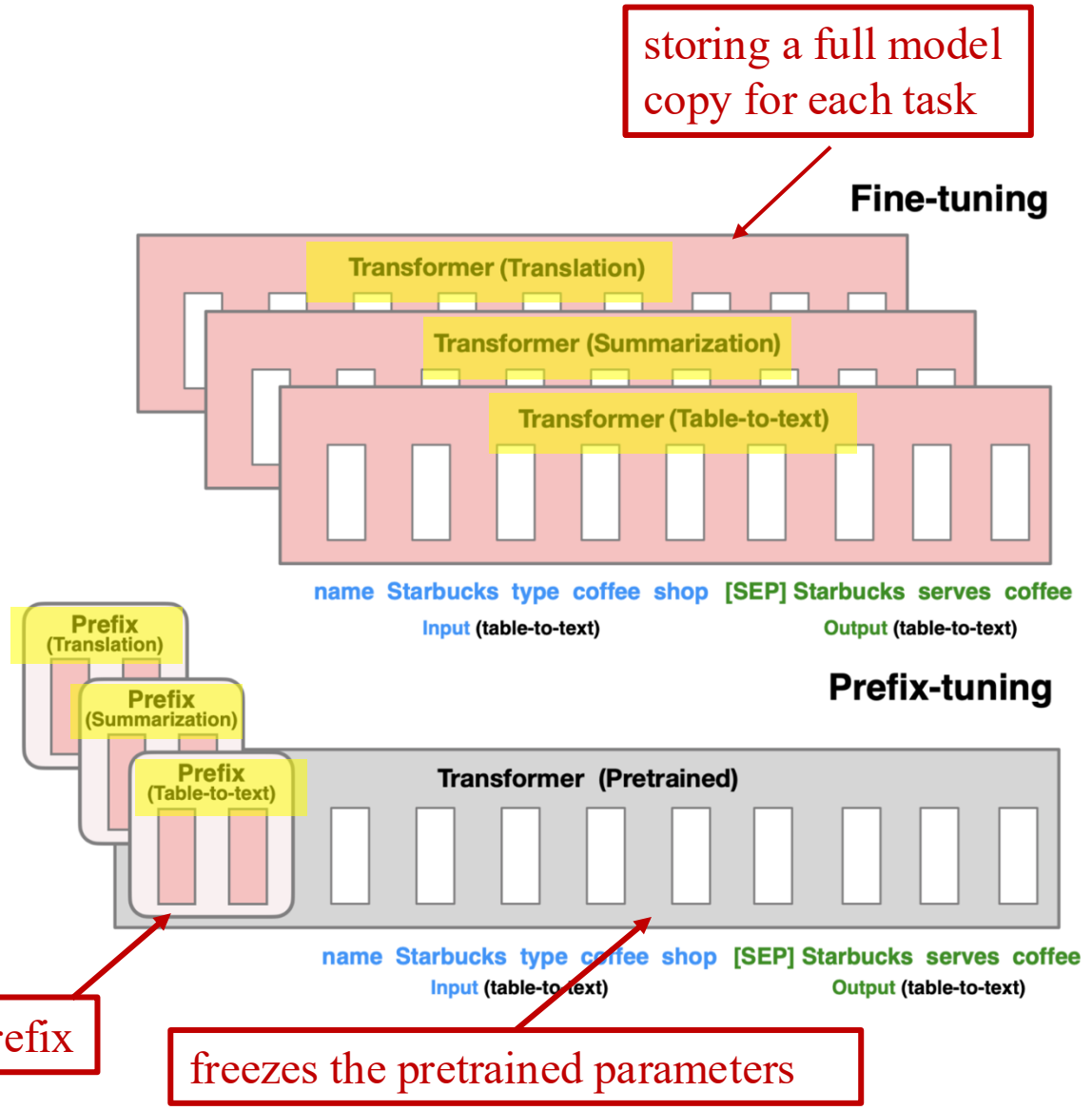
Extending this intuition:

- The right **context (prefix)** can guide the LM's encoding and generation process.
- Instead of **discrete token prompts**, optimize **continuous word embeddings** as prefixes.
- This influences **all Transformer layers**, making it **more expressive than discrete prompting** but **less intrusive than full fine-tuning**.

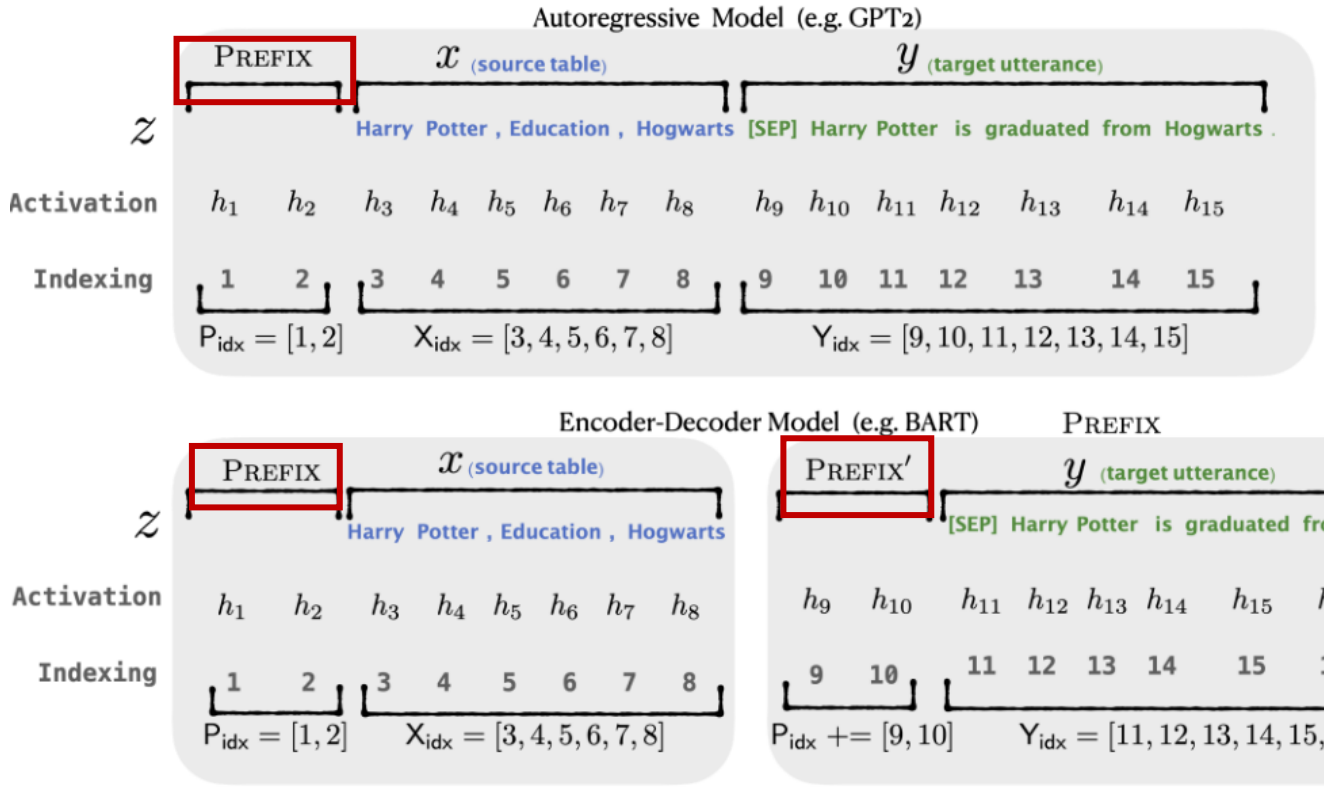
Prefix-Tuning: Optimizing Continuous Prompts for Generation

How It Works:

- Prefix-tuning **prepends a task-specific prefix** to the input sequence while keeping the pre-trained model **frozen**.
- Instead of optimizing discrete tokens, **prefix-tuning optimizes continuous word embeddings**, which are propagated through **all Transformer activation layers** and influence subsequent tokens.
- This method is **more expressive** than discrete token optimization and **less intrusive** than modifying all activation layers, helping to reduce long-range dependencies while keeping fewer trainable parameters.



Li, X. L., & Liang, P. (2020). Prefix-Tuning: Optimizing Continuous Prompts for Generation. *ArXiv*. https://arxiv.org/abs/2101.00190



Summarization Example

```
Article: Scientists at University College London discovered people
tend to think that their hands are wider and their fingers are
shorter than they truly are.They say the confusion may lie in the
way the brain receives information from different parts of the
body.Distorted perception may dominate in some people, leading to
body image problems ... [ignoring 308 words] could be very
motivating for people with eating disorders to know that there was
a biological explanation for their experiences, rather than
feeling it was their fault."

Summary: The brain naturally distorts body image -
a finding which could explain eating disorders like
anorexia, say experts.
```

Table-to-text Example x is a linearized table

```
Table: name[Clowns] customer-
rating[1 out of 5] eatType[coffee
shop] food[Chinese] area[riverside]
near[Clare Hall]

Textual Description: Clowns is a
coffee shop in the riverside area
near Clare Hall that has a rating
1 out of 5 . They serve Chinese
food .
```

y is a textual description

Prefix-tuning initializes a trainable matrix P_θ (parametrized by θ) of dimension $|P_{idx}| \times \dim(h_i)$ to store the prefix parameters.

$$h_i = \begin{cases} P_\theta[i, :], & \text{if } i \in \mathsf{P}_{\mathsf{idx}}, \\ \mathrm{LM}_\phi(z_i, h_{<i}), & \text{otherwise.} \end{cases}$$

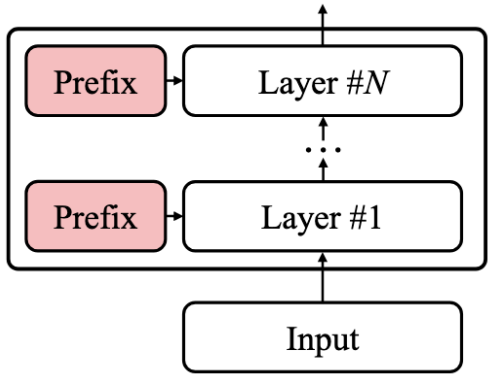
The training objective $\max_\phi \log p_\phi(y \mid x) = \sum_{i \in \mathsf{Y}_{\mathsf{idx}}} \log p_\phi(z_i \mid h_{<i})$.

These additional parameters are usually referred to as **Soft Prompts**. Unlike **manually written** Hard Prompts, soft prompts are dynamically adjusted during training. Essentially, they are **trainable and continuous embeddings**.

Li, X. L., & Liang, P. (2020). Prefix-Tuning: Optimizing Continuous Prompts for Generation. *ArXiv*. https://arxiv.org/abs/2101.00190

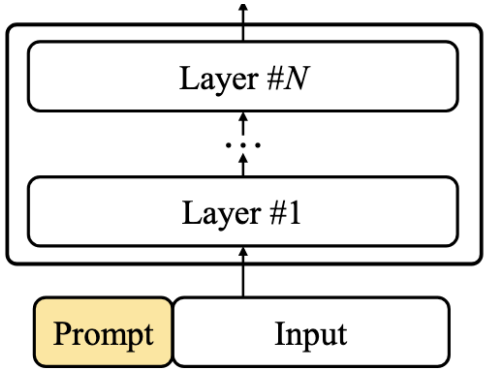
Prefix Tuning vs. Prompt Tuning

Prefix-tuning and Prompt-tuning are very similar, but they handle soft prompts differently.



Prefix-Tuning

◆ Prepend **trainable prefixes** into **both input embeddings and attention modules**.



Prompt-Tuning

◆ **Only** prepend soft prompts to the **input embeddings**.

Average PEFT performance on T5 model

Method	$T5_{LARGE}$	$T5_{3B}$	$T5_{11B}$
Additive methods			
Adapters (Houlsby)	**67.34**$_{\pm 9.58}$	74.66$_{\pm 1.68}$	**76.16**$_{\pm 1.47}$
Adapters (Pfeiffer)	62.93$_{\pm 3.52}$	69.92$_{\pm 5.61}$	50.72$_{\pm 1.69}$
Parallel Adapter	66.78$_{\pm 3.85}$	74.15$_{\pm 0.88}$	68.74$_{\pm 12.73}$
IA3	55.06$_{\pm 1.80}$	41.77$_{\pm 0.50}$	61.05$_{\pm 3.42}$
Prefix Tuning	45.05$_{\pm 3.89}$	48.90$_{\pm 5.37}$	51.93$_{\pm 2.21}$
Prompt Tuning	8.97$_{\pm 30.91}$	8.38$_{\pm 0.50}$	-
Selective methods			
LN Tuning	64.68$_{\pm 4.59}$	72.95$_{\pm 1.38}$	**73.77**$_{\pm 0.93}$
Reparametrization-based methods			
LoRA (q and v)	**67.42**$_{\pm 2.32}$	**75.49**$_{\pm 1.71}$	**76.20**$_{\pm 1.27}$
LoRA (all linear)	**68.76**$_{\pm 1.83}$	**75.22**$_{\pm 1.28}$	**76.58**$_{\pm 2.16}$
KronA	65.68$_{\pm 3.27}$	71.98$_{\pm 0.57}$	72.13$_{\pm 7.30}$
Hybrid methods			
MAM	46.90$_{\pm 6.47}$	45.57$_{\pm 4.67}$	51.49$_{\pm 0.54}$
Compacter	64.48$_{\pm 1.81}$	70.72$_{\pm 0.87}$	**74.33**$_{\pm 1.40}$
Compacter++	64.78$_{\pm 2.23}$	71.00$_{\pm 1.62}$	**74.72**$_{\pm 0.82}$
Unipelt	44.10$_{\pm 15.48}$	47.16$_{\pm 4.84}$	52.29$_{\pm 3.09}$
Full tuning	67.22	74.83	73.25

Lialin, V., Deshpande, V., Yao, X., & Rumshisky, A. (2023). Scaling Down to Scale Up: A Guide to Parameter-Efficient Fine-Tuning. *ArXiv*. https://arxiv.org/abs/2303.15647

Zhao, W. X., Zhou, K., Li, J., Tang, T., Wang, X., Hou, Y., Min, Y., Zhang, B., Zhang, J., Dong, Z., Du, Y., Yang, C., Chen, Y., Chen, Z., Jiang, J., Ren, R., Li, Y., Tang, X., Liu, Z., . . . Wen, J. (2023). A Survey of Large Language Models. *ArXiv*. https://arxiv.org/abs/2303.18223

LoRA Fine-Tuning

Why LoRA?

- **Key Insight:** Neural networks are **over-parameterized** but rely on a **low intrinsic dimension** for task adaptation.
- **Intrinsic Dimension Hypothesis:** When fine-tuning a pre-trained language model, updating parameters only in a **low-dimensional subspace** can achieve performance similar to **full-parameter updates.**

What is LoRA?

Low-rank adaptation methods (LoRA) approximate the original weight update matrix using **low-rank matrices** and only fine-tune the low-rank matrices.

Benefits of LoRA:

1. **Parameter Efficiency**

◆ LoRA only fine-tunes a subset of low-rank parameters, significantly reducing the computational cost of training.

2. **Modularity**

◆ After training, LoRA parameters can be kept separate from the model parameters, enabling a modular approach.

◆ This allows for efficient model sharing and deployment.

3. **Cross-Task Generalization**

◆ When multiple LoRA adapters are available for different tasks, they can be combined to achieve strong cross-task generalization.

LoRA Fine-Tuning

How LoRA Works?

1. Applies **low-rank decomposition** (dimensionality reduction & expansion) to model weight updates.
2. **Freeze pretrained model weights (W_0)**, train only LoRA matrices **(**A & B**):**
   - **A:** Projects input into a **lower-dimensional space**.
   - **B:** Expands back into the **original dimension**.
3. **During inference**, combine LoRA updates with the frozen PLM:
   $h = W_0 x + \Delta W\, x = W_0 x + B\, A\, x$
4. **Initialization Strategy:**
   - **A** initialized with a **Gaussian distribution**.
   - **B** initialized as a **zero matrix** to ensure the LoRA path has no effect at the start of training.

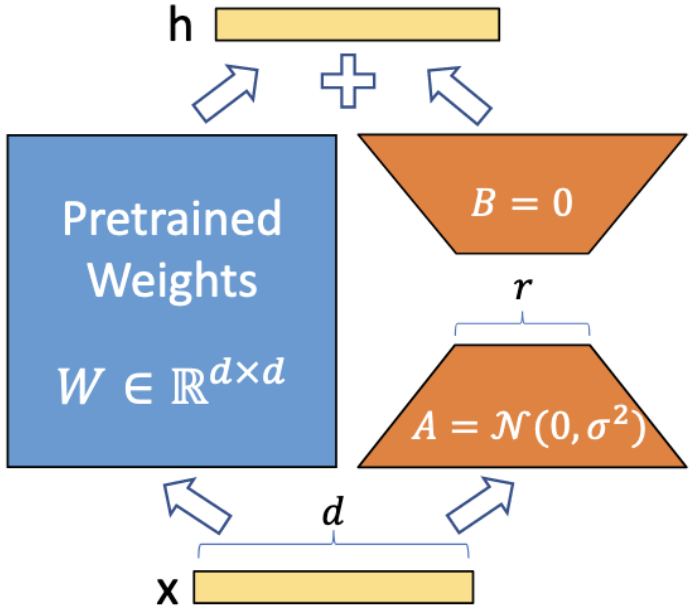


Figure 1: Our reparametrization. We only train A and B.

Hu, E. J., Shen, Y., Wallis, P., Li, Y., Wang, S., Wang, L., & Chen, W. (2021). LoRA: Low-Rank Adaptation of Large Language Models. *ArXiv*. https://arxiv.org/abs/2106.09685

Which Weight Matrices In Transformer Should We Apply LoRA To?

	# of Trainable Parameters = 18M						
Weight Type	W_q	W_k	W_v	W_o	W_q, W_k	W_q, W_v	W_q, W_k, W_v, W_o
Rank r	8	8	8	8	4	4	2
WikiSQL (±0.5%)	70.4	70.0	73.0	73.2	71.4	**73.7**	**73.7**
MultiNLI (±0.1%)	91.0	90.8	91.0	91.3	91.3	91.3	**91.7**

Under the same parameter budget, applying LoRA to different weight matrices results in varying impacts.

Key Findings:

1. Among **single weight matrices**, applying LoRA to W_o yields the best performance.
2. Maintaining the same total number of fine-tuned parameters, the more types of matrices included, the better the performance.

Hu, E. J., Shen, Y., Wallis, P., Li, Y., Wang, S., Wang, L., & Chen, W. (2021). LoRA: Low-Rank Adaptation of Large Language Models. *ArXiv*. https://arxiv.org/abs/2106.09685

Content

1. Introduction to Large Language Models

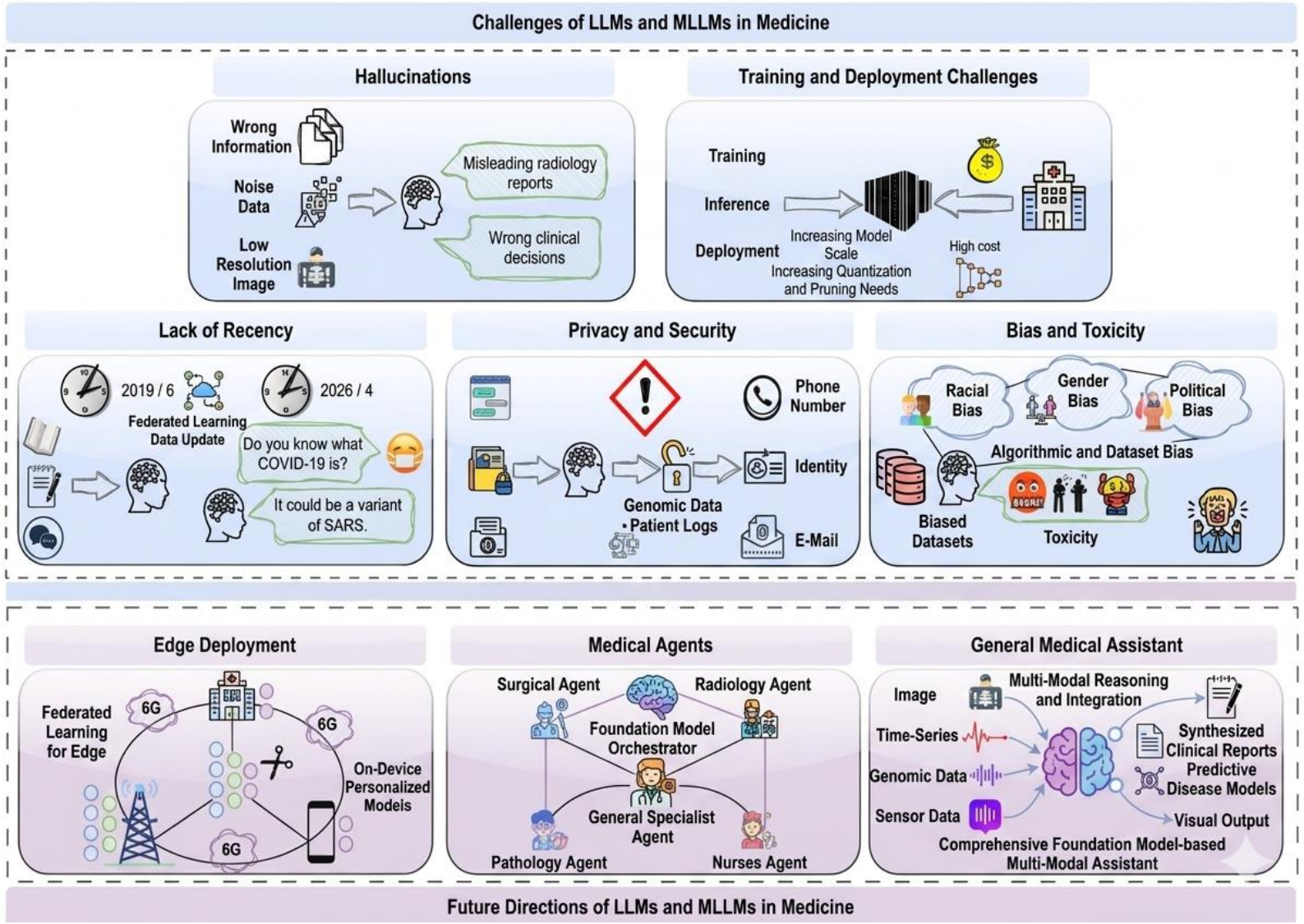
2. Popular Large Language Model Architectures

3. Make LLMs More Suitable for Your Downstream Application

4. Challenges

5. Applications

Challenges and Future Directions for Medical LLMs

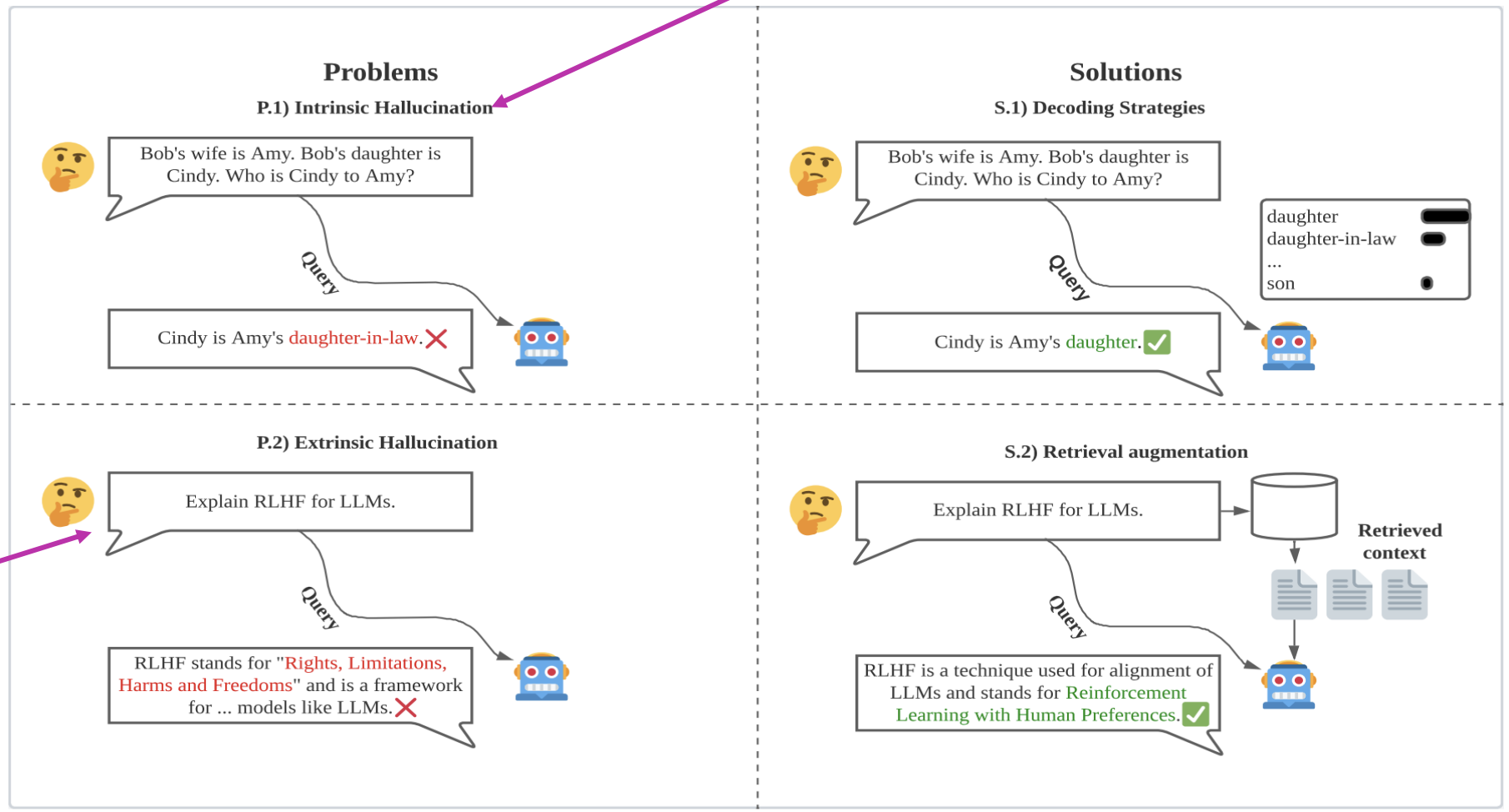


Xiao, H., Zhou, F., Liu, X., Liu, T., Li, Z., Liu, X., & Huang, X. (2025). A comprehensive survey of large language models and multimodal large language models in medicine. *Information Fusion*, *117*, 102888. https://doi.org/10.1016/j.inffus.2024.102888

Major Issue: hallucination

- LLMs are prone to generate untruthful information that either conflicts with the existing source or cannot be verified by the available source.

The LLM gives a conflicting judgment about the relationship between Cindy and Amy, which contradicts the input.



The LLM seems to have an incorrect understanding of the meaning of RLHF (reinforcement learning from human feedback), though it can correctly understand the meaning of LLMs (in this context).

Kaddour, J., Harris, J., Mozes, M., Bradley, H., Raileanu, R., & McHardy, R. (2023). Challenges and Applications of Large Language Models. *ArXiv*. https://arxiv.org/abs/2307.10169

Major Issue: Knowledge recency

Knowledge recency: The parametric knowledge of LLMs is hard to be updated in a timely manner.

Solution: Augmenting LLMs with external knowledge sources is a practical approach to tackling the issue. However, how to effectively update knowledge within LLMs remains an open research problem.

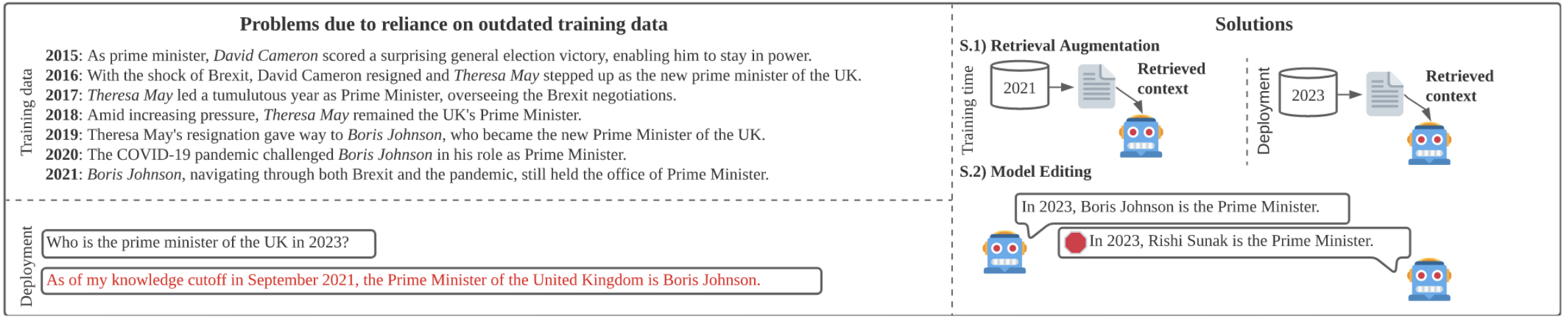


Figure 11: **Outdated knowledge** can be addressed with S.1) retrieval augmentation by hot-swapping an underlying retrieval index with up-to-date knowledge or S.2) by applying model editing techniques.

Kaddour, J., Harris, J., Mozes, M., Bradley, H., Raileanu, R., & McHardy, R. (2023). Challenges and Applications of Large Language Models. *ArXiv*. https://arxiv.org/abs/2307.10169

Major Issues

- **Reasoning Inconsistency**: LLMs may generate the correct answer following an invalid reasoning path, or produce a wrong answer after a correct reasoning process, leading to inconsistency between the derived answer and the reasoning process.

➢ **Solution**: The issue can be alleviated by fine-tuning LLMs with process-level feedback, using an ensemble of diverse reasoning paths, and refining the reasoning process with selfreflection or external feedback.

- **Numerical Computation**: LLMs face difficulties in numerical computation, especially for the symbols that are seldom encountered during pre-training.

➢ **Solution**: In addition to using mathematical tools, tokenizing digits into individual tokens is also an effective design choice for improving the arithmetic ability of LLMs.

Content

1. Introduction to Large Language Models

2. Popular Large Language Model Architectures

3. Make LLMs More Suitable for Your Downstream Application

4. Challenges

5. Applications

Application 1: How to Deploy LLMs Locally

Get up and running with large language models.

Run Llama 3.3, DeepSeek-R1, Phi-4, Mistral, Gemma 2, and other models, locally.

Download ↓

Available for macOS, Linux, and Windows

Step1: Download Ollama

Ollama is a tool that enables local deployment of LLMs easily.

https://ollama.com/

Step 2: Start Ollama

After downloading, install and run Ollama in the terminal:

```
./ollama serve
```

Step3: Choose a suitable LLM and run it （in a separate shell，https://ollama.com/search)

```
ollama run deepseek-r1:7b
```

Step3*: Load a Local Model (Optional)

If you have a pre-downloaded model, load it locally:

```
# Create a file named Modelfile, with
a FROM instruction with the local filepath to the
model you want to import.

FROM ./vicuna-33b.Q4_0.gguf
ollama create example -f
# Run the model
ollama run example 3
```


https://github.com/ollama/ollama

Step 5: Test and Interact

Once the model is running, interact with it in the terminal or integrate it with your application.

Use Case 1: interact with it in the terminal

For example:

```
>>> Tell me a fun fact about LLMs.
<think>
Alright, so I need to figure out how to respond to this user's request for
a fun fact about large language models (LLMs). The user previously asked a
question in French and received an answer in English, so maybe they're
switching languages or just being interactive.

Looking at the query "Tell me a fun fact about LLMs," I should come up
with something interesting and engaging. LLMs are pretty fascinating
because they can perform tasks like writing text, answering questions, and
even creative writing. But there's also more to explore beyond their
capabilities.
.....
</think>

Sure! Here's a fun fact about large language models (LLMs) like OpenAI's
GPT-4: They can sometimes produce grammatically correct but nonsensical or
nonsensical-sounding sentences when generating text. For example, they
might write something like, "The quick brown fox jumps over the lazy dog" —
 which is actually a valid sentence that makes no sense as a coherent
statement about the real world! This phenomenon highlights how LLMs can
sometimes create meaningful-looking output without fully understanding or
contextualizing the meaning of their words.
```

Step 5: Test and Interact

Use Case 2: Integrating the Model into a Python Script (Using Ollama's Python API)

```
import requests
import json

url = "http://localhost:11434/api/generate"

data = {
  "model": "deepseek-r1:32b",
  "prompt": "Tell me a fun fact about LLMs.",
}

response = requests.post(url, json=data, stream=True)
response.raise_for_status()

for line in response.iter_lines():
  if line:
    json_response = json.loads(line.decode('utf-8'))
    if 'response' in json_response:
      print(json_response['response'], end='')
```

You can integrate the output of LLMs into your downstream tasks.

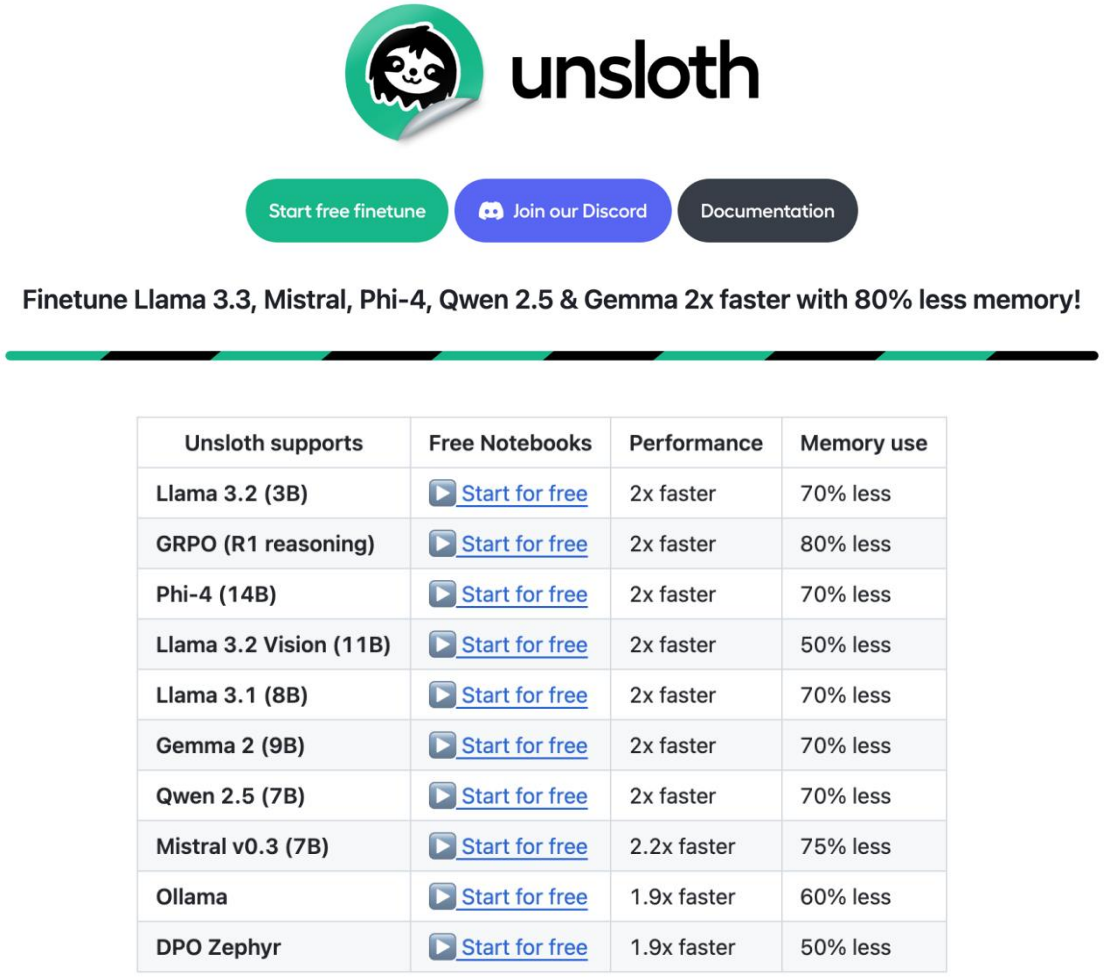
The Answer:

```
<think>
Okay, so I need to come up with a fun fact about
Large Language Models (LLMs). Hmm, let's think
about what makes them interesting or unique. I
remember that LLMs are used for generating text,
but there's more to it.
……
That should give a well-rounded fun fact about LLMs.
</think>

Large Language Models (LLMs) are fascinating due to
their versatility and advanced capabilities. Here's
an engaging overview of some key points:
1. **Vast Training Data**: LLMs are trained on
extensive datasets, often comprising trillions of
words from diverse sources like books, articles,
and websites, enabling them to understand and
generate human-like text.
……
6. **Ethical Considerations**: They contribute to
addressing ethical issues by identifying biases
within datasets or outputs, highlighting their
impact on social dynamics and fairness.

These points underscore the profound influence of
LLMs across various domains ……
```

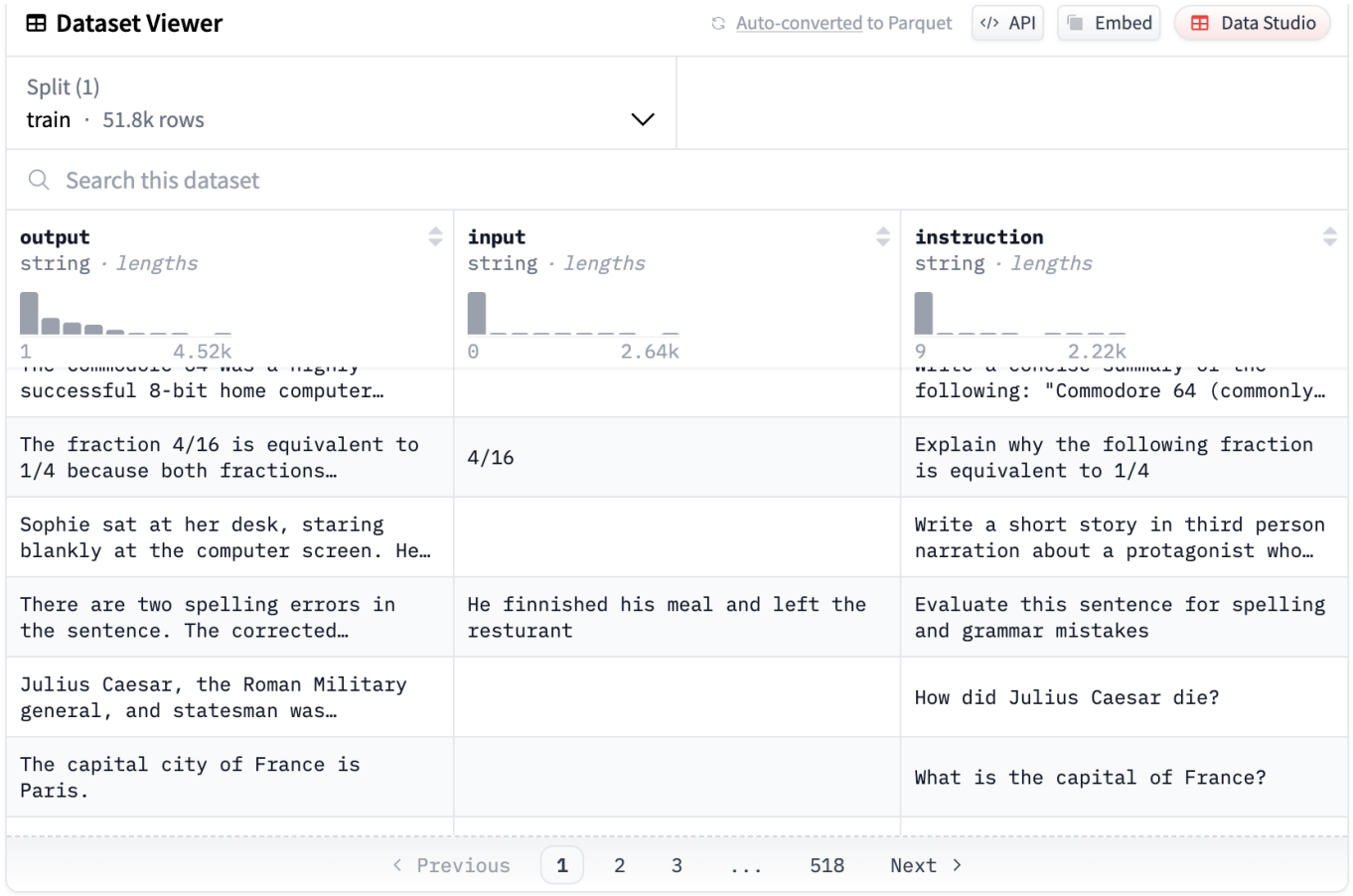
Application 2: How to Fine-Tune LLMs with Our Datasets

You can find code for fine-tuning several major open-source models in **unsloth**. Here, we only demonstrate how to fine-tune **DeepSeek-R1-Distill-Llama-8B**.

We use the **Alpaca**-Cleaned dataset for fine-tuning. (You can also create your own dataset to train the model on the specific capabilities you expect.)



Unsloth supports	Free Notebooks	Performance	Memory use
Llama 3.2 (3B)	Start for free	2x faster	70% less
GRPO (R1 reasoning)	Start for free	2x faster	80% less
Phi-4 (14B)	Start for free	2x faster	70% less
Llama 3.2 Vision (11B)	Start for free	2x faster	50% less
Llama 3.1 (8B)	Start for free	2x faster	70% less
Gemma 2 (9B)	Start for free	2x faster	70% less
Qwen 2.5 (7B)	Start for free	2x faster	70% less
Mistral v0.3 (7B)	Start for free	2.2x faster	75% less
Ollama	Start for free	1.9x faster	60% less
DPO Zephyr	Start for free	1.9x faster	50% less



https://github.com/unslothai/unsloth

Fine-tuning (LoRA)

Step1: Installation & Unsloth

```
%%capture
import os
if "COLAB_" not in "".join(os.environ.keys()):
!pip install unsloth
else:
# Do this only in Colab notebooks! Otherwise use pip
install unsloth
!pip install --no-deps bitsandbytes accelerate
xformers==0.0.29 peft trl triton
!pip install --no-deps cut_cross_entropy unsloth_zoo
!pip install sentencepiece protobuf datasets
huggingface_hub hf_transfer
!pip install --no-deps unsloth

from unsloth import FastLanguageModel
import torch
max_seq_length = 2048 # Choose any! We auto support RoPE
Scaling internally!
dtype = None # None for auto detection. Float16 for Tesla
T4, V100, Bfloat16 for Ampere+
load_in_4bit = True # Use 4bit quantization to reduce
memory usage. Can be False.


model, tokenizer = FastLanguageModel.from_pretrained(
model_name = "unsloth/DeepSeek-R1-Distill-Llama-8B",
max_seq_length = max_seq_length,
dtype = dtype,
load_in_4bit = load_in_4bit,
)
```

Step2: Data Prep

We now use the Alpaca dataset from yahma, which is a filtered version of 52K of the original Alpaca dataset. You can replace this code section with your own data prep.

```
alpaca_prompt = """Below is an instruction that describes a task,
paired with an input that provides further context. Write a
response that appropriately completes the request.
### Instruction:
{}
### Input:
{}
### Response:
{}"""

EOS_TOKEN = tokenizer.eos_token # Must add EOS_TOKEN
def formatting_prompts_func(examples):
instructions = examples["instruction"]
inputs = examples["input"]
outputs = examples["output"]
texts = []
for instruction, input, output in zip(instructions, inputs,
outputs):
# Must add EOS_TOKEN, otherwise your generation will go on
forever!
text = alpaca_prompt.format(instruction, input, output) +
EOS_TOKEN
texts.append(text)
return { "text" : texts, }
pass

from datasets import load_dataset
dataset = load_dataset("yahma/alpaca-cleaned", split = "train")
dataset = dataset.map(formatting_prompts_func, batched = True,)
```

Step3: We now add LoRA adapters so we only need to update 1 to 10% of all parameters!

```
model = FastLanguageModel.get_peft_model(
model,
r = 16, # Choose any number > 0 ! Suggested 8, 16, 32, 64, 128
target_modules = ["q_proj", "k_proj", "v_proj", "o_proj",
"gate_proj", "up_proj", "down_proj",],
lora_alpha = 16,
lora_dropout = 0, # Supports any, but = 0 is optimized
bias = "none", # Supports any, but = "none" is optimized
# [NEW] "unsloth" uses 30% less VRAM, fits 2x larger batch sizes!
use_gradient_checkpointing = "unsloth", # True or "unsloth" for
very long context
random_state = 3407,
use_rslora = False, # We support rank stabilized LoRA
loftq_config = None, # And LoftQ
)
```

```
Unsloth: Will smartly offload gradients to save VRAM!
```

[16/60 01:46 < 05:35, 0.13 it/s, Epoch 0.00/1]

Step	Training Loss
1	1.941900
2	2.375000
3	2.086700
4	2.299100
5	2.106300
6	1.853900
7	1.376600

Step4: Train the model

```
from trl import SFTTrainer
from transformers import TrainingArguments
from unsloth import is_bfloat16_supported

trainer = SFTTrainer(
model = model,
tokenizer = tokenizer,
train_dataset = dataset,
dataset_text_field = "text",
max_seq_length = max_seq_length,
dataset_num_proc = 2,
packing = False, # Can make training 5x faster for
short sequences.
args = TrainingArguments(
per_device_train_batch_size = 2,
gradient_accumulation_steps = 4,
warmup_steps = 5,
# num_train_epochs = 1, # Set this for 1 full training
run.
max_steps = 60,
learning_rate = 2e-4,
fp16 = not is_bfloat16_supported(),
bf16 = is_bfloat16_supported(),
logging_steps = 1,
optim = "adamw_8bit",
weight_decay = 0.01,
lr_scheduler_type = "linear",
seed = 3407,
output_dir = "outputs",
report_to = "none", # Use this for WandB etc
),
)

trainer_stats = trainer.train()
```

Step5: Save

```
model.save_pretrained("lora_model") # Local saving
tokenizer.save_pretrained("lora_model")
```

Step6: Load the LoRA adapters we just saved for inference

```
if False:
from unsloth import FastLanguageModel
model, tokenizer = FastLanguageModel.from_pretrained(
model_name = "lora_model", # YOUR MODEL YOU USED FOR TRAINING
max_seq_length = max_seq_length,
dtype = dtype,
load_in_4bit = load_in_4bit,
)
FastLanguageModel.for_inference(model) # Enable native 2x faster
inference

inputs = tokenizer(
[
alpaca_prompt.format(
"What is a famous tall tower in Paris?", # instruction
"", # input
"", # output - leave this blank for generation!
)
], return_tensors = "pt").to("cuda")

from transformers import TextStreamer
text_streamer = TextStreamer(tokenizer)
_ = model.generate(**inputs, streamer = text_streamer,
max_new_tokens = 128)
```

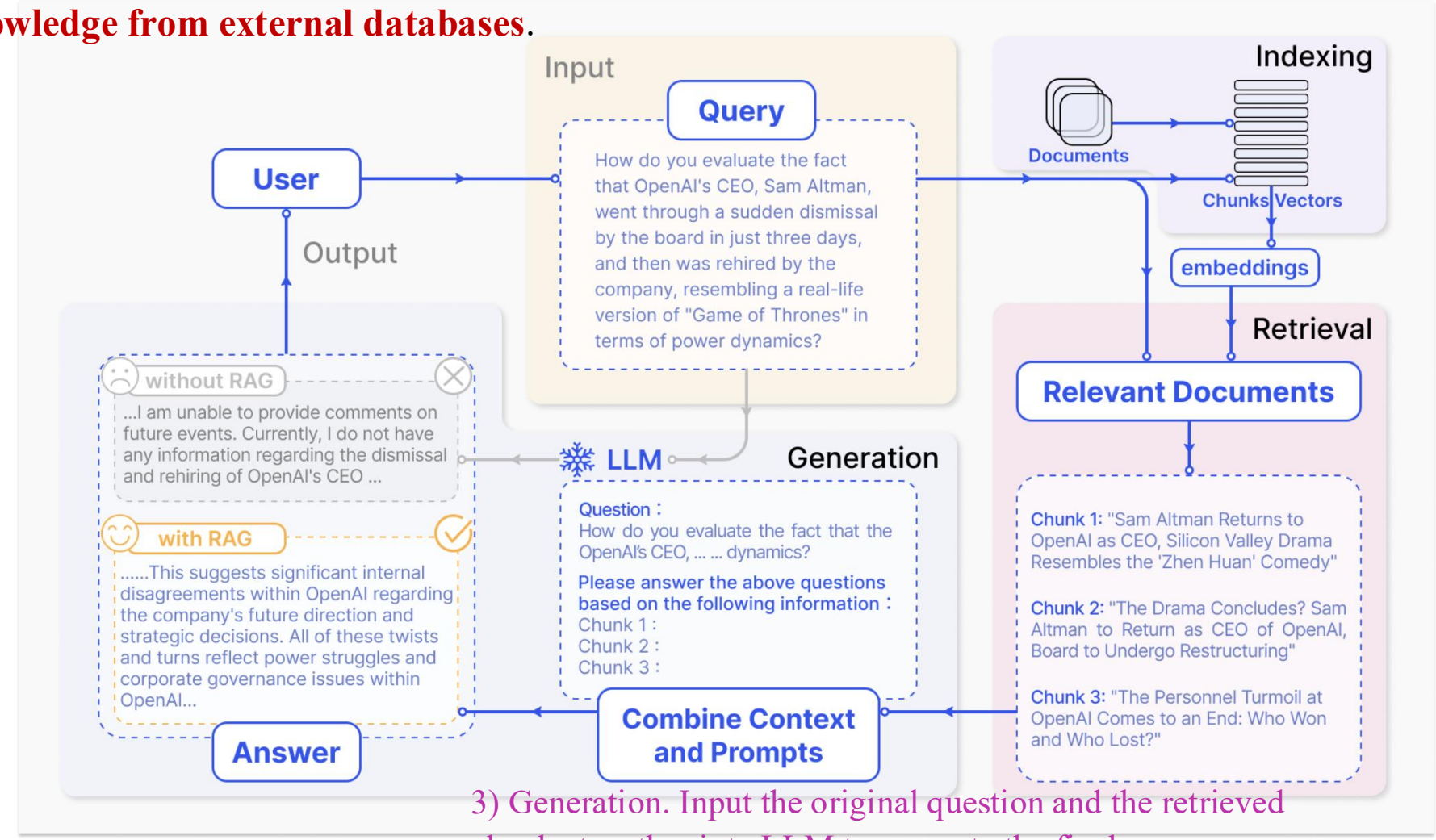
Inference Results:

```
<｜begin▁of▁sentence｜>Below is an instruction
that describes a task, paired with an input that
provides further context. Write a response that
appropriately completes the request.
### Instruction: What is a famous tall tower in
Paris? ###

Input: ###

Response: One of the most famous tall towers in
Paris is the Eiffel Tower, which is one of the
most iconic landmarks of the city. The Eiffel
Tower stands at a height of 324 meters (1,056 feet)
and was completed in 1889. It is located on the
banks of the Seine River and is one of the most
visited tourist attractions in the world.<｜
end▁of▁sentence｜>
```

Application 3: Retrieval-Augmented Generation (RAG)

Why RAG?

LLMs showcase impressive capabilities but encounter challenges **like hallucination, outdated knowledge, and non-transparent, untraceable reasoning processes**. RAG has emerged as a promising solution by **incorporating knowledge from external databases**.



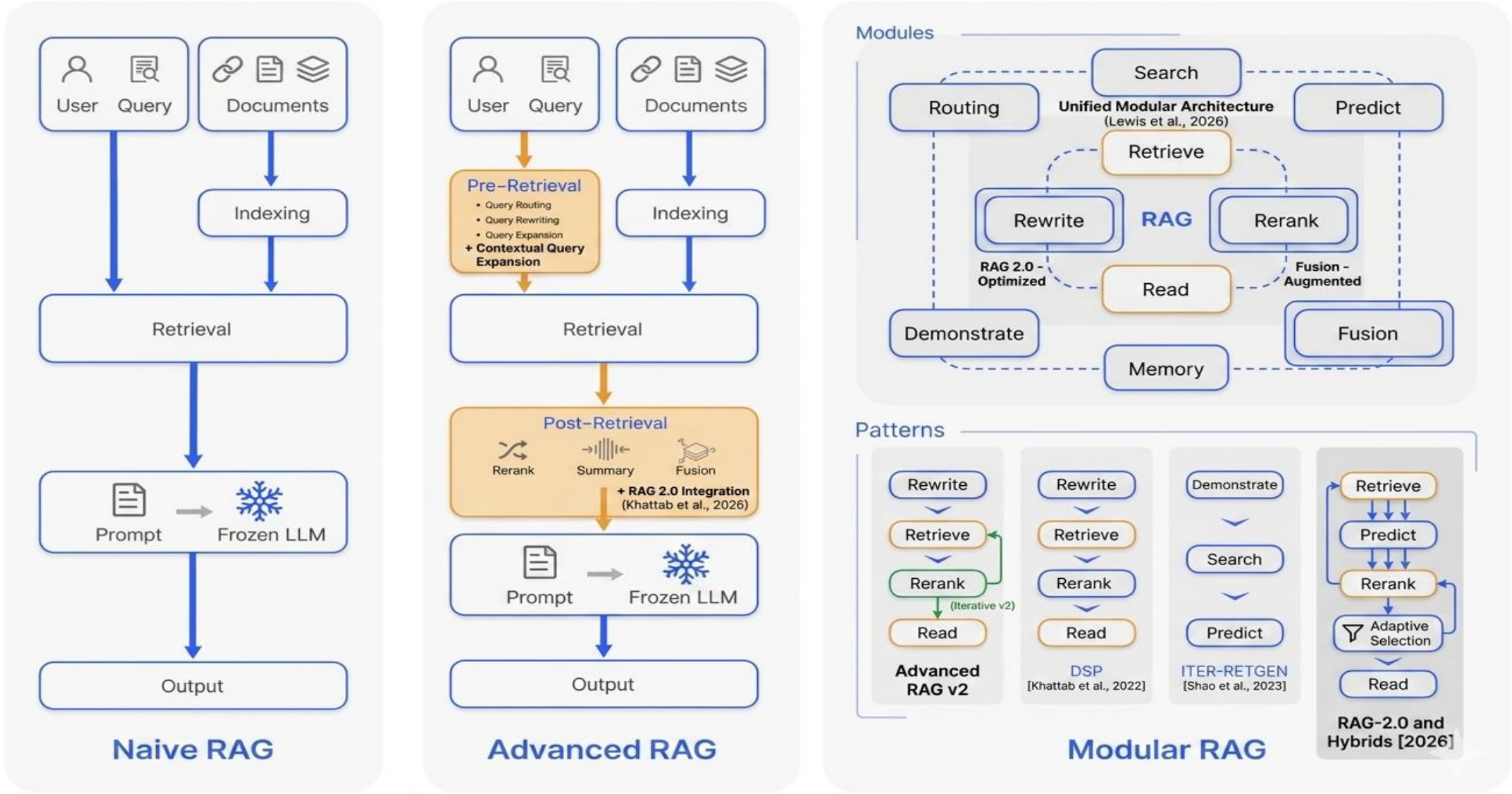
1)Indexing. Documents are split into chunks, encoded into vectors, and stored in a vector database.

2) Retrieval. Retrieve the Top k chunks most relevant to the question based on semantic similarity.

3) Generation. Input the original question and the retrieved chunks together into LLM to generate the final answer.

Gao, Y., Xiong, Y., Gao, X., Jia, K., Pan, J., Bi, Y., Dai, Y., Sun, J., Wang, M., & Wang, H. (2023). Retrieval-Augmented Generation for Large Language Models: A Survey. *ArXiv*. https://arxiv.org/abs/2312.10997

Comparison between the three paradigms of RAG



Gao, Y., Xiong, Y., Gao, X., Jia, K., Pan, J., Bi, Y., Dai, Y., Sun, J., Wang, M., & Wang, H. (2023). Retrieval-Augmented Generation for Large Language Models: A Survey. *ArXiv*. https://arxiv.org/abs/2312.10997

Building a Private Knowledge Base Based on Cherry Studio + Ollama + DeepSeek

- **Step 1:** Deploy the required model locally using Ollama (refer to Task 1).
- **Step 2:** Deploy the embedding model (**nomic-embed-text**, a high-performing open embedding model with a large token context window.)

This model helps retrieve relevant knowledge from the knowledge base.

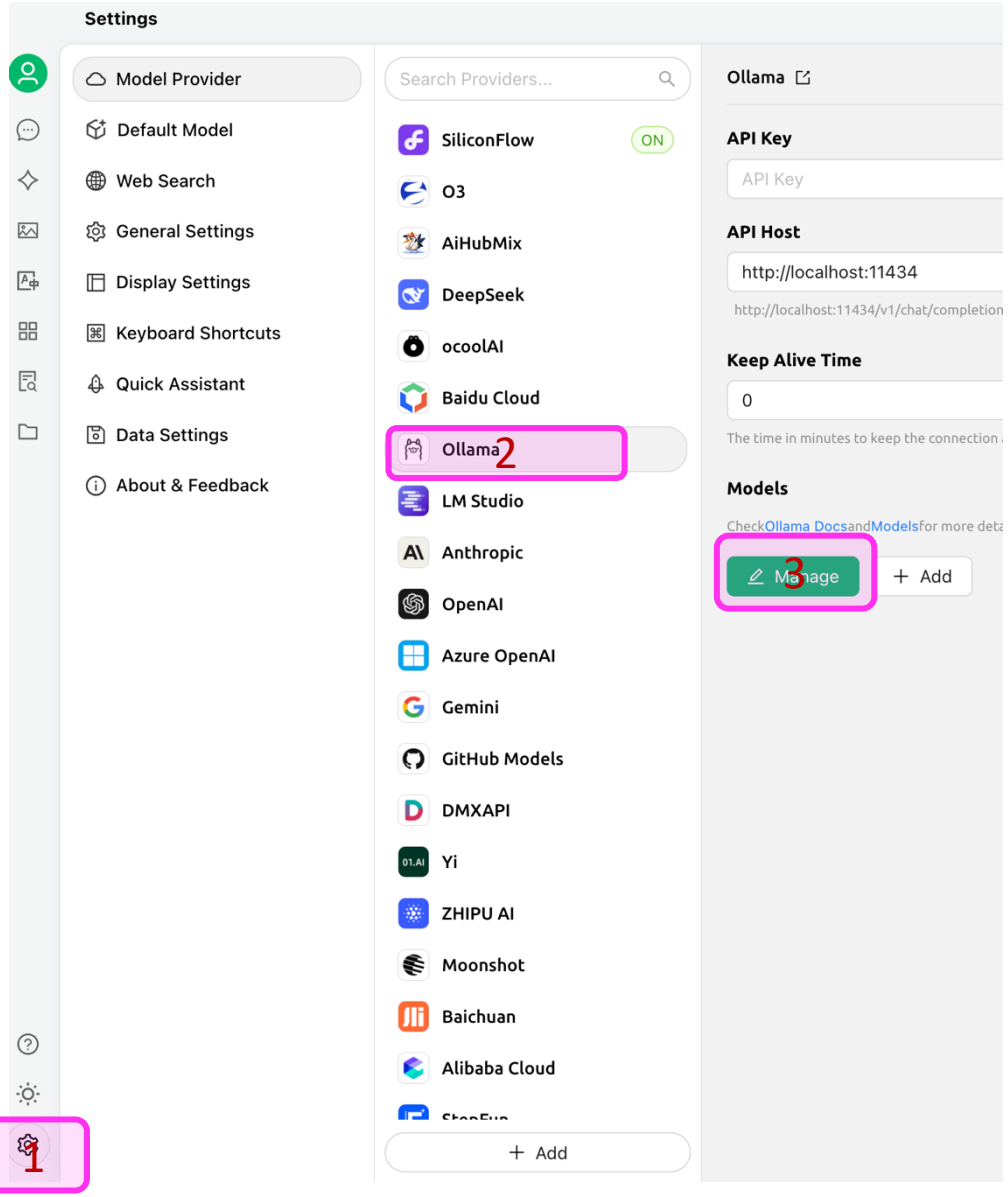
Command: *ollama pull nomic-embed-text*

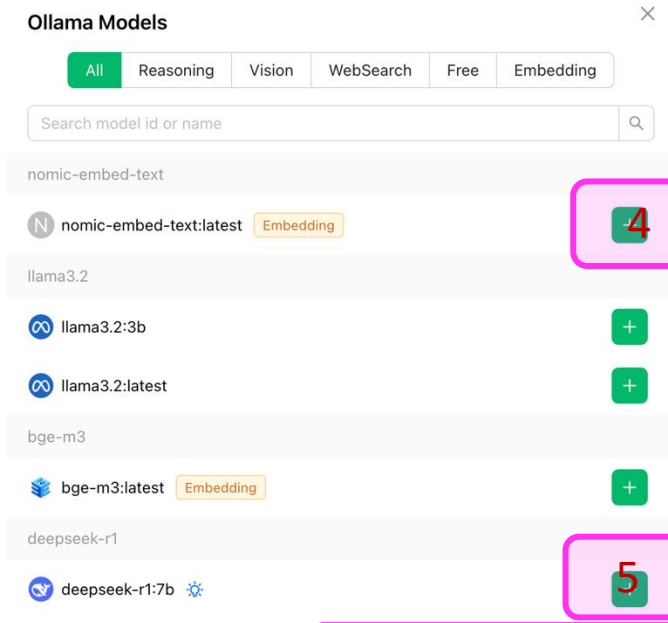
- **Step 3:** Download Cherry Studio: https://cherry-ai.com/download

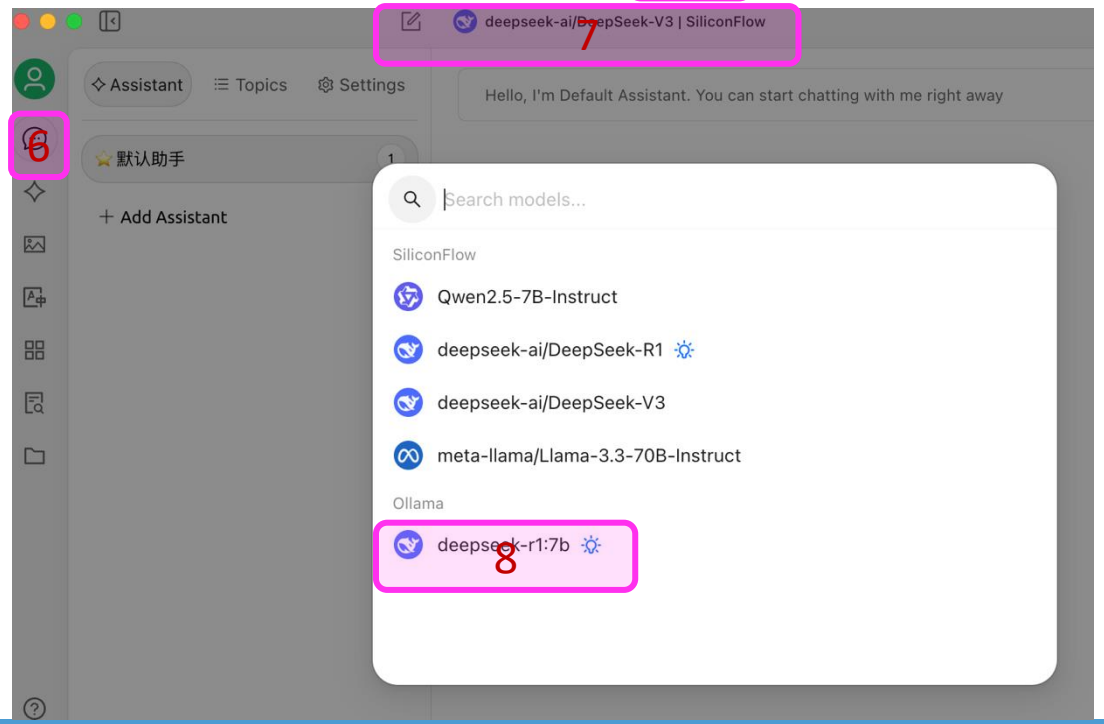
(Cherry Studio is a desktop client that supports multiple LLM providers and is available on Windows, Mac, and Linux.)

https://github.com/CherryHQ/cherry-studio?tab=readme-ov-file

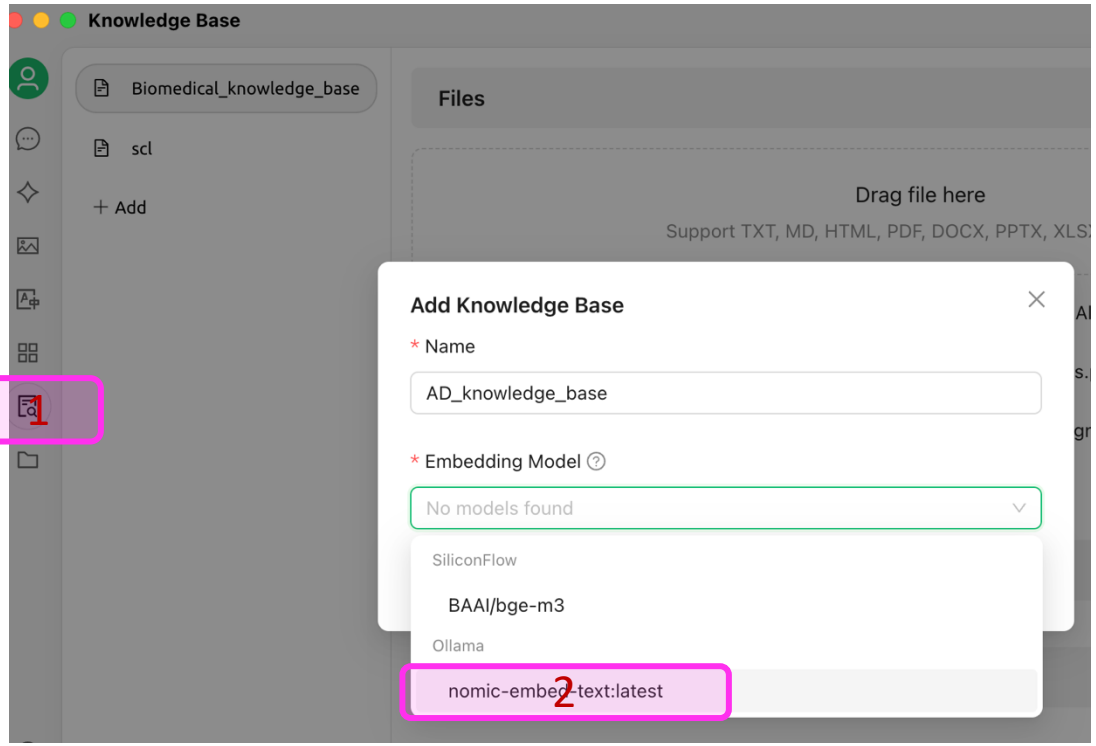
Step 4: Set Up Cherry Studio



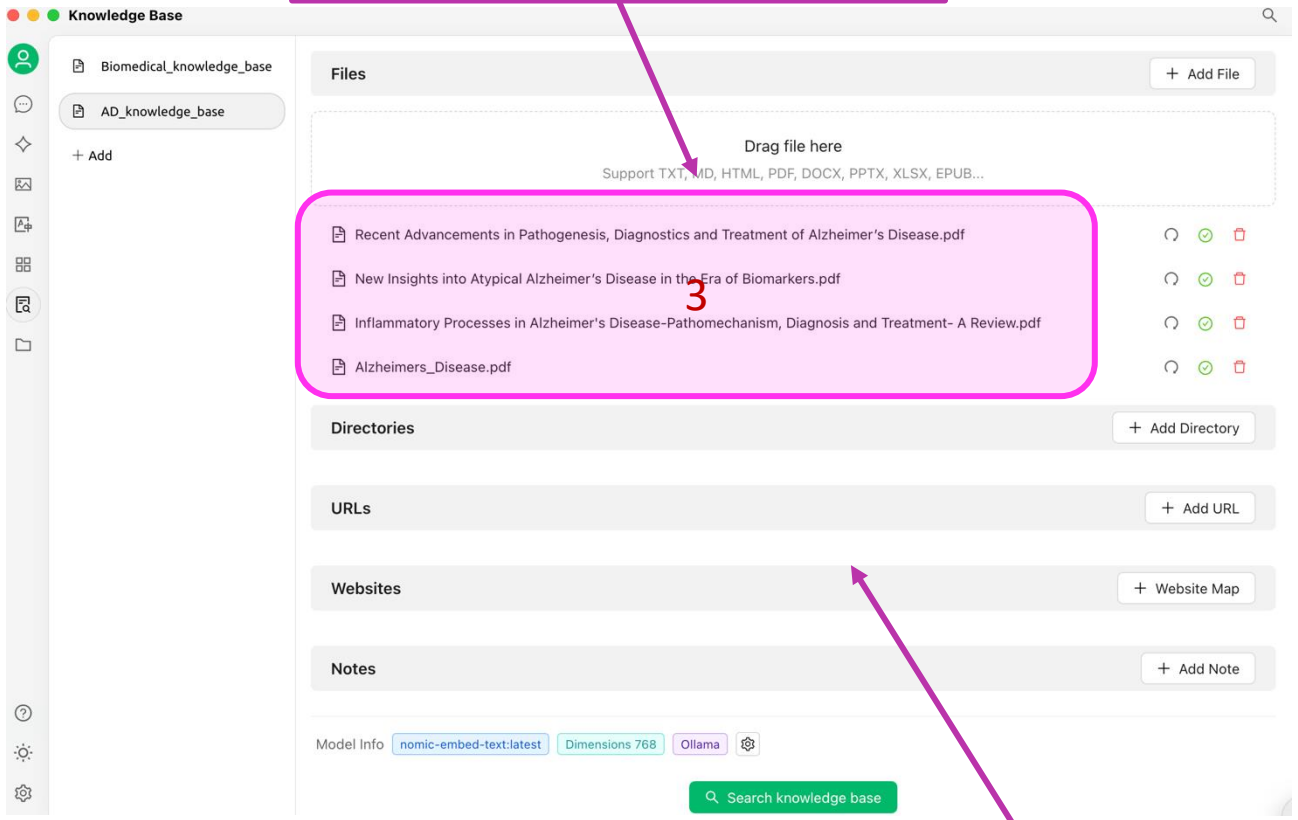




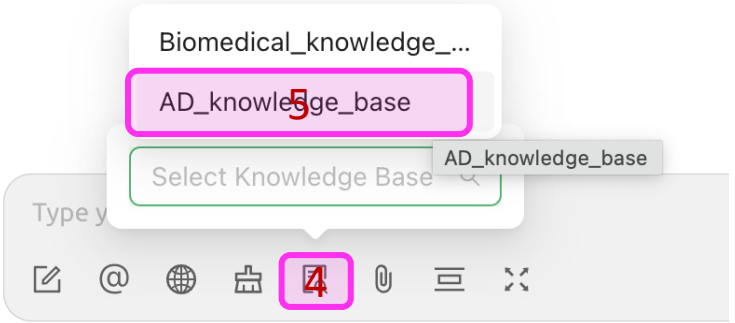
Step5: Add Knowledge Base



We have added four papers on AD



You can add any documents you are interested in (including PPT, PDF, XLSX, etc.), as well as web pages or directories.



After this, the model can **enhance the quality of its responses by querying your knowledge base**.

Step 6: Ask Question 'Why inflammatory cells can cause neuronal degeneration in Alzheimer's Disease?'

DeepSeek-R1-7B's Response:

No knowledge base added

deepseek-r1:7b | Ollama
03/15 00:12

Deeply thought (19.6 seconds)

Inflammatory cells play a significant role in the pathogenesis of Alzheimer's disease (AD) by contributing to neuronal degeneration through several interconnected pathways:

1. **Promotion of Amyloid-β Plaque Formation and Tau Tangles**: Inflammatory cells such as macrophages are attracted to neurons in regions with elevated amyloid-β levels, initiating or exacerbating the formation of amyloid-β plaques. Over time, these plaques lead to tau protein tangles, creating a toxic environment conducive to neuronal damage.
2. **Chronic Neuroinflammation**: Chronic inflammatory responses promote the aggregation and stabilization of amyloid-β plaques, which can kill neurons. This neuroinflammation often leads to a vicious cycle where cytokines, such as IL-1β, attracts more immune cells, further impairing neuronal health.
3. **Neuroprotective Mechanisms Compromised**: Chronic inflammation impairs the brain's natural ability to repair or protect neurons. Inflammation can interfere with processes necessary for neuronal survival and maintenance, making the injury caused by amyloid-beta plaques and tau tangles more severe.
4. **Direct Neuronal Damage by Inflammatory Cytokines**: Certain cytokines released by inflammatory cells activate proteins that bind to neuronal surfaces, directly triggering apoptosis or causing oxidative stress, leading to cell death even in regions unaffected by amyloid-beta or tau changes.
5. **Oxidative Stress**: Chronic inflammation induces elevated levels of reactive oxygen species (ROS), which can cause oxidative damage to neurons. This oxidative stress adds to the cellular stress experienced by neurons near or carrying amyloid-beta plaques, promoting degeneration.

In summary, inflammatory cells exacerbate neuronal degeneration in AD by promoting plaque formation, destabilizing tau tangles, triggering chronic inflammation, impairing protective mechanisms, and causing direct neuronal damage through cytokines and oxidative [illegible]. This multifaceted contribution underscores the critical role of inflammation in the disease's progression.

Before

After adding the knowledge base, DeepSeek-R1-7B's responses **better align** with the **content of the literature and more reliable**!

deepseek-r1:7b | Ollama
03/15 00:11

Deeply thought (14.2 seconds)

In Alzheimer's Disease (AD), neuronal degeneration is closely linked to the activation of inflammatory cells, particularly microglia and astrogli astrocytes. Here's a step-by-step explanation of how this process occurs:

1. **Pathological deposited beta-amyloid plaques**: The progression of AD involves the accumulation of amyloid-beta (Aβ) plaques in the brain. These plaques form around neurons, creating an inflammatory environment.
2. **Activation of microglia**: Normally functioning microglia are part of the immune system protecting the CNS against pathogens or harmful materials. In AD, these cells become activated upon detecting Aβ deposits. Their activation is a critical trigger for further pathological changes.
3. **Role of complement system**: Activated microglia can recruit the complement system through direct cell-cell interaction or cell surface receptors. The Aβ-amyloid binds to the C1 component of this system, leading to its activation, which contributes to the inflammatory state in the brain tissue.
4. **Phagocytic response**: Microglia engage in phagocytosis to clear Aβ deposits from neurons. While this process can sometimes remove pathology, excessive or improper clearing can lead to excitotoxicity—a theory supported by clinical and experimental evidence showing both benefits and damages associated with this pathway.
5. **Damaging effects on neuronal health**: The chronic inflammatory state promotes neuroinflammation, affecting vulnerable neurons and their supporting glial cells. Research indicates increased neuronal activity in areas high in Aβ deposits (Soler et al., 2021), suggesting impaired cognitive function due to these changes.
6. **Secondary neuronal damage factors**: Besides amyloid-beta, activated microglia and other inflammatory cells release pro-inflammatory cytokines that directly or indirectly affect neurons and glial cells involved in information processing.

In conclusion, the interaction between Aβ deposits and activated immune cells leads to chronic neuroinflammation, contributing significantly to neuronal degeneration in AD. This process is not only driven by Aβ but also involves other elements such as complement system activation, phagocytosis efficiency, and cytokine overproduction.

After

Application 4: AI Agents

Why AI Agents?

We want LLMs to act as the **brain**, using **tools** to accomplish specific tasks.

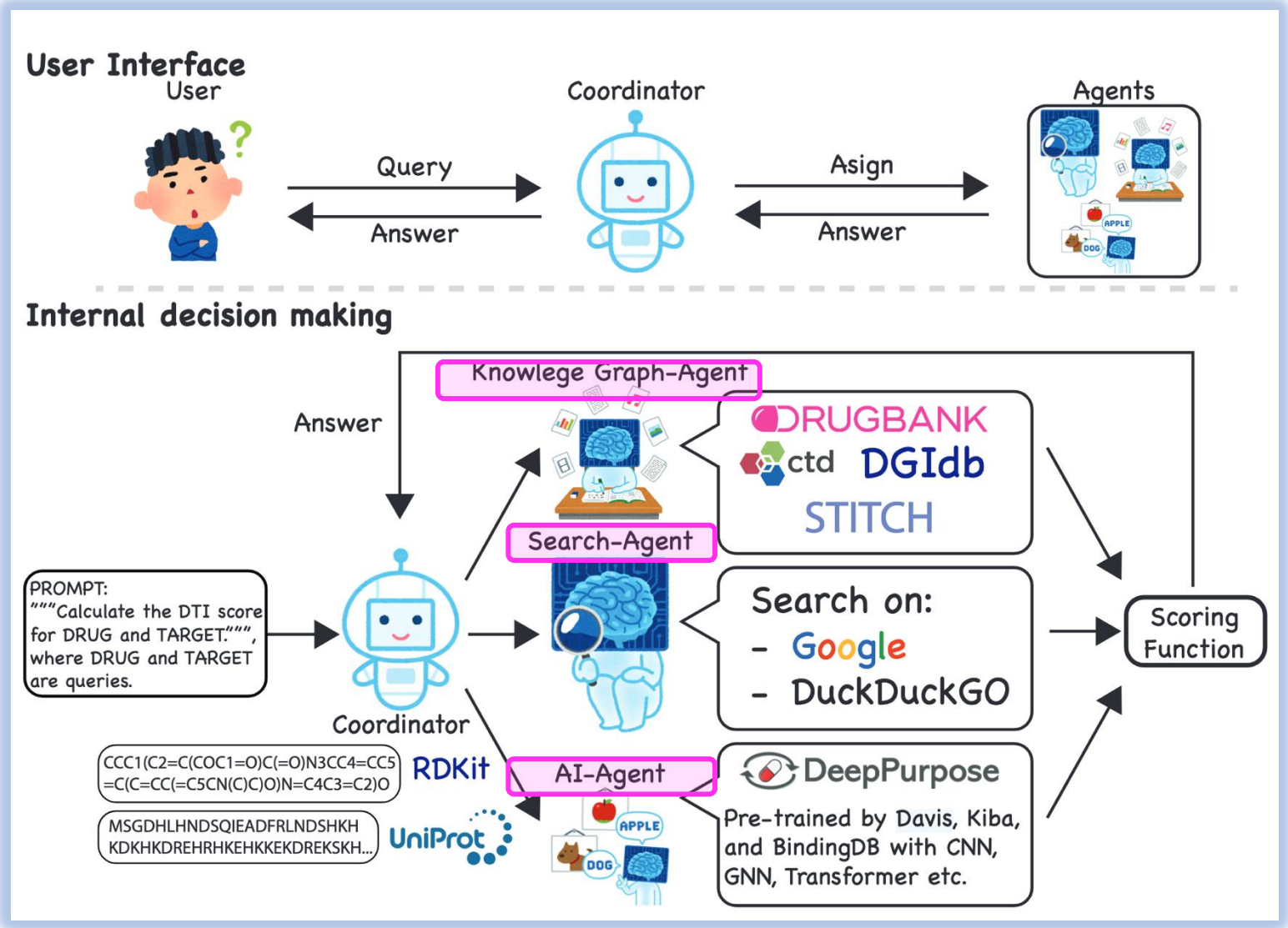
The relationship between LLMs and AI Agents

LLMs serve as the **cognitive core**, providing reasoning, language understanding, and generation.

AI agents extend this by adding **autonomy, decision-making, and multi-modal interactions**.

DrugAgent framework architecture for advanced DTI(drug-target interaction) analysis.
It features a central **Coordinator** managing specialized agents

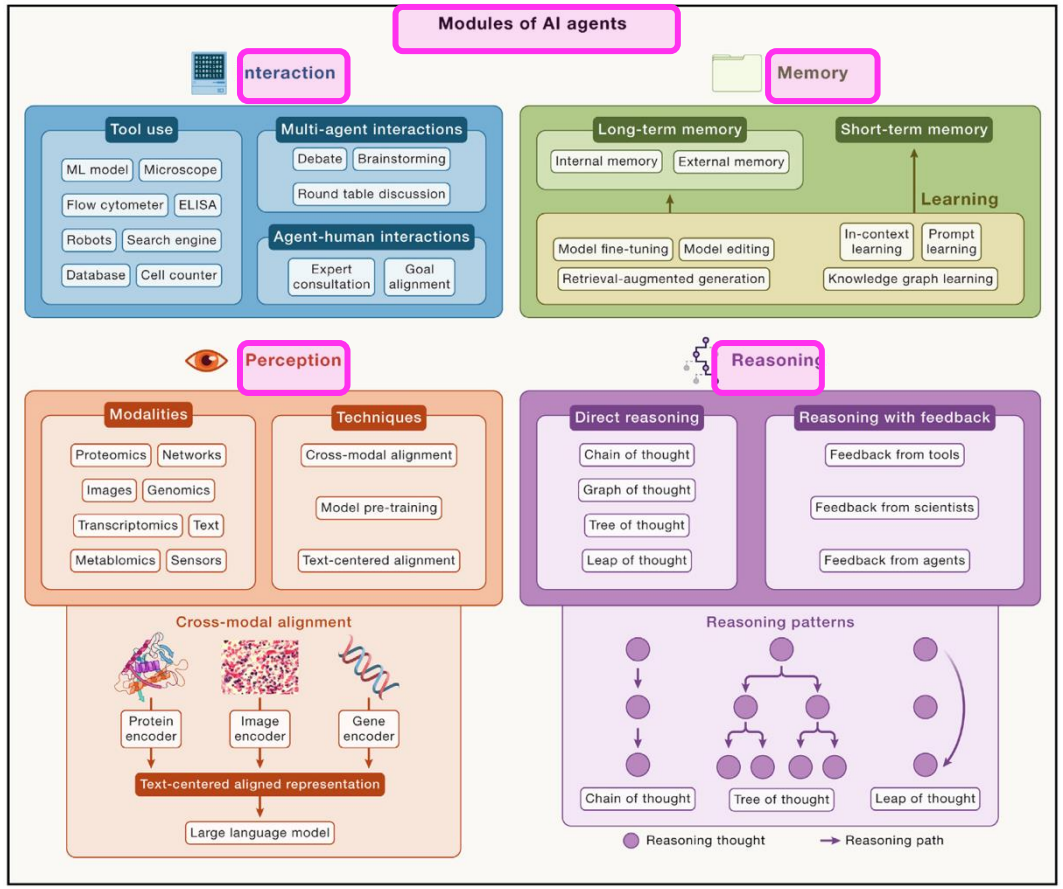
- Knowledge Graph Agent: accessing biomedical databases (DrugBank, CTD, DGIdb, STITCH)
- Search Agent: utilizing web search engines
- AI Agent: employing deep learning models

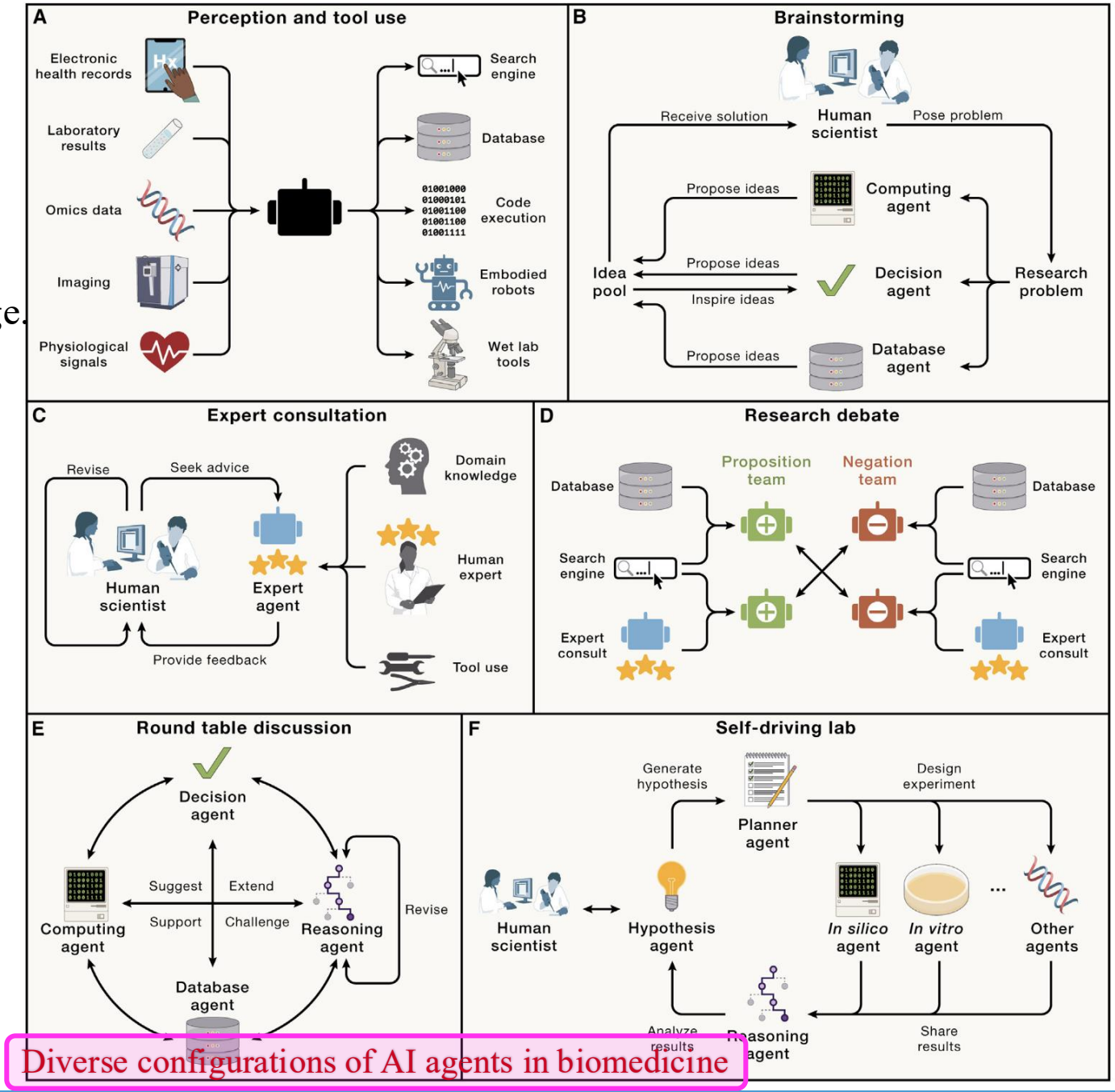


Inoue, Y., Song, T., Wang, X., Luna, A., & Fu, T. (2024). DrugAgent: Multi-Agent Large Language Model-Based Reasoning for Drug-Target Interaction Prediction. *ArXiv*. https://arxiv.org/abs/2408.13378

Empowering Biomedical Discovery with AI Agents

- AI agents are poised to be proficient in various tasks, planning discovery workflows and performing self-assessment to identify and mitigate gaps in their knowledge.





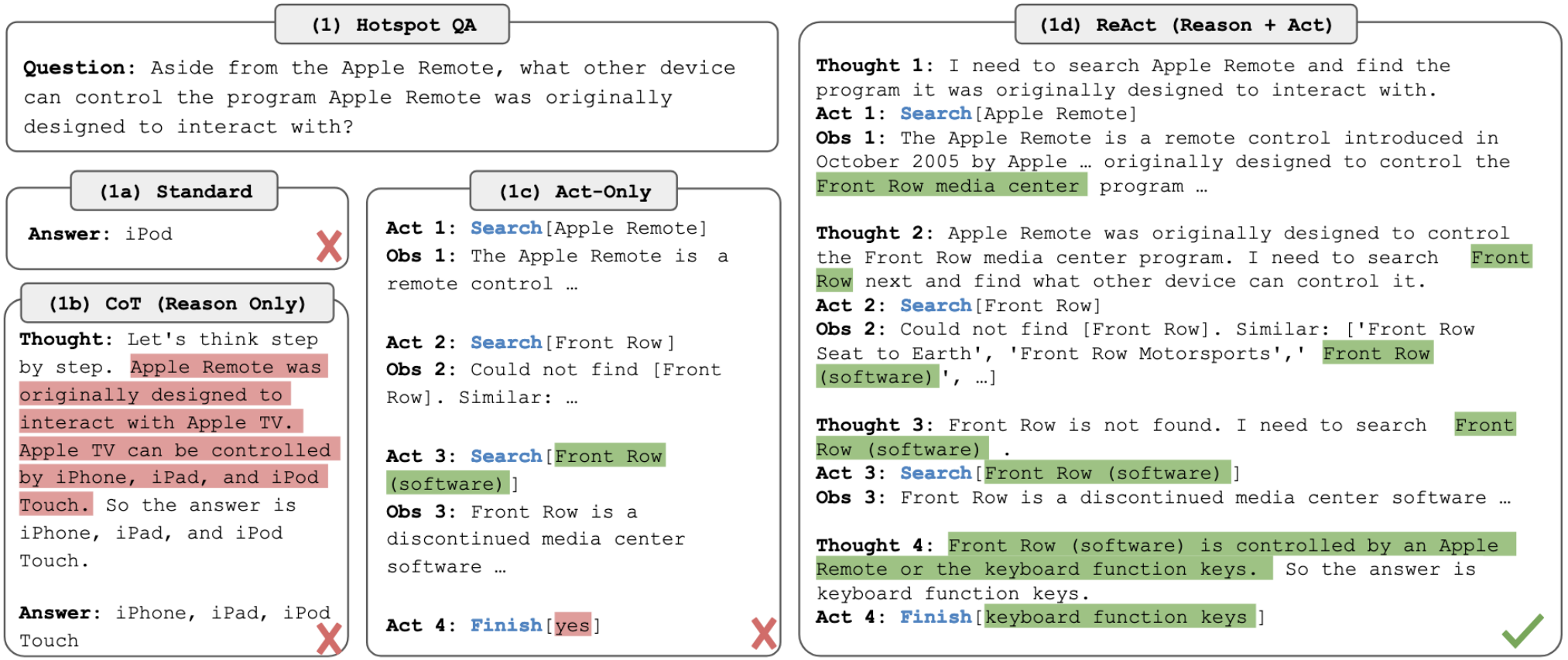
Diverse configurations of AI agents in biomedicine

Gao S, Fang A, Huang Y, Giunchiglia V, Noori A, Schwarz JR, Ektefaie Y, Kondic J, Zitnik M. Empowering biomedical discovery with AI agents. Cell. 2024 Oct 31;187(22):6125-6151. doi: 10.1016/j.cell.2024.09.022.

ReAct: Empowering LLMs to Plan and Act on Tasks

Steps:

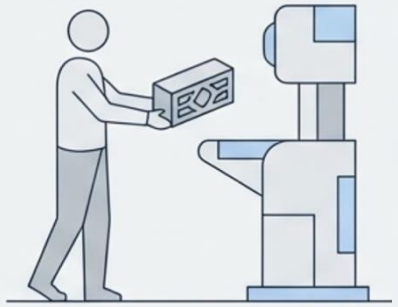
1. Generates verbal reasoning traces and actions for tasks
2. Enables dynamic reasoning and adaptive action planning
3. Interacts with external environments (e.g., Wikipedia) to integrate additional information



Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., & Cao, Y. (2022). ReAct: Synergizing Reasoning and Acting in Language Models. *ArXiv*. https://arxiv.org/abs/2210.03629

Why Agents: From Content Generation to Problem Solving

Predictive AI



Core

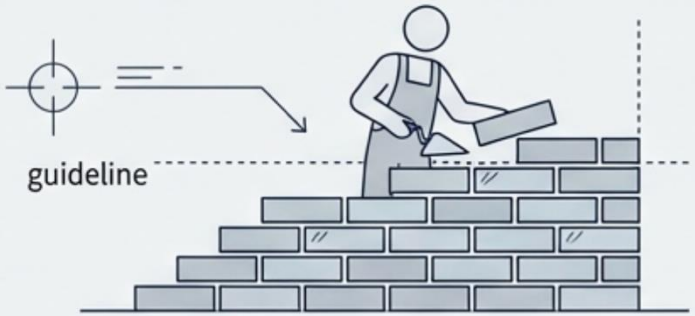
- Passive
- Task-isolated

Capabilities

- Answer questions
- Translate text
- Generate images

Limitations

- Every step requires explicit huma
  instructions



Bricklayer: The developer defines every logical step explicitly.

Autonomous Agents

Core

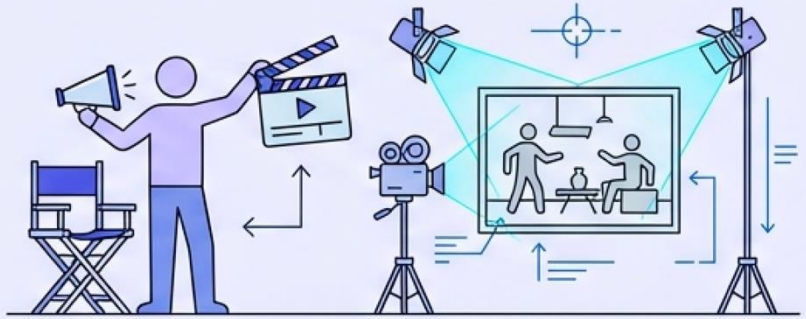
- Proactive
- Goal-driven

Capabilities

- Formulate plans
- Execute multi-step tasks
- Make autonomous decisions

Concept

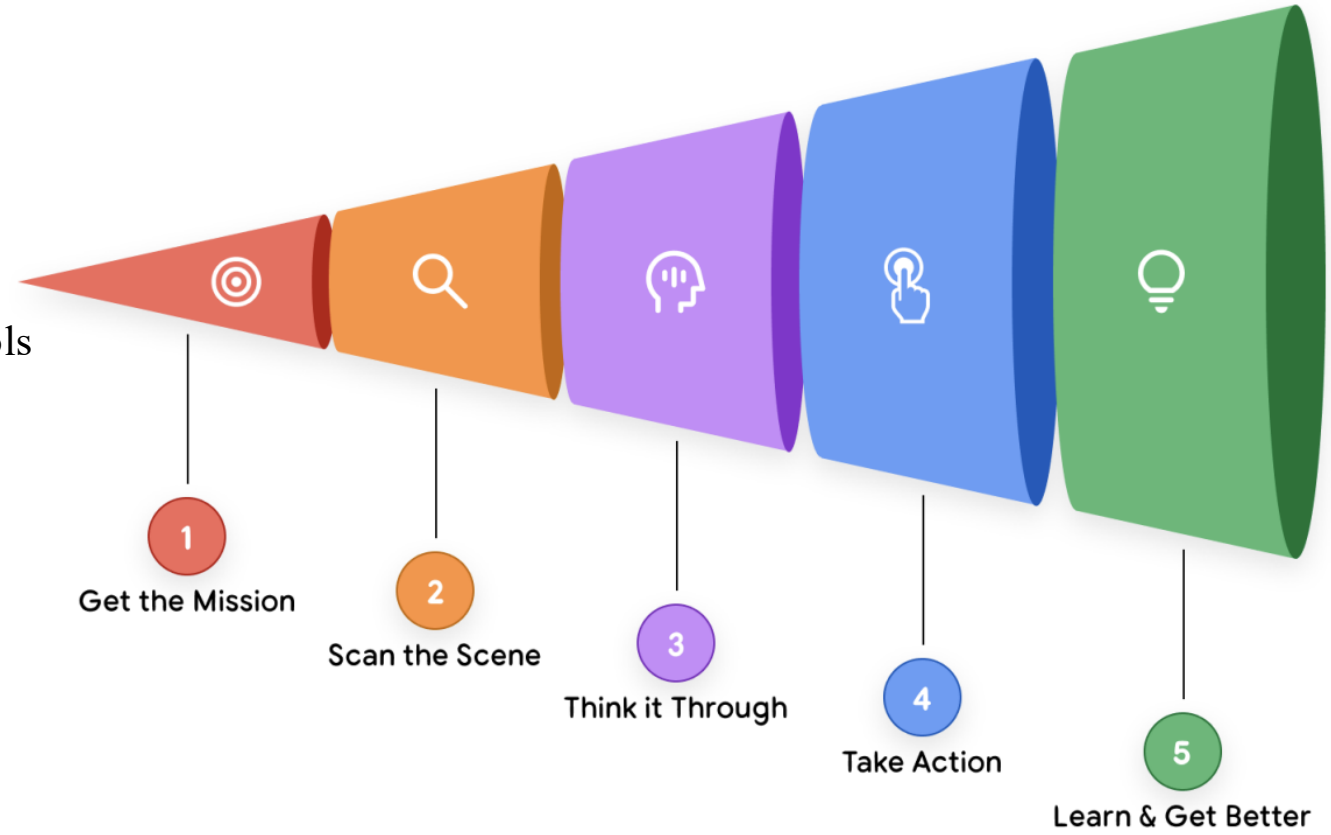
- Agents represent the natural evolution of language models
- into useful systems within software engineering



Director: The developer sets the scene, selects the cast (tools), and provides the necessary context.

What Makes an Agent: The 5-Step Loop

1. Get the Mission —— *(Receive the task)*
A clearly defined high-level goal
(e.g., “Help me track my package”)
2. Scan the Scene —— *(Scan the environment)*
Perceive the environment, access memory and available tools
3. Think It Through —— *(Reasoning)*
Core reasoning loop
4. Take Action —— *(Execute actions)*
Invoke APIs, execute code, or perform queries
5. Observe & Iterate —— *(Observe and iterate)*
Observe results, store them in memory, and repeat the loop

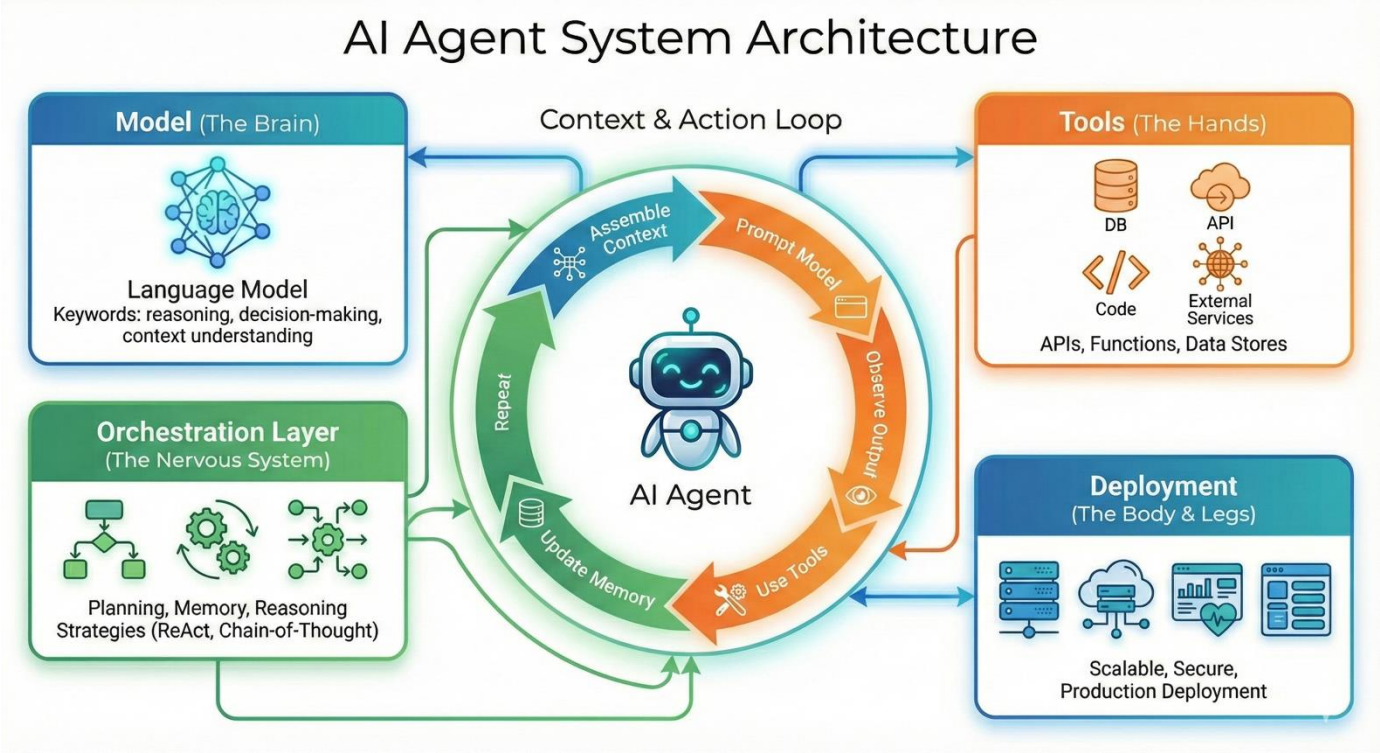


Key Insight
At its core, an agent is a “Curator of the Context Window.”
It continuously assembles context, prompts the model, observes outcomes, and then re-assembles the context for the next step.

Definition: What Is an AI Agent?

Essence

Assemble context → Call the model → Observe results → Re-assemble context, until the goal is achieved.



An Agent is an LM-driven system that can use tools, maintain memory, and iteratively improve its actions to achieve a goal.

What's Inside an Agent: 4 Core Components

AI Agent = Model + Tools + Orchestration Layer + Runtime / Deployment Services
→ Uses a Language Model (LM) to accomplish goals **through an iterative loop**

Model (The "Brain")

- Responsible for understanding, reasoning, and decision-making
- Capabilities depend on the model type (general-purpose, fine-tuned, multimodal)
- Core responsibility: **precisely curating and interpreting the input context**

Tools (The "Hands")

- APIs, functions, and code execution
- Access to real-time data sources such as databases and vector stores
- Transform **reasoning** into **executable actions**

Orchestration Layer (The "Nervous System")

- Manages the agent loop: **plan → act → verify**
- Maintains state and memory (short-term and long-term)
- Decides **when to think** and **when to use tools**

Runtime / Deployment (The "Body")

- Production-grade services (security, scalability, reliability)
- Monitoring, logging, and operational management
- Accessed via UI or APIs (including Agent-to-Agent, A2A)

https://drive.google.com/file/d/1C-HvqgxM7dj4G2kCQLnuMXi1fTpXRdpx/view

Model (Brain): Choose the Most Suitable, Not the Strongest

Core idea: A model = the reasoning core of an agent

Key deciding factors

- Reasoning capability (multi-step reasoning)
- Tool-calling reliability
- Cost & latency

Agent ≠ a single model

- Complex reasoning tasks → large models
- High-frequency, simple tasks → small models
- Common strategies → *Model Routing* / *Model Ensemble*

Counterintuitive but critical point:

- Agent success ≠ highest benchmark score

What an agent really cares about:

- Can it consistently generate executable plans?
- Can it invoke tools correctly and reliably?

Popular providers

Provider	Package	Downloads	Latest version	JS/TS support
OpenAI	langchain-openai	downloads/month 36M	v1.1.7	✅
Google (Vertex AI)	langchain-google-vertexai	downloads/month 29M	v3.2.1	✅
Anthropic (Claude)	langchain-anthropic	downloads/month 7M	v1.3.1	✅
AWS	langchain-aws	downloads/month 7M	v1.2.1	✅
Google (GenAI)	langchain-google-genai	downloads/month 6M	v4.2.0	✅
Ollama	langchain-ollama	downloads/month 2M	v1.0.1	✅
Groq	langchain-groq	downloads/month 1M	v1.1.1	✅
Huggingface	langchain-huggingface	downloads/month 1M	v1.2.0	✅
Databricks	databricks-langchain	downloads/month 1M	v0.13.0	✅
Chroma	langchain-chroma	downloads/month 1M	v1.1.0	✅
Postgres	langchain-postgres	downloads/month 1M	v0.0.16	✅
Pinecone	langchain-pinecone	downloads/month 791k	v0.2.13	✅
MistralAI	langchain-mistralai	downloads/month 669k	v1.1.1	✅
Cohere	langchain-cohere	downloads/month 631k	v0.5.0	✅
Fireworks	langchain-fireworks	downloads/month 595k	v1.1.0	✅
MongoDB	langchain-mongodb	downloads/month 682k	v0.11.0	✅
xAI (Grok)	langchain-xai	downloads/month 541k	v1.2.1	✅
Deepseek	langchain-deepseek	downloads/month 472k	v1.0.1	✅
Perplexity	langchain-perplexity	downloads/month 518k	v1.1.0	✅

Model Selection in Agent Systems (Illustrated with LangChain Providers)

https://drive.google.com/file/d/1C-HvqgxM7dj4G2kCQLnuMXi1fTpXRdpx/view
https://docs.langchain.com/oss/python/integrations/providers/overview

Tools (Hands): Retrieval + Action

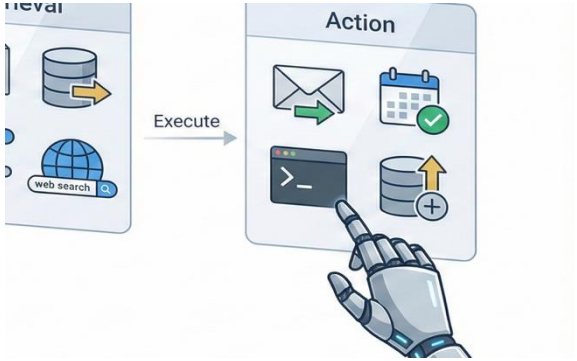
Tools enable an agent to:

- Read information
- Write / Act on the world

Two key categories of tools:

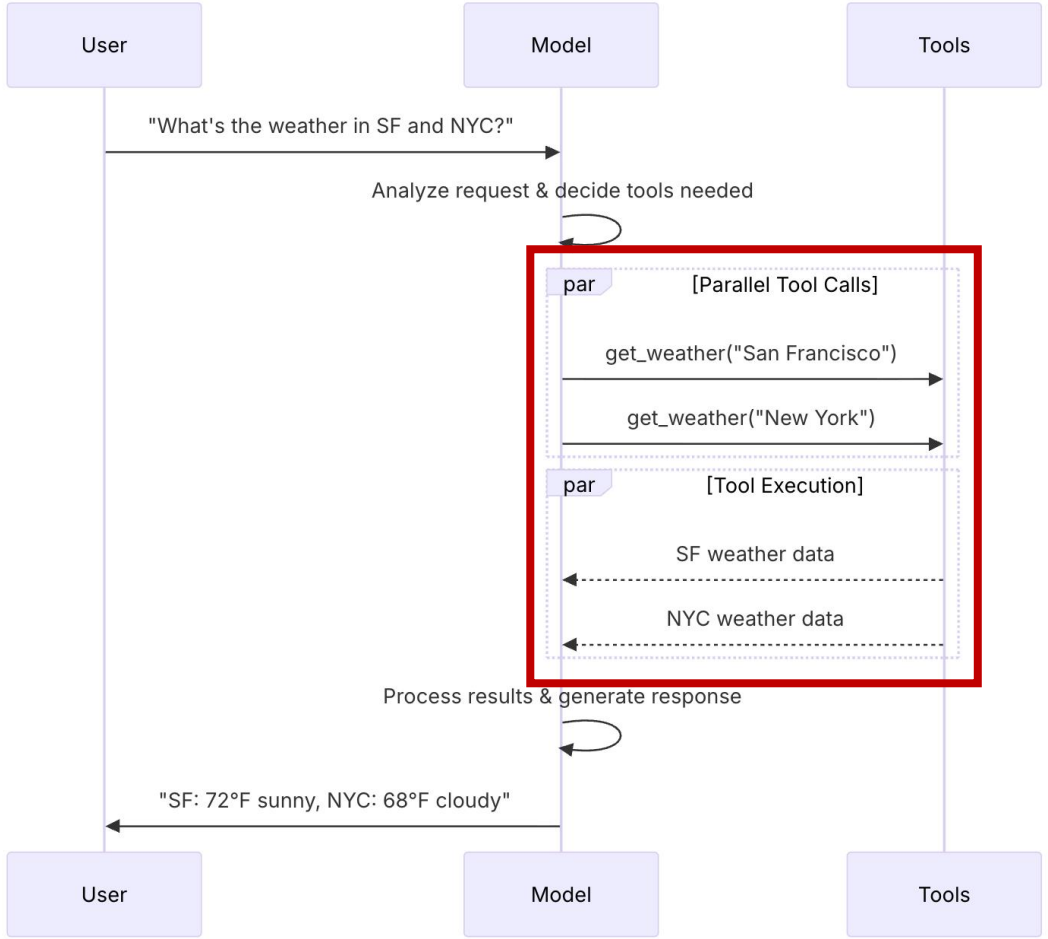
❑ **1. Retrieval / Grounding**

- RAG
- Vector Databases
- Knowledge Graphs
- NL2SQL
- Web / API Search



❑ **2. Action / Execution**

- Calling APIs (sending emails, scheduling meetings, updating databases)
- Generating and executing code (Python / SQL)
- Human-in-the-loop is also a tool



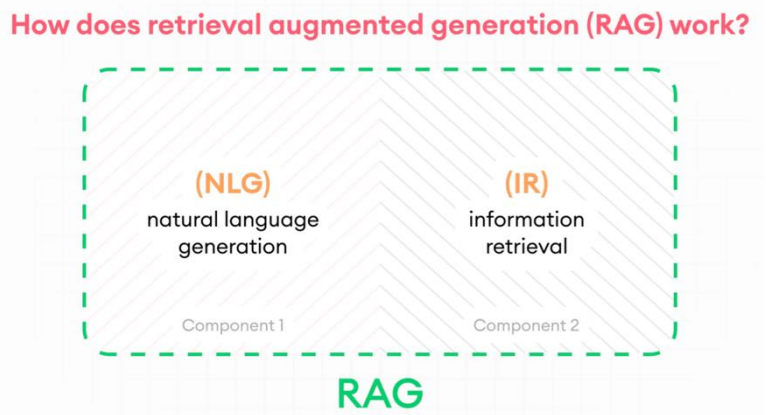
An agent without tools ≈ *"Smart but blind"*
Tools are not plugins — they are part of an agent's capabilities.

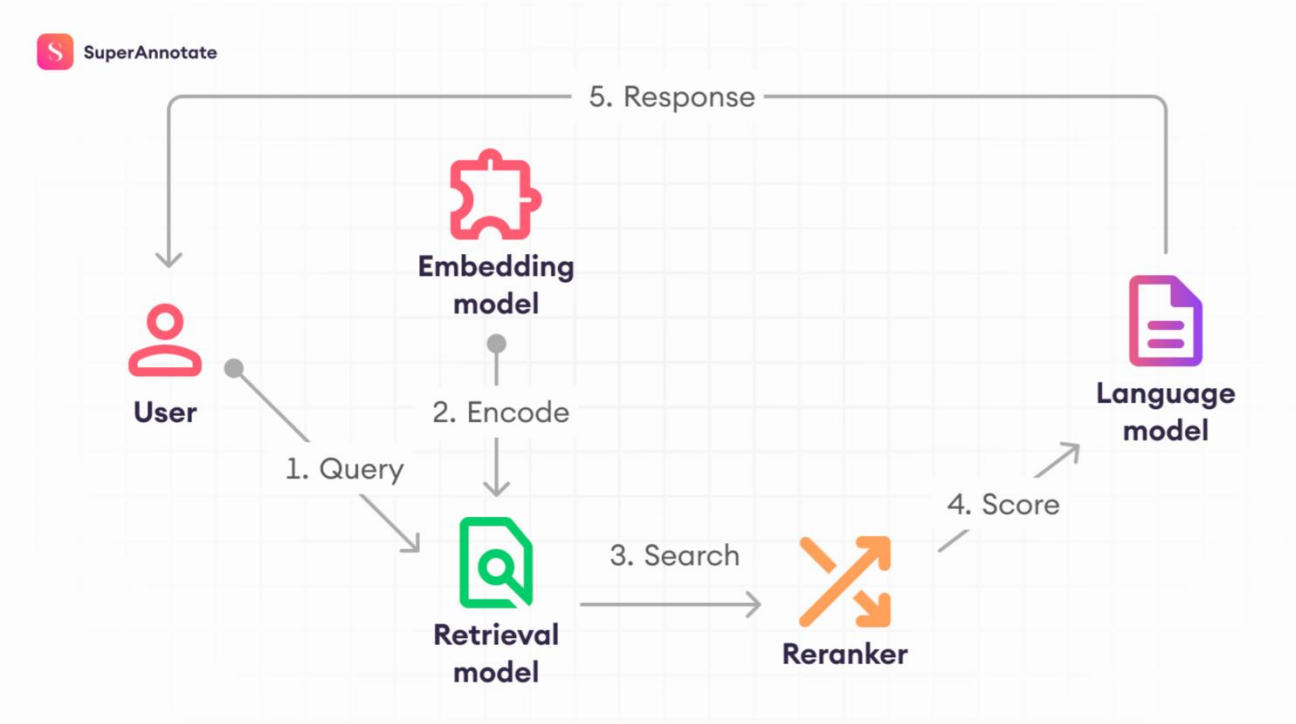
https://drive.google.com/file/d/1C-HvqgxM7dj4G2kCQLnuMXi1fTpXRdpx/view
https://docs.langchain.com/oss/python/langchain/models

Retrieval-Augmented Generation (RAG): Retrieve First, Then Generate

What is RAG?

RAG = Retrieve relevant external knowledge first, then generate answers grounded in that knowledge.





Why do we need RAG?

Key limitations of general-purpose LLMs:

- ❌ **Hallucinations**: confident but incorrect answers
- ❌ **Outdated knowledge**: fixed training cutoff
- ❌ **Lack of domain-specific context**
- ❌ **No built-in fact verification**

RAG consists of four key components:

- **Embedding Model:** Converts documents into vector representations.
- **Retriever:** Searches for the most relevant vectors given a query.
- **Reranker (optional):** Refines retrieved results by relevance.
- **Language Model:** Generates the final answer using the top documents and the query.

https://www.superannotate.com/blog/rag-explained

Orchestration (Nervous System): Control Logic of the Agent

Orchestration is responsible for:

- Controlling the **Think / Act / Observe** loop
- Managing **state** and **memory**
- Deciding **when to call models vs. tools**

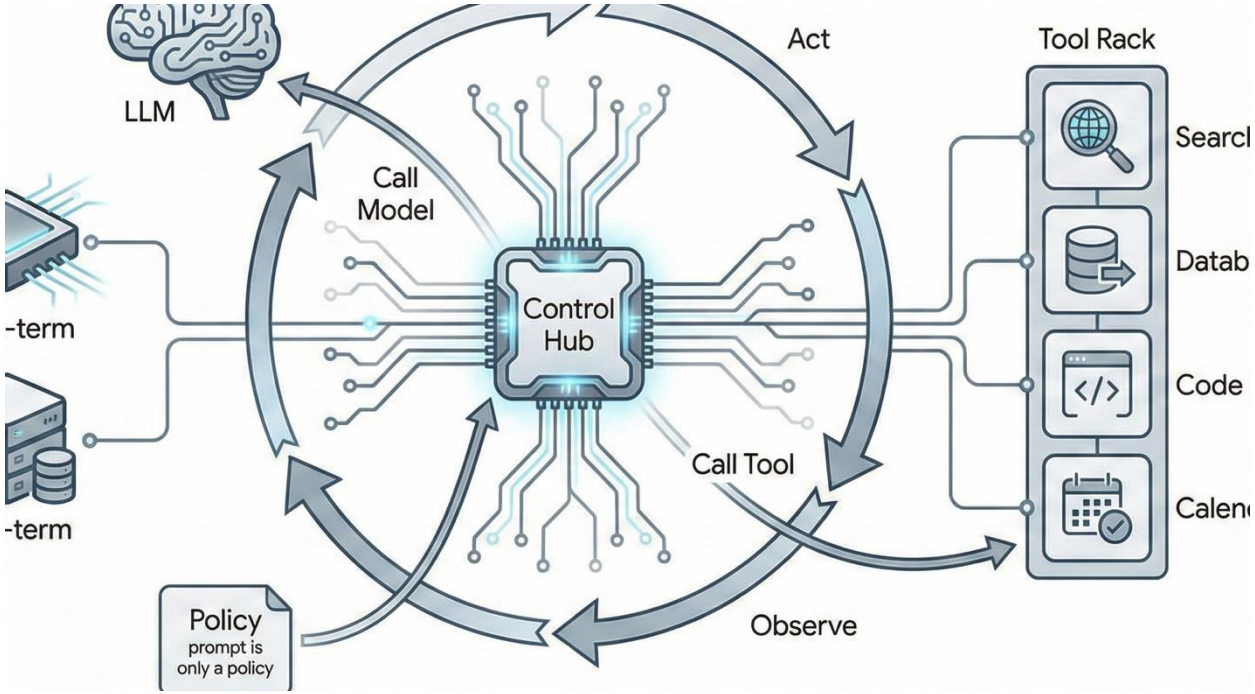
Core mechanisms

- Planning
- Context Engineering
- Memory (short-term & long-term)

A common beginner misconception:
"Agent = prompt + tools"

In reality:

- A prompt is only a policy
- Orchestration is the system logic

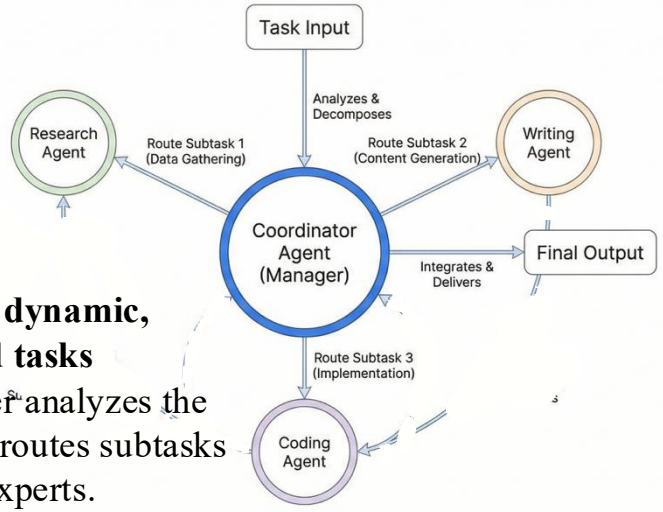


https://drive.google.com/file/d/1C-HvqgxM7dj4G2kCQLnuMXi1fTpXRdpx/view

Multi-Agent Collaboration: 4 Team Patterns

Design Philosophy: Do **not** build an all-purpose *"super agent."* Instead, build a **team of specialized experts**.

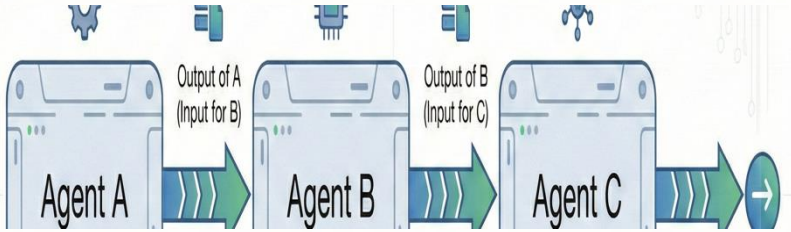
Coordinator Pattern



Use case:

- Suitable for **dynamic, open-ended tasks**
- The manager analyzes the request and routes subtasks to domain experts.

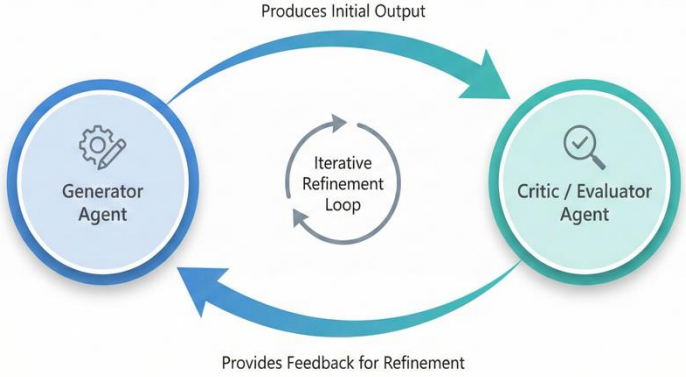
Sequential Pattern



Use case:

- Suitable for **linear workflows** and **data pipelines**

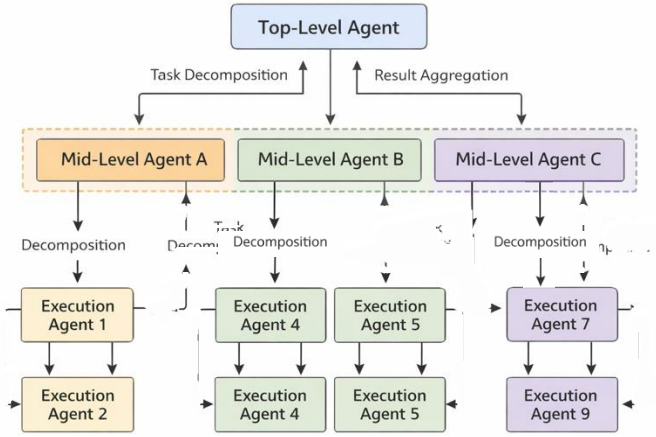
Iterative Refinement



Use case:

- A **generate–evaluate loop**
- Used to **continuously improve quality**

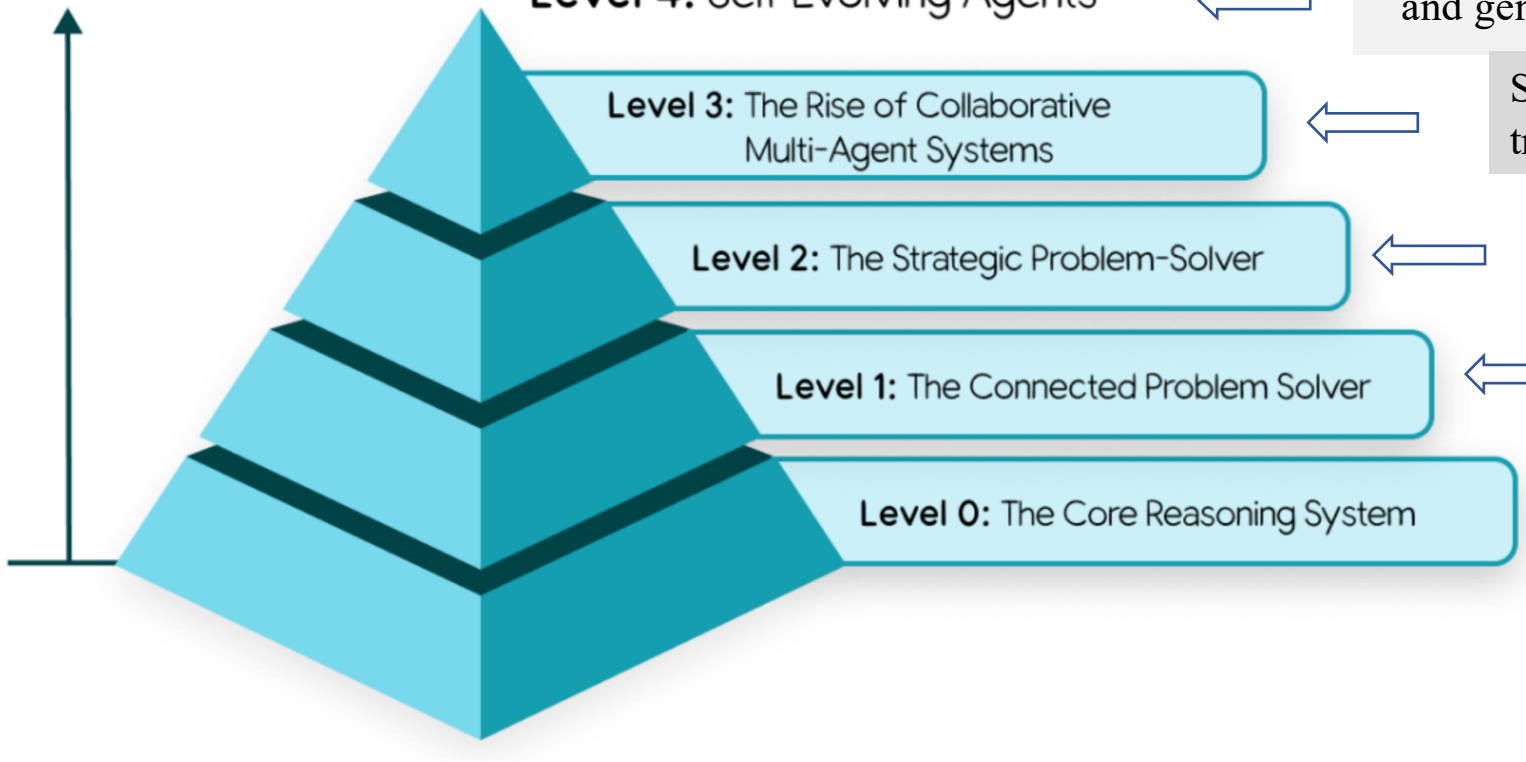
Hierarchical Pattern



Use case:

- Tasks are decomposed top-down
- Results are aggregated bottom-up

Capability Ladder: From Blind LM to Self-Evolving Agents



Autonomously creates tools and generates new intelligent agents

Specialized agents collaborate, treating other agents as tools

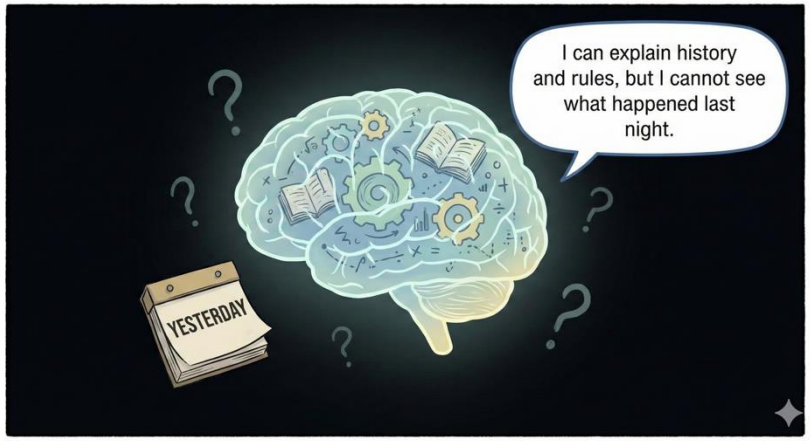
Possesses context-engineering capabilities, supports multi-step reasoning and planning

Uses tools (e.g., search engines, APIs) to retrieve real-time information

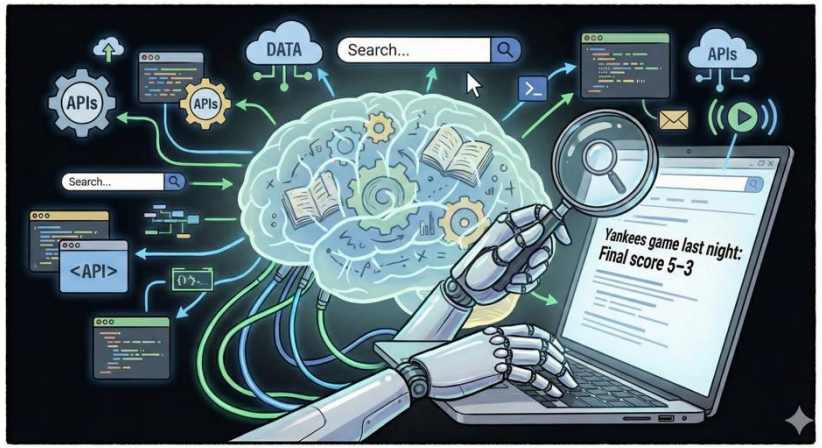
Relies solely on pre-trained knowledge, with no perception of the external world *(Blind to outside events)*

https://drive.google.com/file/d/1C-HvqgxM7dj4G2kCQLnuMXi1fTpXRdpx/view

Levels 0–4: One-Sentence Cheatsheet

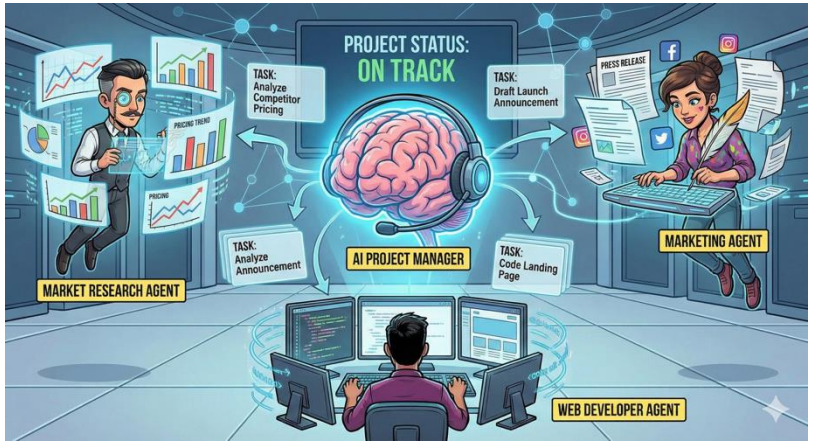


Level 0:
Key idea: Only a *brain*, no eyes, no hands, no tools.
Can **explain concepts**, but **cannot know the latest facts**.

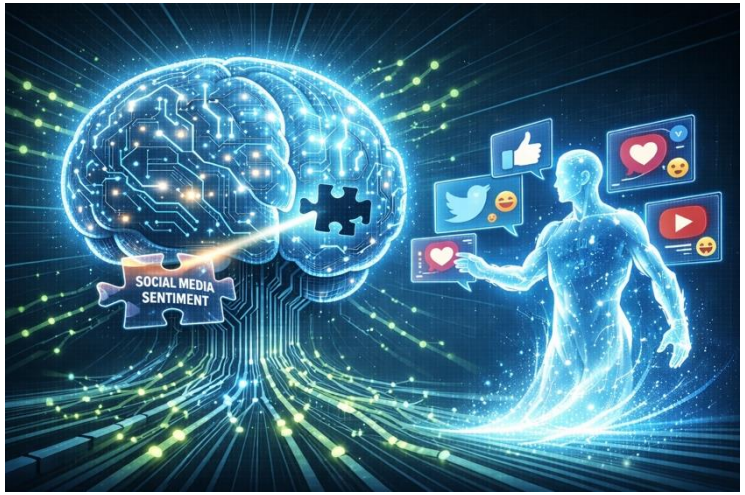


Level 1:
Key idea: Brain + hands = can use tools and access **real-time information**.

Level 2:
Key idea: Not only using tools, but also **planning steps** and **managing context**.



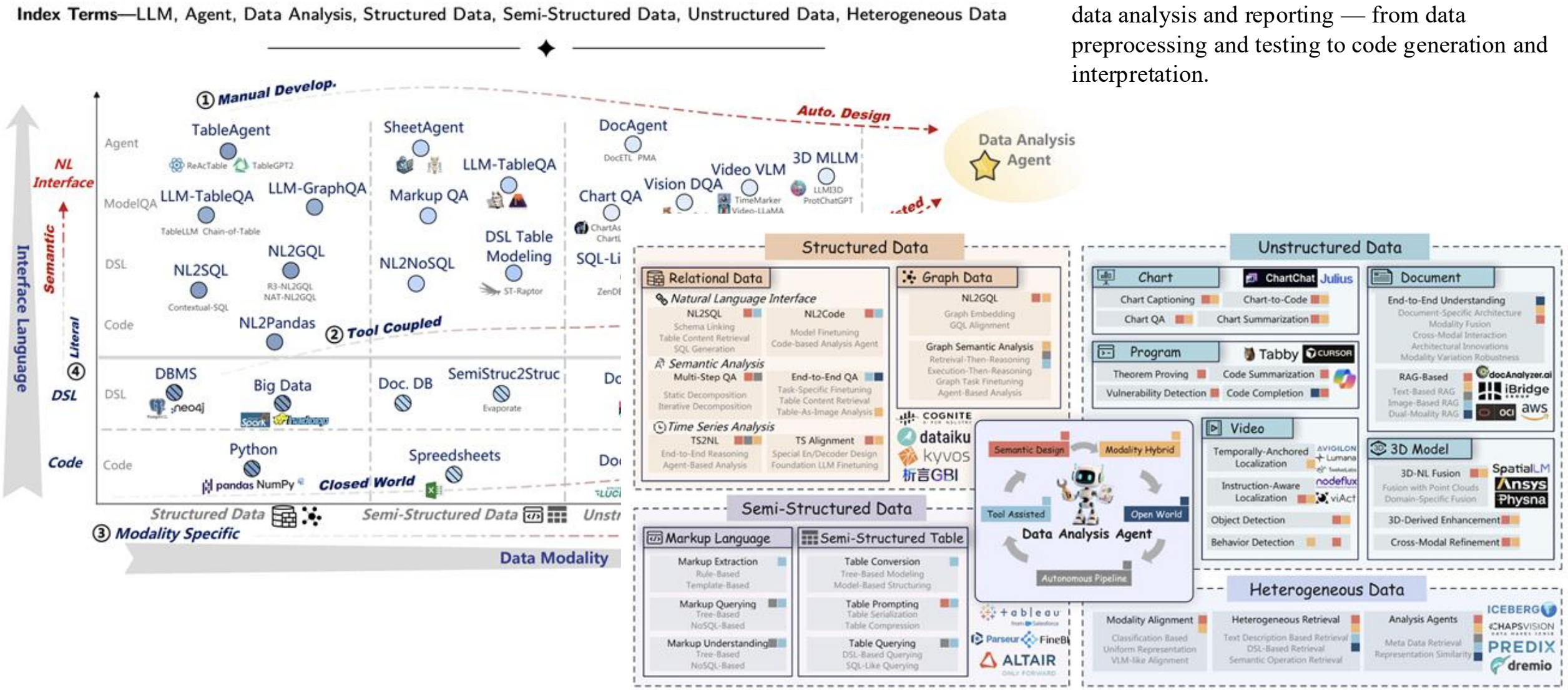
Level 3:
Key idea: A "project manager" AI that **coordinates multiple specialized agents**.



Level 4:
Key idea: The system can **identify what it is missing**,
and then **create new agents or new capabilities by itself**.

Agents as Data Analysts: What They Automate Today

LLM-based agents can already automate routine data analysis and reporting — from data preprocessing and testing to code generation and interpretation.

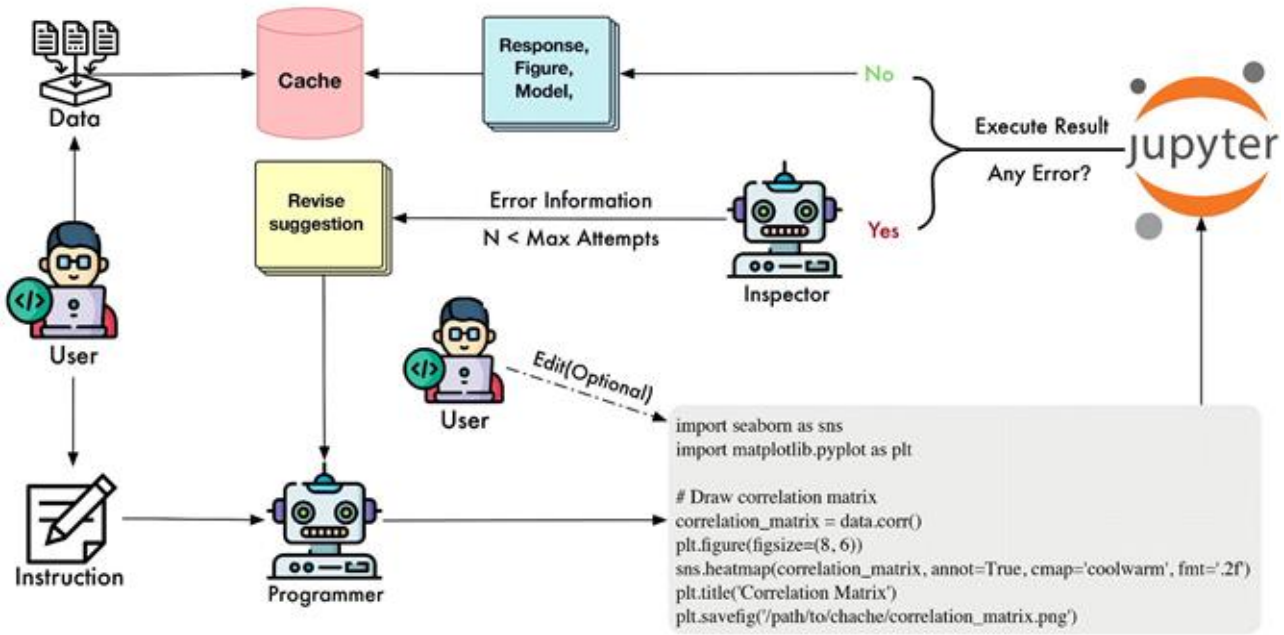


Tang, Z., Wang, W., Zhou, Z., Jiao, Y., Xu, B., Niu, B., Zhou, X., Li, G., He, Y., Zhou, W., Song, Y., Tan, C., Yang, X., Wang, B., He, C., Wang, X., & Wu, F. (2025). LLM/Agent-as-Data-Analyst: A Survey. *ArXiv*. https://arxiv.org/abs/2509.23988

Example: LAMBDA (1)

Figure 9. An example of using LAMBDA for classification analysis with the Wine dataset.

User Intent → Data Understanding → Statistical Testing → Visualization → Report Generation

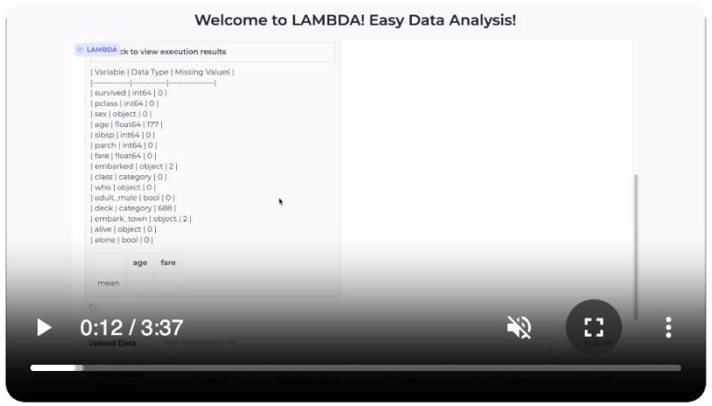


Maojun Sun, Ruijian Han, Binyan Jiang, Houduo Qi, Defeng Sun, Yancheng Yuan & Jian Huang. (2025) A Survey on Large Language Model-based Agents for Statistics and Data Science. *The American Statistician* 0:0, pages 1-14.

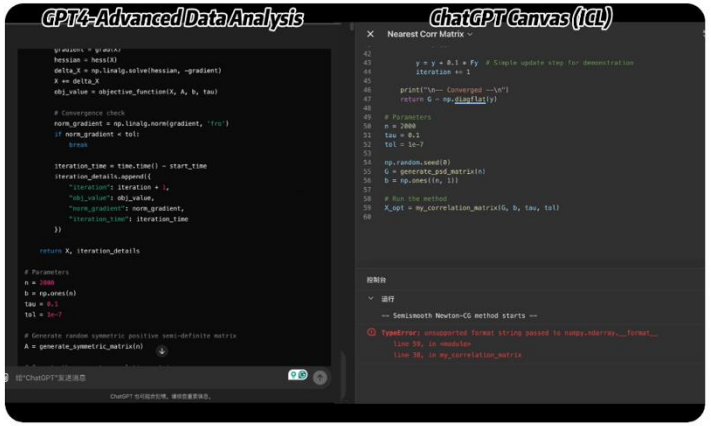
LAMBDA Demos: What the Workflow Looks Like

Demo Videos of LAMBDA

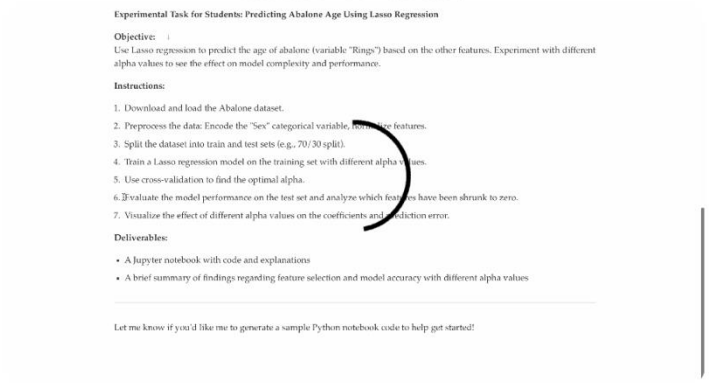
Data Analysis



Integrating Human Intelligence



Data Science Education



Experimental Task for Students: Predicting Abalone Age Using Lasso Regression

Objective:
Use Lasso regression to predict the age of abalone (variable "Rings") based on the other features. Experiment with different alpha values to see the effect on model complexity and performance.

Instructions:

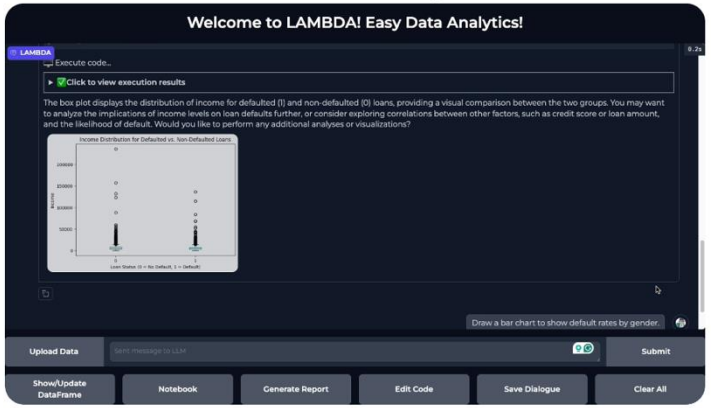
1. Download and load the Abalone dataset.
2. Preprocess the data: Encode the "Sex" categorical variable, [illegible] features.
3. Split the dataset into train and test sets (e.g., 70/30 split).
4. Train a Lasso regression model on the training set with different alpha values.
5. Use cross-validation to find the optimal alpha.
6. Evaluate the model performance on the test set and analyze which features have been shrunk to zero.
7. Visualize the effect of different alpha values on the coefficients and prediction error.

Deliverables:

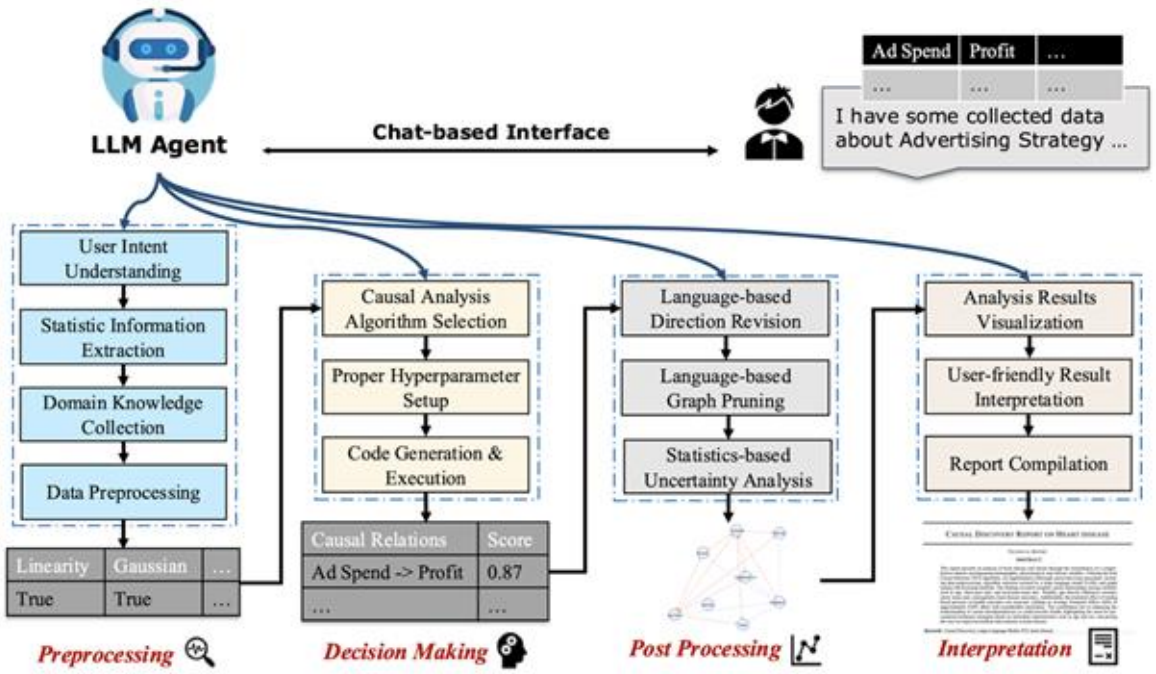
- A Jupyter notebook with code and explanations
- A brief summary of findings regarding feature selection and model accuracy with different alpha values

Let me know if you'd like me to generate a sample Python notebook code to help get started!

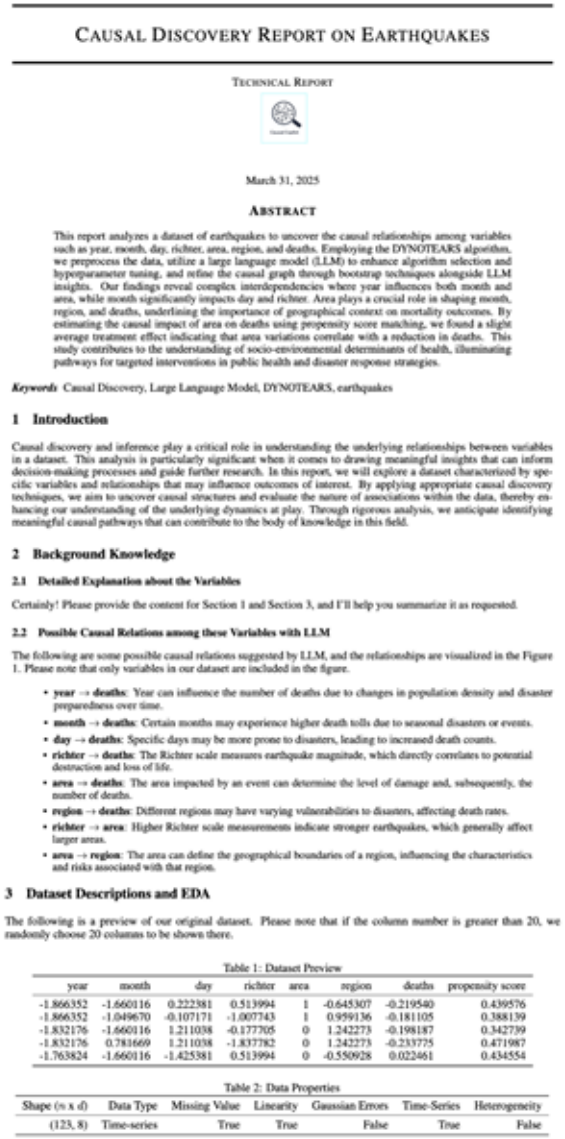
Loan Default Analysis



Causal-Copilot: Autonomous Analysis Beyond Simple Testing



This agent goes beyond simple statistical testing — it
1. Collects background knowledge,
2. Choosing the right algorithm for the right data, and explaining why relationships exist.



CAUSAL DISCOVERY REPORT ON EARTHQUAKES

TECHNICAL REPORT

March 31, 2025

ABSTRACT

This report analyzes a dataset of earthquakes to uncover the causal relationships among variables such as year, month, day, richter, area, region, and deaths. Employing the DYNOTEARS algorithm, we preprocess the data, utilize a large language model (LLM) to enhance algorithm selection and hyperparameter tuning, and refine the causal graph through bootstrap techniques alongside LLM insights. Our findings reveal complex interdependencies where year influences both month and area, while month significantly impacts day and richter. Area plays a crucial role in shaping month, region, and deaths, underlining the importance of geographical context on mortality outcomes. By estimating the causal impact of area on deaths using propensity score matching, we found a slight average treatment effect indicating that area variations correlate with a reduction in deaths. This study contributes to the understanding of socio-environmental determinants of health, illuminating pathways for targeted interventions in public health and disaster response strategies.

Keywords Causal Discovery, Large Language Model, DYNOTEARS, earthquakes

1 Introduction

Causal discovery and inference play a critical role in understanding the underlying relationships between variables in a dataset. This analysis is particularly significant when it comes to drawing meaningful insights that can inform decision-making processes and guide further research. In this report, we will explore a dataset characterized by specific variables and relationships that may influence outcomes of interest. By applying appropriate causal discovery techniques, we aim to uncover causal structures and evaluate the nature of associations within the data, thereby enhancing our understanding of the underlying dynamics at play. Through rigorous analysis, we anticipate identifying meaningful causal pathways that can contribute to the body of knowledge in this field.

2 Background Knowledge

2.1 Detailed Explanation about the Variables

Certainly! Please provide the content for Section 1 and Section 3, and I'll help you summarize it as requested.

2.2 Possible Causal Relations among these Variables with LLM

The following are some possible causal relations suggested by LLM, and the relationships are visualized in the Figure 1. Please note that only variables in our dataset are included in the figure.

- **year → deaths**: Year can influence the number of deaths due to changes in population density and disaster preparedness over time.
- **month → deaths**: Certain months may experience higher death tolls due to seasonal disasters or events.
- **day → deaths**: Specific days may be more prone to disasters, leading to increased death counts.
- **richter → deaths**: The Richter scale measures earthquake magnitude, which directly correlates to potential destruction and loss of life.
- **area → deaths**: The area impacted by an event can determine the level of damage and, subsequently, the number of deaths.
- **region → deaths**: Different regions may have varying vulnerabilities to disasters, affecting death rates.
- **richter → area**: Higher Richter scale measurements indicate stronger earthquakes, which generally affect larger areas.
- **area → region**: The area can define the geographical boundaries of a region, influencing the characteristics and risks associated with that region.

3 Dataset Descriptions and EDA

The following is a preview of our original dataset. Please note that if the column number is greater than 20, we randomly choose 20 columns to be shown here.

Table 1: Dataset Preview

year	month	day	richter	area	region	deaths	propensity score
-1.866352	-1.660116	0.222381	0.513994	1	-0.645307	-0.219540	0.439576
-1.866352	-1.049670	-0.107171	-1.007743	1	0.959136	-0.181105	0.388139
-1.832176	-1.660116	1.211038	-0.177705	0	1.242273	-0.198187	0.342739
-1.832176	0.783669	1.211038	-1.837782	0	1.242273	-0.233775	0.471987
-1.763824	-1.660116	-1.425381	0.513994	0	-0.550928	0.022461	0.434554

Table 2: Data Properties

Shape (n x d)	Data Type	Missing Value	Linearity	Gaussian Errors	Time-Series	Heterogeneity
(123, 8)	Time-series	True	True	False	True	False

Figure 2: Distribution Plots of Variables

Figure 3: Correlation Heatmap of Variables

Figure 4: Initial Graph

The above is the initial result graph produced by our algorithm.

The analysis of the causal relationships among the variables year, month, day, richter, area, region, and deaths reveals a complex interdependence that shapes the impact of various factors on mortality due to seismic events. Notably, year influences both month and area, establishing a temporal and geographic context for the events being analyzed. Month is a critical factor that influences day and richter, highlighting how temporal sequencing may affect the severity of seismic activities and their aftermath. The variable area significantly impacts month, region, and deaths, indicating that the geographical context is vital in shaping the outcomes of earthquake events, such as fatalities. Finally, deaths, which can be linked to various factors, including month, day, area, and region, suggest the pervasive influence of these interconnected variables on mortality rates.

- Year influences month and area, setting a temporal and geographic framework.
- Month plays a pivotal role in affecting both day and richter, indicating a connection between time and seismic activity severity.
- The variable area impacts month, region, and deaths, emphasizing the importance of geographical factors.
- Deaths are influenced by month, day, area, and region, illustrating how interconnected variables collectively determine fatality outcomes in seismic events.

6 Causal Inference Results

6.1 Proposal Overview

In this proposal, we focus on the task of estimating the causal impact of the variable 'area' on the variable 'deaths' to address a critical question in public health and environmental studies: how variations in area—whether measured by size, population density, or geographical features—can influence mortality rates. Understanding this relationship is vital for policymakers and health officials as it can highlight areas that require targeted interventions, resource allocation, or specific health policies. By quantifying this causal effect, our analysis can illuminate potential mechanisms through which area attributes may lead to differences in health outcomes, thereby informing effective public health strategies.

To achieve this, we will employ rigorous causal inference methodologies, such as propensity score matching or instrumental variable analysis, to control for confounding factors that may bias our estimates. This approach allows us to isolate the effect of area on deaths while accounting for other variables that could influence mortality, ensuring that our findings are robust and reliable. Ultimately, this task contributes to a deeper understanding of the socio-environmental determinants of health, offering insights that can improve population health and guide future research initiatives.

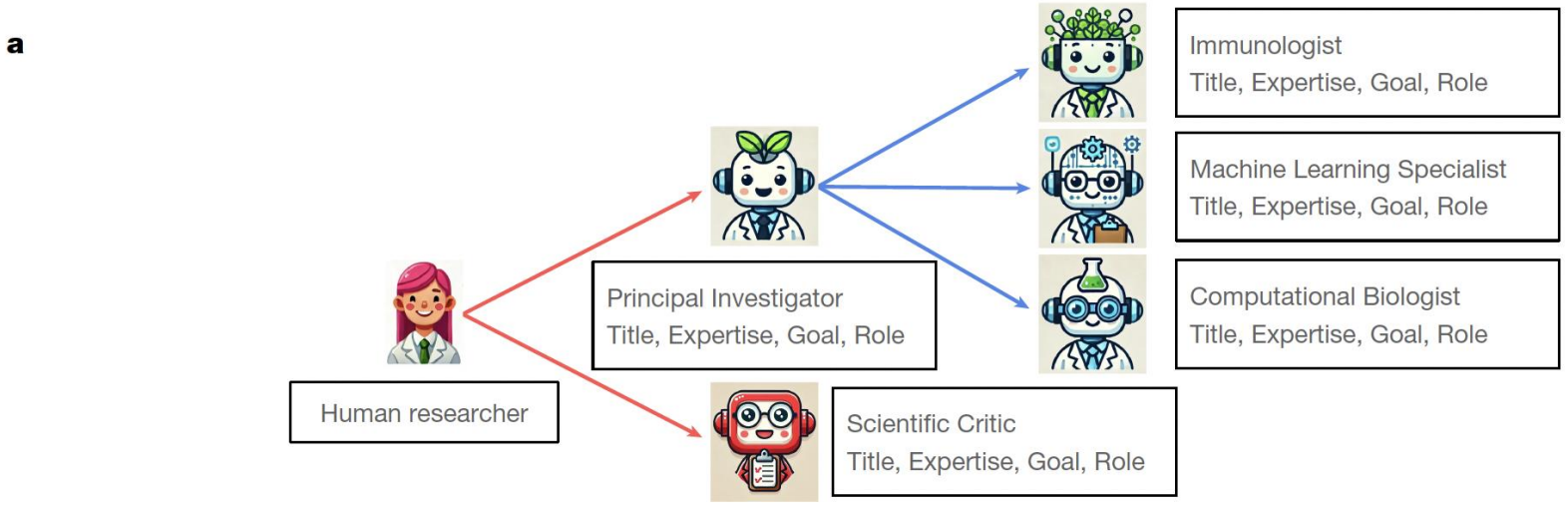
6.3.2 Next Steps Suggestions

To enhance the robustness of the findings, **future analyses should focus on refining the treatment effect estimation methodologies.** Employing advanced techniques such as machine learning approaches for treatment effect estimation, including causal forest or Bayesian additive regression trees, could potentially uncover hidden patterns and interactions that traditional methods might miss. Additionally, conducting sensitivity analyses to assess the impact of unobserved confounding could further validate the results, ensuring that the observed treatment effects are not artifacts of model assumptions. An exploration of subgroup analyses, particularly in identified regions and specific time frames that exhibited significant variability in treatment effects, would also provide deeper insights into the causal mechanisms at play.

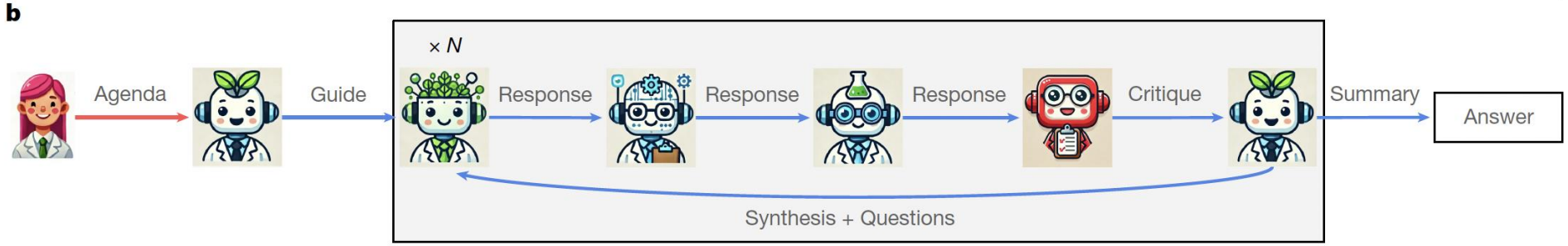
For future research directions, it would be valuable to expand the study by **incorporating longitudinal data to evaluate the persistence of treatment effects over time and to validate these findings across different populations.** Investigating the underlying causal pathways, perhaps through qualitative methods or mixed-methods approaches, can provide context to the quantitative results and aid in developing a nuanced understanding of how and why the treatment interacts with various factors. Additionally, exploring potential moderators and mediators within the treatment effect framework could pave the way for targeted interventions and inform policy decisions aimed at mitigating deaths related to the area treatment. Such extensions not only bolster the generalizability of the findings but also enhance their applicability in real-world settings.

Wang, X., Zhou, K., Wu, W., Singh, H. S., Nan, F., Jin, S., Philip, A., Patnaik, S., Zhu, H., Singh, S., Prashant, P., Shen, Q., & Huang, B. (2025). Causal-Copilot: An Autonomous Causal Analysis Agent. *ArXiv*. https://arxiv.org/abs/2504.13263

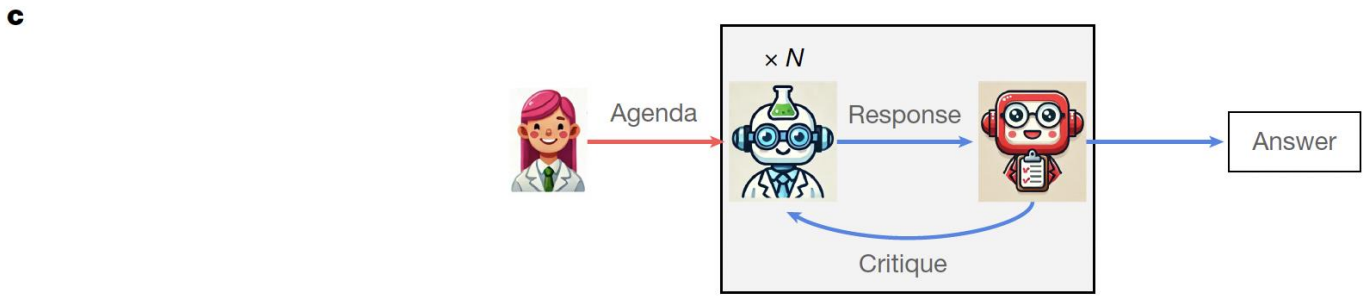
Virtual Lab: A Multi-Agent Scientific Team



a. Human-defined agent roles enable the PI agent to automatically construct a team of specialized scientist agents for the project.



b. In team meetings, the PI agent orchestrates multi-agent discussion with iterative critique to synthesize collective decisions.



c. In individual meetings, a scientist agent iteratively refines its response through feedback from the Scientific Critic to produce an improved solution.

Swanson, K., Wu, W., Bulaong, N.L. *et al.* The Virtual Lab of AI agents designs new SARS-CoV-2 nanobodies. *Nature* **646**, 716–723 (2025). https://doi.org/10.1038/s41586-025-09442-9

Virtual Lab for Nanobody Design

a. Human-in-the-loop orchestration

A human researcher defines agent roles (PI, Scientific Critic) and project goals.

b. Multi-agent scientific collaboration

A PI agent assembles a team of specialized scientist agents to jointly reason and decide.

c. Structured decision-making via meetings

Agents collaboratively determine design strategy (antibodies vs. nanobodies) and tool selection.

d. Tool-augmented scientific workflow

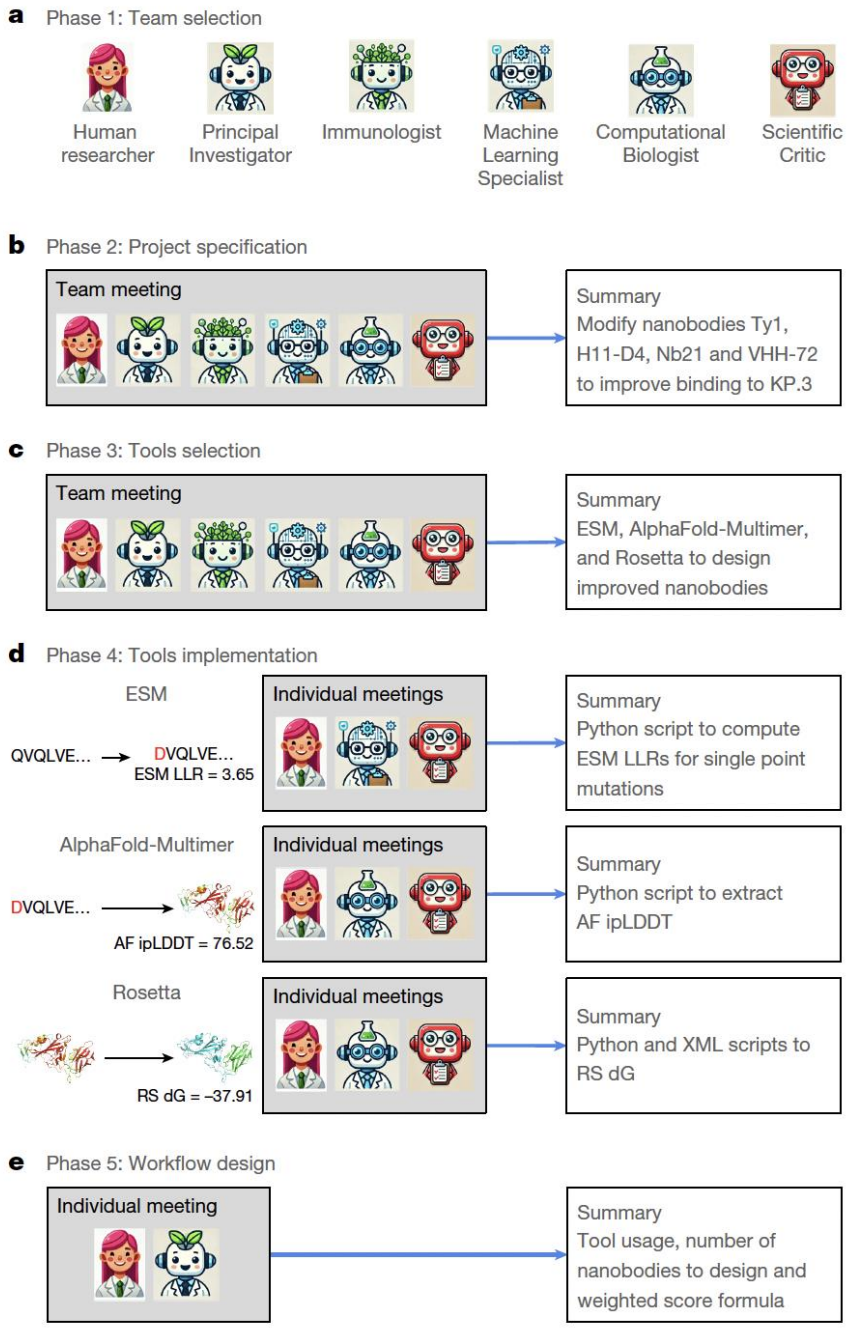
The system integrates state-of-the-art computational tools (ESM, AlphaFold-Multimer, Rosetta).

e. Iterative code development and refinement

ML and computational biology agents implement and iteratively improve code with critic feedback.

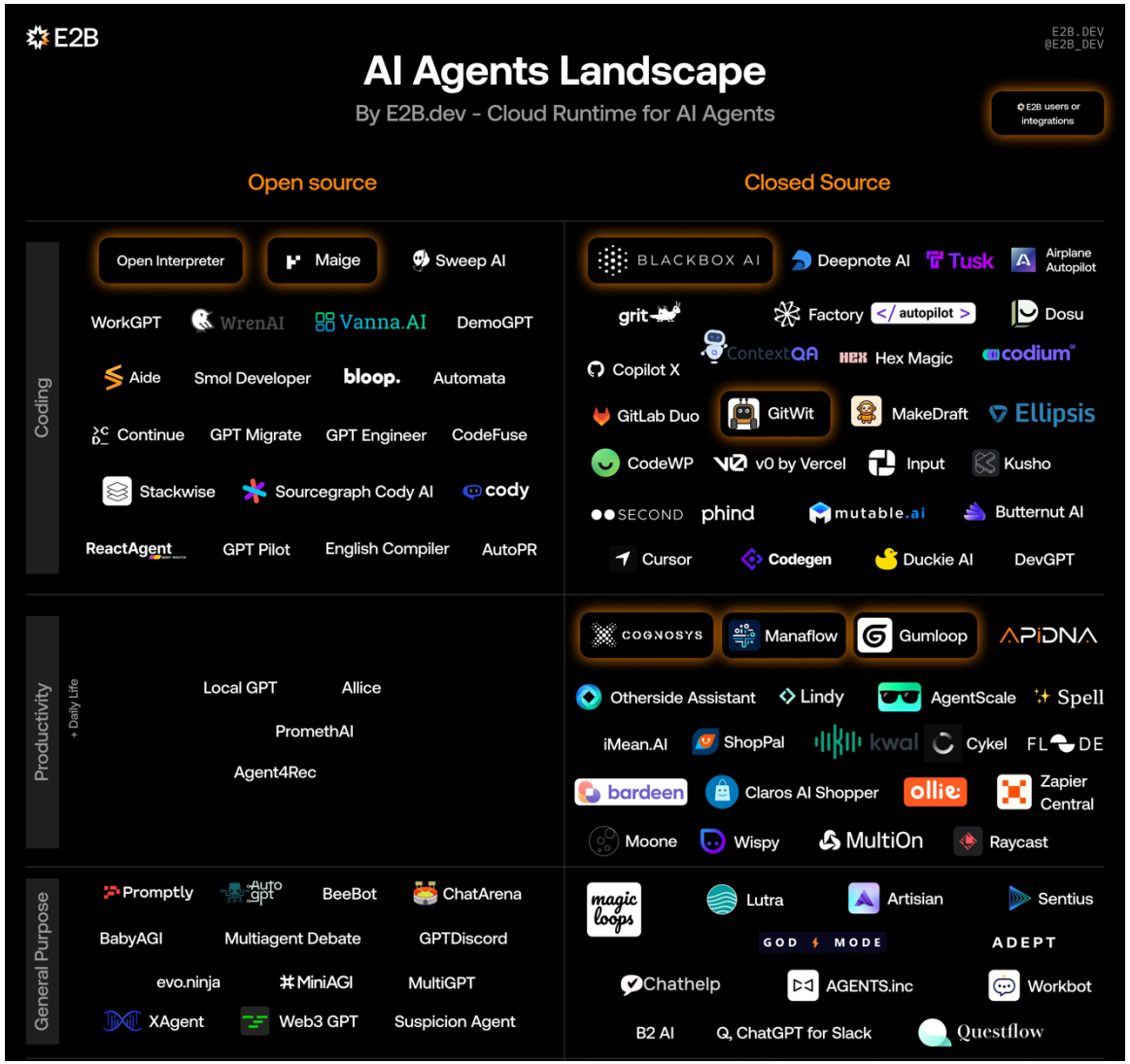
f. Centralized planning and execution

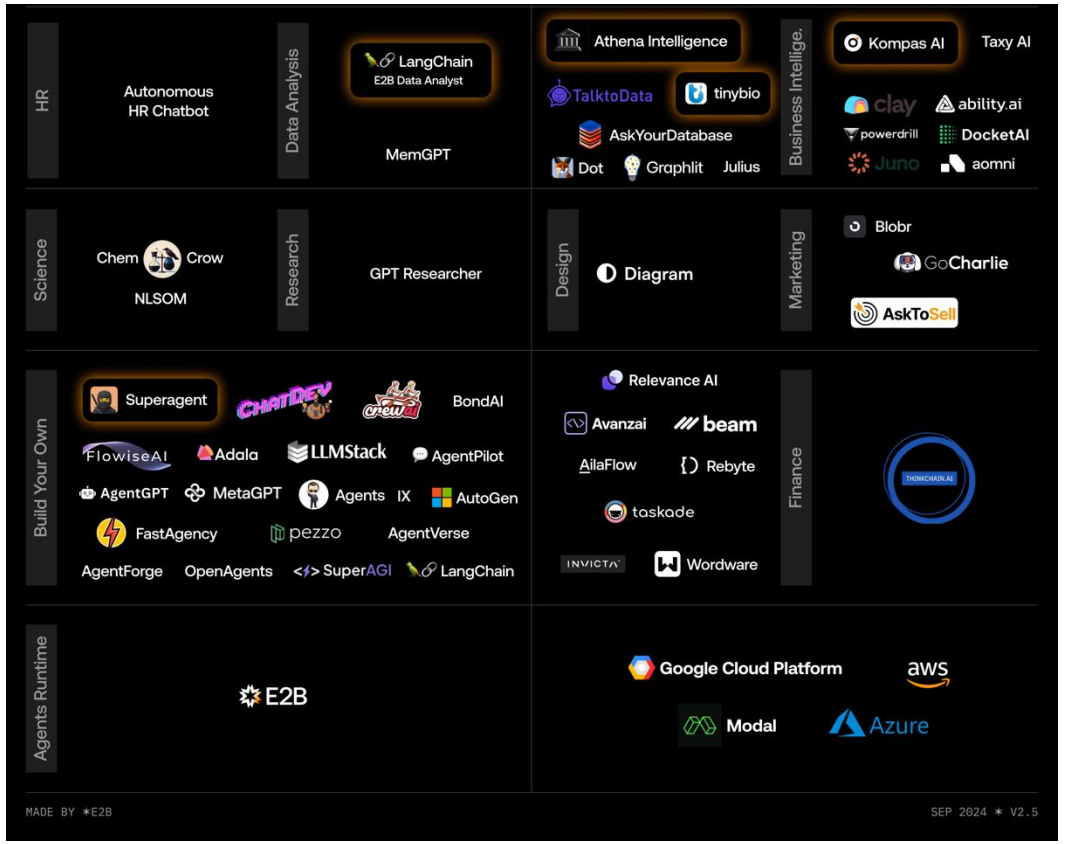
The PI agent orchestrates tool usage and selects optimized nanobody candidates.



Swanson, K., Wu, W., Bulaong, N.L. *et al.* The Virtual Lab of AI agents designs new SARS-CoV-2 nanobodies. *Nature* **646**, 716–723 (2025). https://doi.org/10.1038/s41586-025-09442-9

The AI Agent Ecosystem Is Expanding Rapidly





AI agents now exist as **frameworks, platforms, and end-to-end products**
The ecosystem includes:

- Open-source tools for flexibility and customization
- Closed-source solutions for fast deployment

https://github.com/e2b-dev/awesome-ai-agents

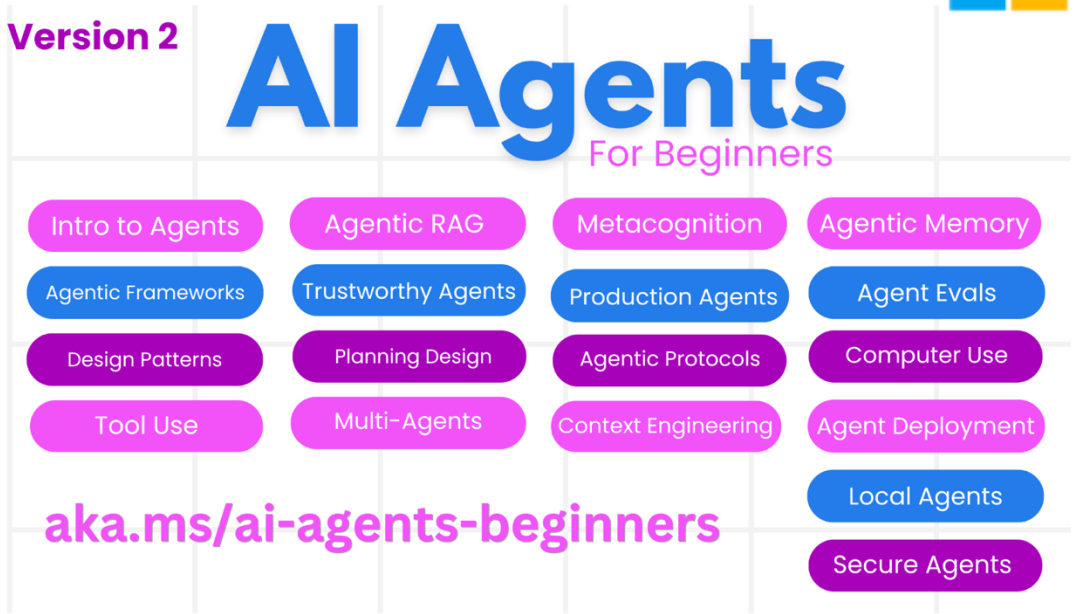
Materials: Agent Use Cases and Learning Resources

🌟 500+ AI Agent Projects / UseCases

500-AI-Agents-Projects UseCase

A curated collection of AI agent use cases across industries, showcasing practical applications and linking to open-source projects for implementation. Explore how AI agents are transforming industries like healthcare, finance, education, and more! 🤖✨

AI Agents for Beginners - A Course



A course teaching everything you need to know to start building AI Agents

license MIT contributors 66 issues 6 open pull requests 0 open PRs welcome

Appendix0: Overview of Existing LLMs —— Open Sources

	Model	Release Time	Size (B)	Base Model	Adaptation IT	Adaptation RLHF	Pre-train Data Scale	Latest Data Timestamp	Hardware (GPUs / TPUs)	Training Time	Evaluation ICL	Evaluation CoT
Publicly Available	T5 [82]	Oct-2019	11	-	-	-	1T tokens	Apr-2019	1024 TPU v3	-	✓	-
	mT5 [83]	Oct-2020	13	-	-	-	1T tokens	-	-	-	✓	-
	PanGu-α [84]	Apr-2021	13*	-	-	-	1.1TB	-	2048 Ascend 910	-	✓	-
	CPM-2 [85]	Jun-2021	198	-	-	-	2.6TB	-	-	-	-	-
	T0 [28]	Oct-2021	11	T5	✓	-	-	-	512 TPU v3	27 h	✓	-
	CodeGen [86]	Mar-2022	16	-	-	-	577B tokens	-	-	-	✓	-
	GPT-NeoX-20B [87]	Apr-2022	20	-	-	-	825GB	-	96 40G A100	-	✓	-
	Tk-Instruct [88]	Apr-2022	11	T5	✓	-	-	-	256 TPU v3	4 h	✓	-
	UL2 [89]	May-2022	20	-	-	-	1T tokens	Apr-2019	512 TPU v4	-	✓	✓
	OPT [90]	May-2022	175	-	-	-	180B tokens	-	992 80G A100	-	✓	-
	NLLB [91]	Jul-2022	54.5	-	-	-	-	-	-	-	✓	-
	CodeGeeX [92]	Sep-2022	13	-	-	-	850B tokens	-	1536 Ascend 910	60 d	✓	-
	GLM [93]	Oct-2022	130	-	-	-	400B tokens	-	768 40G A100	60 d	✓	-
	Flan-T5 [69]	Oct-2022	11	T5	✓	-	-	-	-	-	✓	✓
	BLOOM [78]	Nov-2022	176	-	-	-	366B tokens	-	384 80G A100	105 d	✓	-
	mT0 [94]	Nov-2022	13	mT5	✓	-	-	-	-	-	✓	-
	Galactica [35]	Nov-2022	120	-	-	-	106B tokens	-	-	-	✓	✓
	BLOOMZ [94]	Nov-2022	176	BLOOM	✓	-	-	-	-	-	✓	-
	OPT-IML [95]	Dec-2022	175	OPT	✓	-	-	-	128 40G A100	-	✓	✓
	LLaMA [57]	Feb-2023	65	-	-	-	1.4T tokens	-	2048 80G A100	21 d	✓	-
	Pythia [96]	Apr-2023	12	-	-	-	300B tokens	-	256 40G A100	-	✓	-
	CodeGen2 [97]	May-2023	16	-	-	-	400B tokens	-	-	-	✓	-
	StarCoder [98]	May-2023	15.5	-	-	-	1T tokens	-	512 40G A100	-	✓	✓
	LLaMA2 [99]	Jul-2023	70	-	✓	✓	2T tokens	-	2000 80G A100	-	✓	-
	Baichuan2 [100]	Sep-2023	13	-	✓	✓	2.6T tokens	-	1024 A800	-	✓	-
	QWEN [101]	Sep-2023	14	-	✓	✓	3T tokens	-	-	-	✓	-
	FLM [102]	Sep-2023	101	-	✓	-	311B tokens	-	192 A800	22 d	✓	-
	Skywork [103]	Oct-2023	13	-	-	-	3.2T tokens	-	512 80G A800	-	✓	-

Zhao, W. X., Zhou, K., Li, J., Tang, T., Wang, X., Hou, Y., Min, Y., Zhang, B., Zhang, J., Dong, Z., Du, Y., Yang, C., Chen, Y., Chen, Z., Jiang, J., Ren, R., Li, Y., Tang, X., Liu, Z., . . . Wen, J. (2023). A Survey of Large Language Models. *ArXiv*. https://arxiv.org/abs/2303.18223

Appendix0: Overview of Existing LLMs —— Closed Sources

	Model	Release Time	Size (B)	Base Model	Adaptation IT	Adaptation RLHF	Pre-train Data Scale	Latest Data Timestamp	Hardware (GPUs / TPUs)	Training Time	Evaluation ICL	Evaluation CoT
Closed Source	GPT-3 [55]	May-2020	175	-	-	-	300B tokens	-	-	-	✓	-
	GShard [104]	Jun-2020	600	-	-	-	1T tokens	-	2048 TPU v3	4 d	-	-
	Codex [105]	Jul-2021	12	GPT-3	-	-	100B tokens	May-2020	-	-	✓	-
	ERNIE 3.0 [106]	Jul-2021	10	-	-	-	375B tokens	-	384 V100	-	✓	-
	Jurassic-1 [107]	Aug-2021	178	-	-	-	300B tokens	-	800 GPU	-	✓	-
	HyperCLOVA [108]	Sep-2021	82	-	-	-	300B tokens	-	1024 A100	13.4 d	✓	-
	FLAN [67]	Sep-2021	137	LaMDA-PT	✓	-	-	-	128 TPU v3	60 h	✓	-
	Yuan 1.0 [109]	Oct-2021	245	-	-	-	180B tokens	-	2128 GPU	-	✓	-
	Anthropic [110]	Dec-2021	52	-	-	-	400B tokens	-	-	-	✓	-
	WebGPT [81]	Dec-2021	175	GPT-3	-	✓	-	-	-	-	✓	-
	Gopher [64]	Dec-2021	280	-	-	-	300B tokens	-	4096 TPU v3	920 h	✓	-
	ERNIE 3.0 Titan [111]	Dec-2021	260	-	-	-	-	-	-	-	✓	-
	GLaM [112]	Dec-2021	1200	-	-	-	280B tokens	-	1024 TPU v4	574 h	✓	-
	LaMDA [68]	Jan-2022	137	-	-	-	768B tokens	-	1024 TPU v3	57.7 d	-	-
	MT-NLG [113]	Jan-2022	530	-	-	-	270B tokens	-	4480 80G A100	-	✓	-
	AlphaCode [114]	Feb-2022	41	-	-	-	967B tokens	Jul-2021	-	-	-	-
	InstructGPT [66]	Mar-2022	175	GPT-3	✓	✓	-	-	-	-	✓	-
	Chinchilla [34]	Mar-2022	70	-	-	-	1.4T tokens	-	-	-	✓	-
	PaLM [56]	Apr-2022	540	-	-	-	780B tokens	-	6144 TPU v4	-	✓	✓
	AlexaTM [115]	Aug-2022	20	-	-	-	1.3T tokens	-	128 A100	120 d	✓	✓
	Sparrow [116]	Sep-2022	70	-	-	✓	-	-	64 TPU v3	-	✓	-
	WeLM [117]	Sep-2022	10	-	-	-	300B tokens	-	128 A100 40G	24 d	✓	-
	U-PaLM [118]	Oct-2022	540	PaLM	-	-	-	-	512 TPU v4	5 d	✓	✓
	Flan-PaLM [69]	Oct-2022	540	PaLM	✓	-	-	-	512 TPU v4	37 h	✓	✓
	Flan-U-PaLM [69]	Oct-2022	540	U-PaLM	✓	-	-	-	-	-	✓	✓
	GPT-4 [46]	Mar-2023	-	-	✓	✓	-	-	-	-	✓	✓
	PanGu-Σ [119]	Mar-2023	1085	PanGu-α	-	-	329B tokens	-	512 Ascend 910	100 d	✓	-
	PaLM2 [120]	May-2023	16	-	✓	-	100B tokens	-	-	-	✓	✓

Zhao, W. X., Zhou, K., Li, J., Tang, T., Wang, X., Hou, Y., Min, Y., Zhang, B., Zhang, J., Dong, Z., Du, Y., Yang, C., Chen, Y., Chen, Z., Jiang, J., Ren, R., Li, Y., Tang, X., Liu, Z., . . . Wen, J. (2023). A Survey of Large Language Models. *ArXiv*. https://arxiv.org/abs/2303.18223

Appendix 0: Overview of LLMs in Biomedicine

Model	Date	Parameters	Base Model	Fine-tuning	Tasks and Purpose Description	Unimodal	Open Source
GatorTron [81]	2022.12	8.9B/3.9B/345M	BERT	From scratch	Clinical NLP tasks	✓	✓
BianQue [20]	2023.12	6B	ChatGLM	Full parameter	Health advice, multi-turn dialogue	✓	✓
ChatDoctor [11]	2023.06	7B	LLaMA-7B	Instruction tuning	Medical dialogue	✓	✓
DISC-MedLLM [21]	2023.08	13B	Baichuan-13B-Base	Supervised fine-tuning	Medical consultation	✓	✓
DNABERT-S [82]	2024.02	-	BERT	-	DNA sequence analysis	✓	✓
GeneCompass [34]	2023.09	>100M	T5	From scratch	Genomic data analysis	✓	✓
GenePT [83]	2024.03	-	-	-	Gene and cell representation	✓	✓
BenTsao [12]	2023.04	7B	LLaMA	Instruction tuning	Chinese biomedical tasks	✓	✓
HuatuoGPT [10]	2023.05	7B	Bloomz-7b1-mt	Supervised fine-tune, RLAIF	Medical exams, research queries	✓	✓
Med-PaLM [29]	2022.12	540B	Flan-PaLM	Prompt tuning	Medical knowledge evaluation	✓	-
MedChatZH [84]	2024.03	7B	BaiChuan	Prompt tuning	Chinese medical dialogue	✓	✓
Radiology-GPT [85]	2024.03	7B	Alpaca-7B	Instruction tuning, LoRA	Radiology report generation	✓	✓
RadOnc-GPT [86]	2023.11	-	LLaMA2	Instruction tuning, LoRA	Radiation treatment planning	✓	-
scBERT [9]	2022.09	-	BERT	From scratch	Single-cell RNA analysis	✓	✓
scGPT [87]	2024.02	-	Transformer	From scratch	Single-cell multi-omics analysis	✓	✓
Taiyi [88]	2024.02	7B	Qwen-7B-base	Supervised fine-tuning	Multilingual biomedical NLP	✓	✓
OARA [89]	2024.02	7B	Vicuna v1.5	LoRA	Surgical/anesthetic education	✓	-
Med-PaLM 2 [8]	2023.05	340B	PaLm2	Instruction tuning, LoRA	Advanced medical Q&A	✓	✓
Hypnos [90]	2024.03	7B	LLaMA	LoRA	Anesthesiology tasks	✓	-
VetLLM [91]	2023.12	7B	Alpaca-7B	LoRA	Veterinary diagnosis	✓	✓
BioMedLM [19]	2024.03	2.7B	GPT-2	From scratch	Biomedical Q&A	✓	✓
CancerGPT [28]	2023.04	124M	GPT	K-SHOT	Drug synergy prediction	✓	-
ESM-2 [92]	2023.03	15B	-	From scratch	Protein structure prediction	✓	✓
HuatuoGPT II [93]	2023.11	7/13B	Baichuan2-7/13B-Base	Instruction tuning	TCM tasks	✓	✓
DoctorGLM [94]	2023.04	6B	ChatGLM	LoRA	Chinese medical Q&A	✓	✓
MedCPT [95]	2023.11	-	-	-	Biomedical information retrieval	✓	✓
BioGPT [27]	2023.04	-	GPT-2	From scratch	Biomedical text generation	✓	✓
GatorTronGPT [17]	2023.11	5B/20B	GPT-3	From scratch	Medical text synthesis	✓	✓
MEDITRON [96]	2023.11	7B/70B	LLaMA-2	Instruction tuning	Medical text comprehension	✓	✓
ClinicalGPT [97]	2023.06	7B	BLOOM-7B	LoRA	Clinical tasks	✓	-
Qilin-Med [98]	2024.04	-	Baichuan	-	Multi-stage medical training	✓	-
MedAlpaca [99]	2023.01	7/13B	LLaMA	LoRA	Open-source medical LLM	✓	-
Alpacare [100]	2024.05	-	-	Instruction tuning	Medical instruction following	✓	✓
Zhongjing [71]	2023.12	13B	Ziya-LLaMA-13B-v13	Supervised fine-tuning	TCM Q&A	✓	✓
Cpllm [101]	2024.05	13B/2.7B	LLaMA2, PubMedGPT	LoRA	Clinical prediction	✓	✓
MMedLM 2 [68]	2024.02	7B	InternLM	LoRA	Multilingual medical Q&A	✓	✓
AlphaFold 3 [102]	2024.05	-	-	-	Protein structure prediction	✓	-
Bingo [103]	2023.11	15B	ESM-2	From scratch	Protein-coding gene prediction		✓
BiomedCLIP [104]	2024.01	-	CLIP	-	Multimodal biomedical tasks		✓
Med-PaLm M [66]	2023.07	12B/84B/562B	Palm-E	Instruction tuning	Multimodal medical analysis		✓
MONET [105]	2024.04	-	CLIP	From scratch	Medical image annotation		✓
XrayGPT [106]	2023.06	-	MedCLIP, Vicuna	Modality alignment	Chest X-ray analysis		✓
Med-MLLM [107]	2023.12	-	-	Multi-stage training	X-ray representation learning		-
EchoCLIP [108]	2024.04	-	OpenCLIP	From scratch	Echocardiogram interpretation		✓
OphGLM [109]	2023.06	6B	ChatGLM	Instruction tuning	Ophthalmology diagnosis		✓
ClinicalBLIP [110]	2024.02	3B	InstructBLIP	LoRA	Radiology report generation		-
Med-Gemini [111]	2024.04	-	Gemini	Instruction tuning	Multimodal medical analysis		-
BioMedGPT [112]	2023.08	Instruction tuning	LLaMA2	-	Biomedical question answering		✓

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Appendix 1: Representative basic and advanced abilities and corresponding representative datasets for evaluating

Level	Ability	Task	Dataset
Basic	Language Generation	Language Modeling	Penn Treebank [540], WikiText-103 [541], the Pile [166], LAMBADA [252]
		Conditional Text Generation	WMT'14,16,19,20,21,22 [542–547], Flores-101 [548], DiaBLa [549], CNN/DailyMail [550], XSum [551], WikiLingua [552] OpenDialKG [553]
		Code Synthesis	APPS [376], HumanEval [105], MBPP [223], CodeContest [114], MTPB [86], DS-1000 [554], ODEX [555]
	Knowledge Utilization	Closed-Book QA	Natural Questions [556], ARC [557], TruthfulQA [558], Web Questions [559], TriviaQA [560], PIQA [561], LC-quad2.0 [562], GrailQA [563], KQApro [564], CWQ [565], MKQA [566], ScienceQA [567]
		Open-Book QA	Natural Questions [556], OpenBookQA [568], ARC [557], TriviaQA [560], Web Questions [559], MS MARCO [569], QASC [570], SQuAD [571], WikiMovies [572]
		Knowledge Completion	WikiFact [573], FB15k-237 [574], Freebase [575], WN18RR [576], WordNet [577], LAMA [578], YAGO3-10 [579], YAGO [580]
	Complex Reasoning	Knowledge Reasoning	CSQA [506], StrategyQA [199], HotpotQA [581], ARC [557], BoolQ [582], PIQA [561], SIQA [583], HellaSwag [584], WinoGrande [585], COPA [586], OpenBookQA [568], ScienceQA [567], proScript [587], ProPara [588], ExplaGraphs [589], ProofWriter [590], EntailmentBank [591], ProOntoQA [592]
		Symbolic Reasoning	CoinFlip [33], ReverseList [33], LastLetter [33], Boolean Assignment [593], Parity [593], Colored Object [70], Penguins in a Table [70], Repeat Copy [436], Object Counting [436]
		Mathematical Reasoning	MATH [362], GSM8k [198], SVAMP [594], MultiArith [595], ASDiv [505], MathQA [596], AQUA-RAT [597], MAWPS [598], DROP [599], NaturalProofs [600], PISA [601], miniF2F [602], ProofNet [603]

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Appendix 1: Representative basic and advanced abilities and corresponding representative datasets for evaluating

Advanced	Human Alignment	Honestness	TruthfulQA [558], HaluEval [604]
		Helpfulness	HH-RLHF [183]
		Harmlessness	HH-RLHF [183], Crows-Pairs [605] WinoGender [606], RealToxicityPrompts [607]
	Interaction with External Environment	Household	VirtualHome [608], BEHAVIOR [609], ALFRED [610],ALFWorld [611]
		Website Environment	WebShop [612], Mind2Web [613]
		Open World	MineRL [614], MineDojo [615]
	Tool Manipulation	Search Engine	HotpotQA [581], TriviaQA [560], Natural Questions [556]
		Code Executor	GSM8k [198], TabMWP [616], Date Understanding [70]
		Calculator	GSM8k [198], MATH [362], CARP [617]
		Model Interface	GPT4Tools [618], Gorilla [619]
		Data Interface	WebQSP [620], MetaQA [621], WTQ [622] WikiSQL [623], TabFact [624], Spider [625]

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Appendix 2: Benchmarks and Evaluation

Method	Evaluation	Model Types	Abilities/Domain	Data Source
Benchmark	MMLU [362]	Base/Fine-tuned/Specialized	General	Human exam/practice
	BIG-bench [70]	Base/Fine-tuned/Specialized	General	Human annotation
	HELM [522]	Base/Fine-tuned/Specialized	General	Benchmark collection
	Open LLM Leaderboard [709]	Base/Fine-tuned/Specialized	General	Benchmark collection
	AGIEval [710]	Base/Fine-tuned/Specialized	General	Human exam/practice
	MMCU [711]	Base/Fine-tuned/Specialized	General	Human exam/practice
	M3KE [712]	Base/Fine-tuned/Specialized	General	Human exam/practice
	C-Eval [713]	Base/Fine-tuned/Specialized	General	Human exam/practice
	Xiezhi [714]	Base/Fine-tuned/Specialized	General	Human exam/practice
	OpenCompass [715]	Base/Fine-tuned/Specialized	General	Benchmark collection
	Chain-of-Thought Hub [716]	Base/Fine-tuned	General	Benchmark collection
	KoLA [717]	Base/Fine-tuned	Knowledge utilization	Web
	ARB [718]	Fine-tuned	Complex reasoning	Human exam/practice
	APIBench [719]	Base/Fine-tuned	Tool manipulation	Web
	APIBank [720]	Fine-tuned	Tool manipulation	Synthesis
	ToolAlpaca [721]	Base/Fine-tuned	Tool manipulation	Synthesis
	T-Bench [722]	Fine-tuned	Tool manipulation	Synthesis
	ToolBench [723]	Fine-tuned	Tool manipulation	Synthesis
	BOLAA [724]	Base/Fine-tuned	Environment interaction	Benchmark collection
	AgentBench [725]	Base/Fine-tuned	Environment interaction	Human annotation/Synthesis
	HaluEval [604]	Base/Fine-tuned	Human alignment	Human annotation/Synthesis
	PromptBench [726]	Base/Fine-tuned	Robustness	Benchmark collection
	HumanEval [105]	Base/Fine-tuned/Specialized	Code synthesis	Human annotation
	MultiMedQA [354]	Specialized	Healthcare	Benchmark collection
	FLUE [727]	Specialized	Finance	Benchmark collection
	LegalBench [728]	Specialized	Legal	Human annotation
Human	Chatbot Arena [729]	Base/Fine-tuned/Specialized	Human Alignment	Human annotation
	SciBench [730]	Fine-tuned	Complex reasoning	Human exam/practice
Model	AlpacaEval [731]	Fine-tuned	Instruction following	Synthesis
	MT-bench [729]	Fine-tuned	Human alignment	Human annotation
	TrustGPT [732]	Base/Fine-tuned	Human alignment	Benchmark collection
	LMExamQA [733]	Base/Fine-tuned	Knowledge utilization	Synthesis
	ChatEval [734]	Base/Fine-tuned	Knowledge utilization	Benchmark collection

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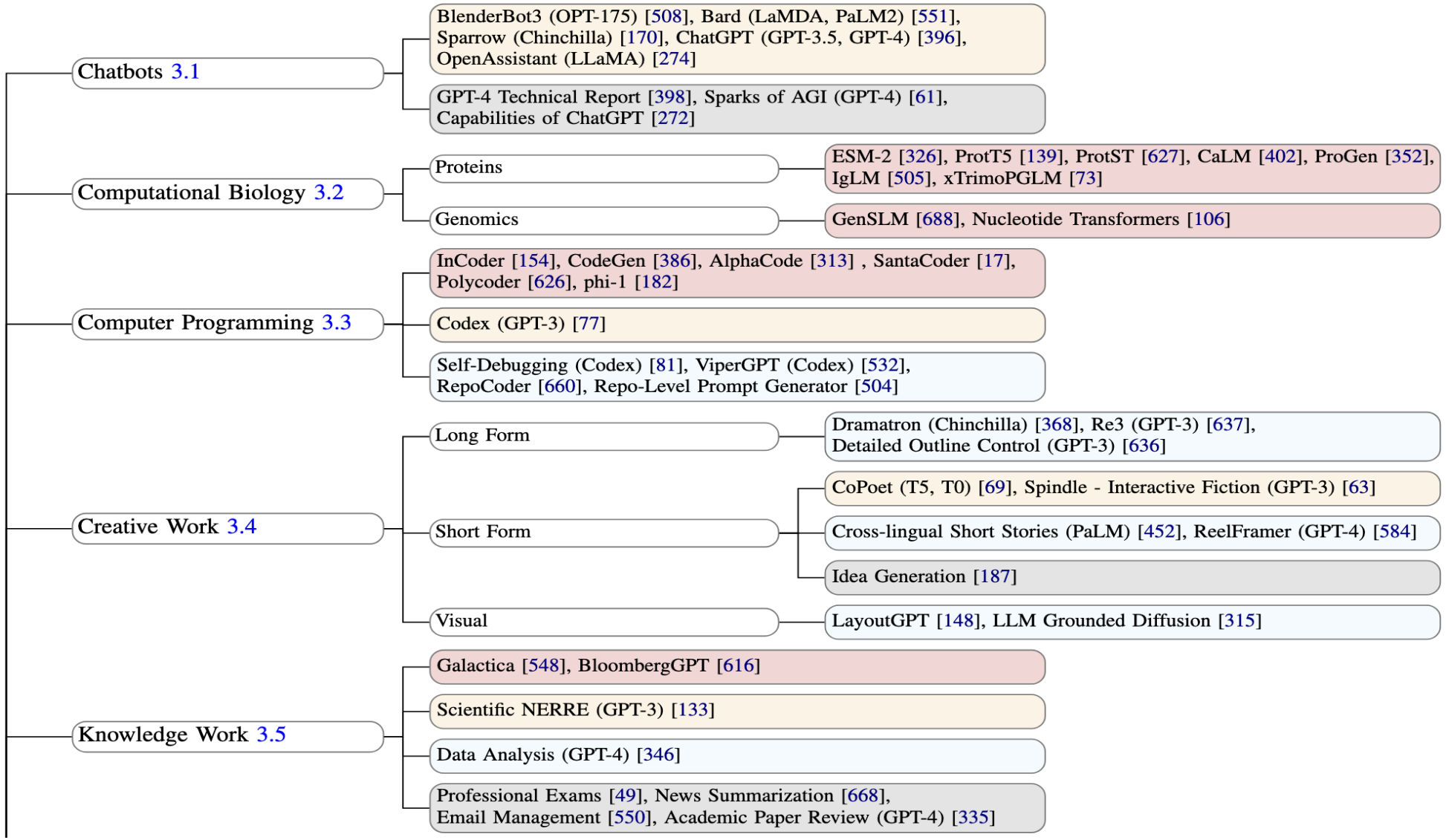
Appendix 3: LLM Applications



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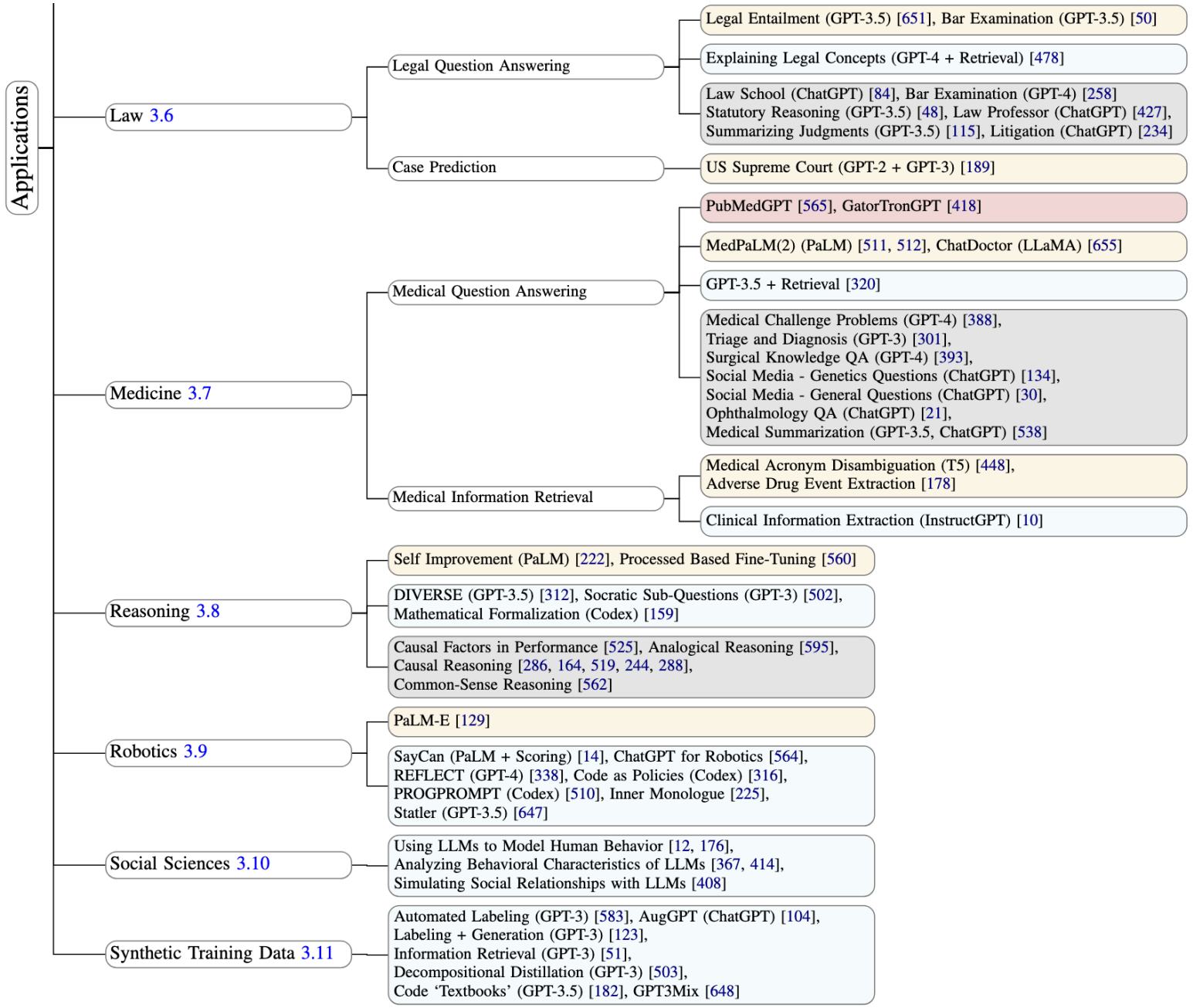
Applications
Law 3.6
Legal Question Answering
Legal Entailment (GPT-3.5) [651], Bar Examination (GPT-3.5) [50]
Explaining Legal Concepts (GPT-4 + Retrieval) [478]
Law School (ChatGPT) [84], Bar Examination (GPT-4) [258]
Statutory Reasoning (GPT-3.5) [48], Law Professor (ChatGPT) [427],
Summarizing Judgments (GPT-3.5) [115], Litigation (ChatGPT) [234]
Case Prediction
US Supreme Court (GPT-2 + GPT-3) [189]
Medicine 3.7
Medical Question Answering
PubMedGPT [565], GatorTronGPT [418]
MedPaLM(2) (PaLM) [511, 512], ChatDoctor (LLaMA) [655]
GPT-3.5 + Retrieval [320]
Medical Challenge Problems (GPT-4) [388],
Triage and Diagnosis (GPT-3) [301],
Surgical Knowledge QA (GPT-4) [393],
Social Media - Genetics Questions (ChatGPT) [134],
Social Media - General Questions (ChatGPT) [30],
Ophthalmology QA (ChatGPT) [21],
Medical Summarization (GPT-3.5, ChatGPT) [538]
Medical Information Retrieval
Medical Acronym Disambiguation (T5) [448],
Adverse Drug Event Extraction [178]
Clinical Information Extraction (InstructGPT) [10]
Reasoning 3.8
Self Improvement (PaLM) [222], Processed Based Fine-Tuning [560]
DIVERSE (GPT-3.5) [312], Socratic Sub-Questions (GPT-3) [502],
Mathematical Formalization (Codex) [159]
Causal Factors in Performance [525], Analogical Reasoning [595],
Causal Reasoning [286, 164, 519, 244, 288],
Common-Sense Reasoning [562]
Robotics 3.9
PaLM-E [129]
SayCan (PaLM + Scoring) [14], ChatGPT for Robotics [564],
REFLECT (GPT-4) [338], Code as Policies (Codex) [316],
PROGPROMPT (Codex) [510], Inner Monologue [225],
Statler (GPT-3.5) [647]
Social Sciences 3.10
Using LLMs to Model Human Behavior [12, 176],
Analyzing Behavioral Characteristics of LLMs [367, 414],
Simulating Social Relationships with LLMs [408]
Synthetic Training Data 3.11
Automated Labeling (GPT-3) [583], AugGPT (ChatGPT) [104],
Labeling + Generation (GPT-3) [123],
Information Retrieval (GPT-3) [51],
Decompositional Distillation (GPT-3) [503],
Code 'Textbooks' (GPT-3.5) [182], GPT3Mix [648]

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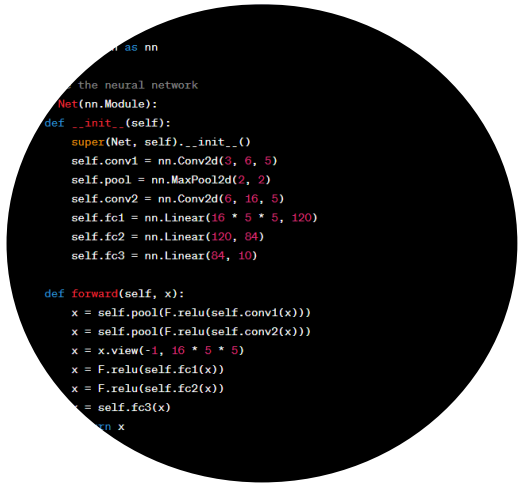
How to succeed in this course?



Practice

Explore

Visualize

Discuss

Ask